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AIUB, MATH-4, Notes.

Final Term
Review

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Eigenvalues and Eigenvectors

Chioy

II Eigenvalues and Eigenvectors:-

Let

$A \rightarrow$ Square matrix $n \times n$

$V \rightarrow$ nonzero column matrix $n \times 1$

$\lambda \rightarrow$ is a scalar.

Such that

$$\boxed{AV = \lambda V}$$

where

$\lambda =$ eigenvalues.

$V =$ eigenvector.

Example:-

$\Rightarrow$ The vector $V = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ is an eigenvector of $A = \begin{bmatrix} 3 & 0 \\ 8 & -1 \end{bmatrix}$ corresponding to the eigenvalue $\lambda = 3$, prove that.

$\Rightarrow$ we know:-

$$AV = \lambda V$$

$$\therefore AV = \begin{bmatrix} 3 & 0 \\ 8 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$= \begin{bmatrix} 3+0 \\ 8-2 \end{bmatrix}$$

$$= \begin{bmatrix} 3 \\ 6 \end{bmatrix}$$

$$= 3 \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \lambda V$$

λ

V

Proved

Characteristic Matrix:-

Procedure:-

⇒ We know:-

$$AV = \lambda V$$

$$\Rightarrow AV - \lambda V = 0$$

$$\Rightarrow (A - \lambda I)V = 0$$

Where
I = Identity Matrix

Formula:-

① Characteristic Matrix : $A - \lambda I$

② Characteristic Polynomial : $|A - \lambda I|$

③ Characteristic Equation : $|A - \lambda I| = 0$

④ Now Solving the C.E to find the eigenvalues (λ)

⑤ And Solving the $(A - \lambda I)V = 0$ to find the eigenvectors (V)

Example 1:-

⇒ Find the eigenvalues and eigenvectors for $A = \begin{bmatrix} 1 & 2 & -1 \\ 0 & -2 & 0 \\ 0 & 5 & 2 \end{bmatrix}$

⇒ We know:-

$$\text{C.E Matrix} = A - \lambda I$$

$$= \begin{bmatrix} 1 & 2 & -1 \\ 0 & -2 & 0 \\ 0 & 5 & 2 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1-\lambda & 2 & -1 \\ 0 & -2-\lambda & 0 \\ 0 & 5 & 2-\lambda \end{bmatrix}$$

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Now :- @.H polynomial $= |A - \lambda I|$

$$= \begin{vmatrix} 1-\lambda & 2 & -1 \\ 0 & -2-\lambda & 0 \\ 0 & 5 & 2-\lambda \end{vmatrix}$$

$$= (1-\lambda)(-4+\lambda^2-2\lambda+\lambda^2) - 2(0-0) - 1(0-0)$$

$$= -4 + \lambda^2 + 4\lambda - \lambda^3$$

$$= -\lambda^3 + \lambda^2 + 4\lambda - 4$$

$$= (\lambda-1)(\lambda+2)(\lambda-2)$$

Now :- @.H Equation $\Rightarrow |A - \lambda I| = 0$

$$(\lambda-1)(\lambda+2)(\lambda-2) = 0$$

$$\therefore \lambda = 1, 2, -2$$

$\therefore$ The eigenvalues :- $\lambda = 1, 2, -2$

Now :- The eigenvectors

We know

$$(A - \lambda I)V = 0$$

For $\lambda = 1$:-

$$= (A - \lambda I)$$

$$= \begin{bmatrix} 1-1 & 2 & -1 \\ 0 & -2-1 & 0 \\ 0 & 5 & 2-1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 2 & -1 \\ 0 & -3 & 0 \\ 0 & 5 & 1 \end{bmatrix}$$

In matrix:-

$$AX = B$$

So, the Augmented matrix:-

$$[A|B] = \left[\begin{array}{ccc|c} 0 & 2 & -1 & 0 \\ 0 & -3 & 0 & 0 \\ 0 & 5 & 1 & 0 \end{array} \right] \quad \begin{array}{l} \tilde{r}_2 = 2r_2 + 3r_1 \\ \tilde{r}_3 = 2r_3 - 5r_1 \end{array}$$

$$= \left[\begin{array}{ccc|c} 0 & 2 & -1 & 0 \\ 0 & 0 & -3 & 0 \\ 0 & 0 & 7 & 0 \end{array} \right] \quad \tilde{r}_2 = 7r_2 + 3r_3$$

$$= \left[\begin{array}{ccc|c} 0 & 2 & -1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 7 & 0 \end{array} \right] \quad r_2 \leftrightarrow r_3$$

$$= \left[\begin{array}{ccc|c} 0 & 2 & -1 & 0 \\ 0 & 0 & 7 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right] \quad \tilde{r}_2 = \frac{r_2}{7}$$

$$= \left[\begin{array}{ccc|c} 0 & 2 & -1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

here v_1 is free Variable

Let,

$$v_1 = a$$

$$v_3 = 0$$

$$v_2 = 0$$

$$\therefore \text{eigenvector } v = \begin{bmatrix} a \\ 0 \\ 0 \end{bmatrix}$$

where $(a \neq 0)$

Ans:-

Note: Similar way
Solved for
other eigenvalues.
Like, -2, 2

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Sample Exercise :- 4.1

⇒ Find the eigenvalues and eigenvectors of the following matrices. For Any Eigenvalues find one eigenvector.

(a) $A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$

⇒ We know:-

For eigenvalues:-

$$|A - \lambda I| = 0$$

$$\begin{aligned} \therefore (A - \lambda I) &= \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 1-\lambda & 2 \\ 2 & 1-\lambda \end{bmatrix} \end{aligned}$$

$$\begin{aligned} \text{Now, } |A - \lambda I| &= \begin{vmatrix} 1-\lambda & 2 \\ 2 & 1-\lambda \end{vmatrix} \\ &= (1-\lambda)(1-\lambda) - 4 \\ &= \lambda^2 - 2\lambda - 3 \end{aligned}$$

$$= (\lambda + 1)(\lambda - 3)$$

∴ eigenvalues:-

$$|A - \lambda I| = 0$$

$$(\lambda + 1)(\lambda - 3) = 0$$

$$\therefore \lambda = -1, 3$$

We know:-

For Eigen Vector $(A - \lambda I) \cdot V = 0$

For $\lambda = 3$:-

$$= (A - \lambda I)$$

$$= \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix} - 3 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1-3 & 2 \\ 2 & 1-3 \end{bmatrix}$$

$$= \begin{bmatrix} -2 & 2 \\ 2 & -2 \end{bmatrix}$$

So, Augmented matrix:-

$$[A|b] = \left[\begin{array}{cc|c} -2 & 2 & 0 \\ 2 & -2 & 0 \end{array} \right] \quad R_2 = R_1 + R_2$$

$$= \left[\begin{array}{cc|c} -2 & 2 & 0 \\ 0 & 0 & 0 \end{array} \right]$$

hence v_2 is free variable.

Let, $v_2 = a$

$$\therefore v_1 = \frac{-2a}{-2} = a$$

$$\therefore V = \begin{bmatrix} a \\ a \end{bmatrix}$$

Ans:-

⑥ $A = \begin{bmatrix} -1 & 0 \\ 3 & 2 \end{bmatrix}$

⇒ we know For Eigen Value:-

$$|A - \lambda I| = 0$$

$$\begin{aligned} \therefore A - \lambda I &= \begin{bmatrix} -1 & 0 \\ 3 & 2 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} -1-\lambda & 0 \\ 3 & 2-\lambda \end{bmatrix} \end{aligned}$$

$$\therefore |A - \lambda I| = \begin{vmatrix} -1-\lambda & 0 \\ 3 & 2-\lambda \end{vmatrix} = 0$$

$$\Rightarrow -2 + \lambda - 2\lambda + \lambda^2 = 0$$

$$\Rightarrow \lambda^2 - \lambda - 2 = 0$$

$$\Rightarrow (\lambda + 1)(\lambda - 2) = 0$$

$$\therefore \lambda = -1, 2$$

we know

For Eigen Vectors:-

$$(A - \lambda I)V = 0$$

For $\lambda = -1$:-

$$= A - \lambda I$$

$$= \begin{bmatrix} -1+1 & 0 \\ 3 & 2+1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 \\ 3 & 3 \end{bmatrix}$$

Now:- Augmented Matrix:-

$$[A|B] = \left[\begin{array}{cc|c} 3 & 3 & 0 \\ 0 & 0 & 0 \end{array} \right]$$

here, v_2 is free variable.

so, let, $v_2 = a$

$$\therefore v_1 = \frac{-3a}{3}$$

$$= -a$$

$$\therefore V = \begin{bmatrix} -a \\ a \end{bmatrix}$$

Ans:-

© $A = \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix}$

$\Rightarrow$ we know:-

For Eigen Values:-

$$|A - \lambda I| = 0$$

Now-

$$A - \lambda I = \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1-\lambda & 4 \\ 2 & 3-\lambda \end{bmatrix}$$

now

$$|A - \lambda I| = \begin{vmatrix} 1-\lambda & 4 \\ 2 & 3-\lambda \end{vmatrix} = 0$$

$$\Rightarrow 3 - \lambda - 8\lambda + \lambda^2 = 0$$

$$\Rightarrow \lambda^2 - 9\lambda - 5 = 0$$

$$\Rightarrow (\lambda + 1)(\lambda - 5) = 0$$

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$$\therefore A = -1, 5$$

We know:-

For Eigenvector

$$(A - \lambda I)V = 0$$

For $\lambda = 1$:-

$$= (A - \lambda I)$$

$$= \begin{bmatrix} 1+1 & 4 \\ 2 & 3+1 \end{bmatrix}$$

$$= \begin{bmatrix} 2 & 4 \\ 2 & 4 \end{bmatrix}$$

Now:- Augmented Matrix:-

$$= \left[\begin{array}{cc|c} 2 & 4 & 0 \\ 2 & 4 & 0 \end{array} \right] R_1 \leftrightarrow R_2$$

$$= \left[\begin{array}{cc|c} 2 & 4 & 0 \\ 2 & 4 & 0 \end{array} \right]$$

$$= \left[\begin{array}{cc|c} 2 & 2 & 0 \\ 0 & 4 & 0 \end{array} \right] \tilde{R}_2 = R_1 - R_2$$

$$= \left[\begin{array}{cc|c} 2 & 4 & 0 \\ 0 & 0 & 0 \end{array} \right]$$

hence

v_2 is free variable

let, $v_2 = a$

$$\therefore v_1 = \frac{4a}{2} = -2a$$

$$\therefore V = \begin{bmatrix} -2a \\ a \end{bmatrix}$$

Ans:-

① $A = \begin{bmatrix} 1 & 2 & -1 \\ 0 & -2 & 0 \\ 0 & -5 & 2 \end{bmatrix}$

we know:-

For Eigen Value:-

$$|A - \lambda I| = 0$$

$$\begin{aligned} \therefore A - \lambda I &= \begin{bmatrix} 1 & 2 & -1 \\ 0 & -2 & 0 \\ 0 & -5 & 2 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 1-\lambda & 2 & -1 \\ 0 & -2-\lambda & 0 \\ 0 & -5 & 2-\lambda \end{bmatrix} \end{aligned}$$

$$\therefore |A - \lambda I| = \begin{vmatrix} 1-\lambda & 2 & -1 \\ 0 & -2-\lambda & 0 \\ 0 & -5 & 2-\lambda \end{vmatrix} = 0$$

$$\Rightarrow (1-\lambda)(-4 + 2\lambda - 2\lambda + \lambda^2) = 0$$

$$\Rightarrow -\lambda^2 + \lambda^2 + 4\lambda - 4 = 0$$

$$\Rightarrow (\lambda - 1)(\lambda - 2)(\lambda + 2) = 0$$

$$\therefore \lambda = 1, -2, 2$$

we know:-

For Eigen Vector:-

$$(A - \lambda I)V = 0$$

For $\lambda = 2$:-

$$\begin{aligned} &= A - \lambda I \\ &= \begin{bmatrix} 1-2 & 2 & -1 \\ 0 & -2-2 & 0 \\ 0 & -5 & 2-2 \end{bmatrix} = \begin{bmatrix} -1 & 2 & -1 \\ 0 & -4 & 0 \\ 0 & -5 & 0 \end{bmatrix} \end{aligned}$$

Now:- Augmented Matrix:-

$$= \left[\begin{array}{ccc|c} -1 & 2 & 1 & 0 \\ 0 & -4 & 0 & 0 \\ 0 & -5 & 0 & 0 \end{array} \right]$$

$$R_3 = 4R_2 + 5R_1$$

$$= \left[\begin{array}{ccc|c} -1 & 2 & 1 & 0 \\ 0 & -4 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

$\therefore V_3$ is free variable

Let,

$$V_3 = a$$

$$\therefore V_2 = 0$$

$$V_1 = -a$$

$$\therefore V = \begin{bmatrix} -a \\ 0 \\ a \end{bmatrix}$$

Ans:-

$$(e) \begin{bmatrix} 2 & 3 & 3 \\ 1 & 3 & 2 \\ -1 & -4 & -3 \end{bmatrix}$$

$\Rightarrow$ we know:-

For Eigenvalue:-

$$A - \lambda I = 0$$

$$\therefore A - \lambda I = \begin{bmatrix} 2 & 3 & 3 \\ 1 & 3 & 2 \\ -1 & -4 & -3 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 2-\lambda & 3 & 3 \\ 1 & 3-\lambda & 2 \\ -1 & -4 & -3-\lambda \end{bmatrix}$$

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$$\therefore |A - \lambda I| = \begin{vmatrix} 2-\lambda & 3 & 3 \\ 1 & 3-\lambda & 2 \\ -1 & -4 & -2-\lambda \end{vmatrix} = 0$$

$$\Rightarrow (2-\lambda)(-1-3\lambda+2\lambda+4) - 3(-3-\lambda+2) + 3(-4+3-\lambda) = 0$$

$$\Rightarrow -18 + 2\lambda^2 + 0\lambda - \lambda^3 + 3 + 3\lambda - 3 - 3\lambda + 16 - 8\lambda = 0$$

$$\Rightarrow -\lambda^3 + 2\lambda^2 + 0\lambda - 18 + 16 - 8\lambda = 0$$

$$\Rightarrow (\lambda-2)(\lambda+4)(\lambda-4) = 0$$

$$\therefore \lambda = 2, 4, -4$$

We know:-

For Eigen Vector:-

$$(A - \lambda I)V = 0$$

For $\lambda = 2$:-

$$\begin{aligned} &= (A - \lambda I) \\ &= \begin{bmatrix} 2-2 & 3 & 3 \\ 1 & 3-2 & 2 \\ -1 & -4 & -2-2 \end{bmatrix} \\ &= \begin{bmatrix} 0 & 3 & 3 \\ 1 & 1 & 2 \\ -1 & -4 & -4 \end{bmatrix} \end{aligned}$$

Now:- Augmented matrix:-

$$= \left[\begin{array}{ccc|c} 0 & 3 & 3 & 0 \\ 1 & 1 & 2 & 0 \\ -1 & -4 & -5 & 0 \end{array} \right] \begin{array}{l} r_3 \leftrightarrow r_1 \\ \tilde{r}_2 = r_2 + r_3 \end{array}$$

$$= \left[\begin{array}{ccc|c} -1 & -4 & -5 & 0 \\ 0 & -3 & -3 & 0 \\ 0 & 3 & 3 & 0 \end{array} \right] \tilde{r}_3 = r_3 + r_2$$

$$= \left[\begin{array}{ccc|c} -1 & -4 & -5 & 0 \\ 0 & -3 & -3 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right] \tilde{r}_2 = \frac{r_2}{-3}$$

$$= \left[\begin{array}{ccc|c} -1 & -4 & -5 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right] \tilde{r}_1 = r_1 + 4r_2$$

$$= \left[\begin{array}{ccc|c} -1 & 0 & -1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

here, V_3 is Free Variable,

Let, $V_3 = a$
 $\therefore V_2 = -a$
 $V_1 = -a$

$$\therefore V = \begin{bmatrix} -a \\ -a \\ a \end{bmatrix}$$

Ans:-

Cayley-Hamilton Theorem:-Condition:-

⇒ Every square matrix satisfies its characteristic equation.

Let,

$$A \longrightarrow n \times n$$

$$\therefore \lambda^n + a_{n-1}\lambda^{n-1} + \dots + a_1\lambda + a_0 = 0$$

Now:-

According to C-H theorem:-

$$A^n + a_{n-1}A^{n-1} + \dots + a_1A + a_0I = 0$$

Examples:-

① Verify the Cayley-Hamilton theorem for the matrix $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & -1 & 1 \\ 3 & 1 & 1 \end{bmatrix}$

② Find A^{-1} using C-H theorem.

⇒ ①

We know:-

$$\text{Characteristic Matrix} = A - \lambda I = \begin{bmatrix} 1-\lambda & 2 & 3 \\ 2 & -1-\lambda & 1 \\ 3 & 1 & 1-\lambda \end{bmatrix}$$

Now, Characteristic polynomial = $|A - \lambda I|$

$$= \begin{vmatrix} 1-\lambda & 2 & 3 \\ 2 & -1-\lambda & 1 \\ 3 & 1 & 1-\lambda \end{vmatrix}$$

$$= (1-\lambda)(-1+\lambda-\lambda+1) - 2(2-2-3) + 3(2+3+3\lambda)$$

$$= -1 + \lambda^2 + \lambda - \lambda^3 + 2 + 4\lambda + 15 + 9\lambda - 1 + \lambda$$

$$= -\lambda^3 + \lambda^2 + 14\lambda + 16 - 1 + \lambda$$

$$= -\lambda^3 + \lambda^2 + 15\lambda + 15$$

$$= \lambda^3 - \lambda^2 - 15\lambda - 15$$

So, the characteristic equation:-

$$\Rightarrow |A - \lambda I| = 0$$

$$\therefore \lambda^3 - \lambda^2 - 15\lambda - 15 = 0$$

$\therefore$ According to C-H theorem:-

$$A^3 - A^2 - 15A - 15I = 0$$

①

$$\therefore A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & -1 & 1 \\ 3 & 1 & 1 \end{bmatrix}$$

$$\therefore A^2 = A \cdot A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & -1 & 1 \\ 3 & 1 & 1 \end{bmatrix} \times \begin{bmatrix} 1 & 2 & 3 \\ 2 & -1 & 1 \\ 3 & 1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1+4+9 & 2-2+3 & 3+2+3 \\ 2-2+3 & 4+1+1 & 6-1+1 \\ 3+2+3 & 6-1+1 & 9+1+1 \end{bmatrix} = \begin{bmatrix} 14 & 3 & 8 \\ 3 & 6 & 6 \\ 8 & 6 & 11 \end{bmatrix}$$

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$$\therefore A^2 \cdot A = A^3 = \begin{bmatrix} 14 & 3 & 8 \\ 3 & 6 & 6 \\ 8 & 6 & 11 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 2 & 1 & 1 \\ 3 & 1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 14+6+24 & 28-3+8 & 42+3+8 \\ 3+12+18 & 6-6+6 & 9+6+6 \\ 8+12+33 & 16-6+11 & 24+6+11 \end{bmatrix}$$

$$= \begin{bmatrix} 44 & 33 & 53 \\ 33 & 6 & 21 \\ 53 & 21 & 41 \end{bmatrix}$$

Now put All the value in eq (1) =

$$A^3 - A^2 - 15A - 15I = 0$$

L.H.S

$$= \begin{bmatrix} 44 & 33 & 53 \\ 33 & 6 & 21 \\ 53 & 21 & 41 \end{bmatrix} - \begin{bmatrix} 14 & 3 & 8 \\ 3 & 6 & 6 \\ 8 & 6 & 11 \end{bmatrix} - 15 \begin{bmatrix} 1 & 2 & 3 \\ 2 & 1 & 1 \\ 3 & 1 & 1 \end{bmatrix} - 15 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 44-14-15-15 & 33-3-30-0 & 53-8-45-0 \\ 33-3-30-0 & 6-6+15-15 & 21-6-15-0 \\ 53-8-45-0 & 21-6-15-0 & 41-11-15-15 \end{bmatrix}$$

$$= 0$$

= R.H.S

(Proved)

(11)

Formula:-

⇒ For finding A^{-1} using C-H theorem then
Multiply C-H theorem by A^{-1} then solve the equation
For A^{-1} .

we have:-

$$A^3 - A^2 - 15A - 15I = 0 \quad [\text{multiply } A^{-1}]$$

$$\Rightarrow A^2 - A - 15AA^{-1} - 15IA^{-1} = 0$$

$$\Rightarrow A^2 - A - 15I - 15IA^{-1} = 0 \quad [\because AA^{-1} = I]$$

$$\Rightarrow -15A^{-1} = -A^2 + A + 15I \quad [\because A^{-1}I = A^{-1}]$$

$$\Rightarrow A^{-1} = \frac{1}{15} [A^2 - A - 15I]$$

$$= \frac{1}{15} \begin{bmatrix} 14 & 3 & 8 \\ 3 & 6 & 6 \\ 8 & 6 & 11 \end{bmatrix} - \begin{bmatrix} 1 & 2 & 3 \\ 2 & 1 & 1 \\ 3 & 1 & 1 \end{bmatrix} - \begin{bmatrix} 15 & 0 & 0 \\ 0 & 15 & 0 \\ 0 & 0 & 15 \end{bmatrix}$$

$$= \frac{1}{15} \begin{bmatrix} 14-1-15 & 3-2-0 & 8-3-0 \\ 3-2-0 & 6+1-15 & 6-1-0 \\ 8-3+0 & 6-1-0 & 11-1-15 \end{bmatrix}$$

$$= \frac{1}{15} \begin{bmatrix} -2 & 1 & 5 \\ 1 & -8 & 5 \\ 5 & 5 & -5 \end{bmatrix}$$

Ans:-

Sample - Exercise 84.2

Q: ① Verify the Cayley-Hamilton theorem and

② Find A^{-1} / Inverse matrix

For the following

① $A = \begin{bmatrix} 2 & 5 \\ 1 & 3 \end{bmatrix}$

①

We know:-Ch-equation:-

$$|A - \lambda I| = 0$$

$$\begin{aligned} \therefore A - \lambda I &= \begin{bmatrix} 2 & 5 \\ 1 & 3 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 2-\lambda & 5 \\ 1 & 3-\lambda \end{bmatrix} \end{aligned}$$

$$\therefore \text{Now, } |A - \lambda I| = \begin{vmatrix} 2-\lambda & 5 \\ 1 & 3-\lambda \end{vmatrix} = 0$$

$$\Rightarrow 6 - 2\lambda - 3\lambda + \lambda^2 = 5 = 0$$

$$\Rightarrow \lambda^2 - 5\lambda - 5 = 0$$

$$\Rightarrow \lambda^2 - 5\lambda + 1 = 0$$

According to C-H theorem:-

$$A^2 - 5A + I = 0 \dots \dots \textcircled{1}$$

$$\begin{aligned}\therefore A \cdot A &= A^2 = \begin{bmatrix} 2 & 5 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} 2 & 5 \\ 1 & 3 \end{bmatrix} \\ &= \begin{bmatrix} 4+5 & 10+15 \\ 2+3 & 5+9 \end{bmatrix} \\ &= \begin{bmatrix} 9 & 25 \\ 5 & 14 \end{bmatrix}\end{aligned}$$

Now put the value in eqⁿ (1) $\Rightarrow$

$$\begin{aligned}\text{L.H.S} \Rightarrow A^2 - 5A + I &= 0 \\ \Rightarrow \begin{bmatrix} 9 & 25 \\ 5 & 14 \end{bmatrix} - \begin{bmatrix} 10 & 25 \\ 5 & 15 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 9-10+1 & 25-25+0 \\ 5-5+0 & 14-15+1 \end{bmatrix} \\ &= \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = 0 \\ &= \text{R.H.S}\end{aligned}$$

Proved

(11)

Now,

Multiply A^{-1} in e-H theorem

$$A - 5AA^{-1} + IA^{-1} = 0$$

$$A - 5I + A^{-1} = 0$$

$$A^{-1} = 5I - A$$

$$A^{-1} = \begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix} - \begin{bmatrix} 2 & 5 \\ 1 & 3 \end{bmatrix} = \begin{bmatrix} 3 & -5 \\ -1 & 2 \end{bmatrix}$$

Ans:

$$\textcircled{b} \quad A = \begin{bmatrix} 1 & 2 & 2 \\ 3 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$$

①

 $\Rightarrow$ We know:-ch-Equation:-

$$|A - \lambda I| = 0$$

$$\therefore A - \lambda I = \begin{bmatrix} 1 & 2 & 2 \\ 3 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1-\lambda & 2 & 2 \\ 3 & 1-\lambda & 0 \\ 1 & 1 & 1-\lambda \end{bmatrix}$$

$$\therefore |A - \lambda I| = \begin{vmatrix} 1-\lambda & 2 & 2 \\ 3 & 1-\lambda & 0 \\ 1 & 1 & 1-\lambda \end{vmatrix} = 0$$

$$\Rightarrow (1-\lambda)(1-\lambda-\lambda+\lambda^2) - 2(3-3\lambda) + 2(3-1+\lambda) = 0$$

$$\Rightarrow (1-\lambda)(1-2\lambda+\lambda^2) - 6 + 6\lambda + 6 - 2 + 2\lambda = 0$$

$$\Rightarrow 1 - 2\lambda + \lambda^2 - \lambda + 2\lambda^2 - \lambda^3 - 6 + 6\lambda + 4 + 2\lambda = 0$$

$$\Rightarrow -\lambda^3 + 3\lambda^2 + 5\lambda - 1 = 0$$

$$\Rightarrow \lambda^3 - 3\lambda^2 - 5\lambda + 1 = 0$$

According to C-H theorem:-

$$A^3 - 3A^2 - 5A + I = 0 \quad \text{--- ①}$$

$$\therefore A \cdot A = A^2 = \begin{bmatrix} 1 & 2 & 2 \\ 3 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 2 \\ 3 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 9 & 6 & 4 \\ 8 & 7 & 6 \\ 5 & 4 & 3 \end{bmatrix}$$

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$$A^2 \cdot A = A^3 = \begin{bmatrix} 9 & 6 & 4 \\ 6 & 7 & 6 \\ 5 & 4 & 3 \end{bmatrix} \times \begin{bmatrix} 1 & 2 & 2 \\ 3 & 1 & 0 \\ 2 & 1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 31 & 28 & 22 \\ 33 & 25 & 18 \\ 20 & 17 & 13 \end{bmatrix}$$

Now, put the value of A^2, A^3 in eq. (1) $\Rightarrow$

$$A^3 - 3A^2 - 5A + I = 0$$

$\therefore$ L.H.S. -

$$= \begin{bmatrix} 31 & 28 & 22 \\ 33 & 25 & 18 \\ 20 & 17 & 13 \end{bmatrix} - \begin{bmatrix} 27 & 18 & 12 \\ 18 & 21 & 18 \\ 15 & 12 & 9 \end{bmatrix} - \begin{bmatrix} 5 & 10 & 10 \\ 15 & 5 & 0 \\ 5 & 5 & 5 \end{bmatrix} + \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 31-27-5+1 & 28-18-10+0 & 22-12-10+0 \\ 33-18-15+0 & 25-21-5+1 & 18-18-0+0 \\ 20-15-5+0 & 17-12-5+0 & 13-9-5+1 \end{bmatrix}$$

$$= 0$$

$$= \text{P.H.S.}$$

(Proved)

(ii)

Now multiply A^{-1} in C-H theorem:-

$$A^{-1}(A^3 - 3A^2 - 5A + I) = 0$$

$$\Rightarrow A^2 - 3A - 5AA^{-1} + IA^{-1} = 0$$

$$\Rightarrow A^{-1} = +3A - A^2 - 5I$$

$$= \begin{bmatrix} 3 & 6 & 6 \\ 0 & 3 & 0 \\ 3 & 3 & 3 \end{bmatrix} - \begin{bmatrix} 0 & 6 & 4 \\ 6 & 7 & 6 \\ 5 & 4 & 3 \end{bmatrix} - \begin{bmatrix} 5 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{bmatrix}$$

$$= \begin{bmatrix} -1 & 0 & 2 \\ 3 & 1 & -6 \\ -2 & -1 & 5 \end{bmatrix}$$

Ans:-

⑩ $A = \begin{bmatrix} 2 & 0 & 1 \\ -2 & 3 & 4 \\ -5 & 5 & 6 \end{bmatrix}$

①

We know:-

C-H equation:-

$$|A - \lambda I| = 0$$

$$\therefore A - \lambda I = \begin{bmatrix} 2-\lambda & 0 & 1 \\ -2 & 3-\lambda & 4 \\ -5 & 5 & 6-\lambda \end{bmatrix}$$

$$\therefore |A - \lambda I| = \begin{vmatrix} 2-\lambda & 0 & 1 \\ -2 & 3-\lambda & 4 \\ -5 & 5 & 6-\lambda \end{vmatrix} = 0$$

$$\Rightarrow \lambda^3 - 11\lambda^2 + 21\lambda - 1 = 0$$

Now:- According to C-H theorem:-

$$A^3 - 11A^2 + 21A - I = 0 \quad \text{--- (1)}$$

$$\therefore A \cdot A = A^2 = \begin{bmatrix} 2 & 0 & 1 \\ -2 & 3 & 4 \\ -5 & 5 & 6 \end{bmatrix} \begin{bmatrix} 2 & 0 & 1 \\ -2 & 3 & 4 \\ -5 & 5 & 6 \end{bmatrix}$$

$$= \begin{bmatrix} -1 & 5 & 8 \\ -30 & 20 & 34 \\ -50 & 45 & 51 \end{bmatrix}$$

$$\therefore A^2 \cdot A = A^3 = \begin{bmatrix} -1 & 5 & 8 \\ -30 & 20 & 34 \\ -50 & 45 & 51 \end{bmatrix} \begin{bmatrix} 2 & 0 & 1 \\ -2 & 3 & 4 \\ -5 & 5 & 6 \end{bmatrix}$$

$$= \begin{bmatrix} -52 & 55 & 67 \\ -288 & 257 & 290 \\ -445 & 390 & 436 \end{bmatrix}$$

Now put All the value in eq (1) :-

$$A^3 - 11A^2 + 21A - I = 0$$

$$\text{L.H.S} = \begin{bmatrix} -52 & 55 & 67 \\ -288 & 257 & 290 \\ -445 & 390 & 436 \end{bmatrix} - \begin{bmatrix} -1 & 5 & 8 \\ -30 & 310 & 374 \\ -550 & 495 & 561 \end{bmatrix} + \begin{bmatrix} 42 & 0 & 21 \\ -42 & 63 & 84 \\ -105 & 105 & 126 \end{bmatrix} - \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} -52+1+42-1 & 55-5+0-0 & 67-8+21-0 \\ -288+330-42-0 & 257-310+63-1 & 290-374+84-0 \\ -445+550-105+0 & 390-495+105-0 & 436-561+126-1 \end{bmatrix}$$

$$= 0$$

$$= \text{R.H.S}$$

Proved

(ii)

Now Multiply A^{-1} in e-H theorem:-

$$(1) \quad A^{-1}(A^3 - 11A^2 + 21A - I) = 0$$

$$A^2 - 11A + 21AA^{-1} - IA^{-1} = 0$$

$$A^2 - 11A + 21I - A^{-1} = 0$$

$$\therefore A^{-1} = A^2 - 11A + 21I$$

$$= \begin{bmatrix} -1 & 5 & 8 \\ -30 & 29 & 34 \\ -50 & 45 & 51 \end{bmatrix} - \begin{bmatrix} 22 & 0 & 21 \\ -22 & 33 & 44 \\ -55 & 55 & 66 \end{bmatrix} + \begin{bmatrix} 21 & 0 & 0 \\ 0 & 21 & 0 \\ 0 & 0 & 21 \end{bmatrix}$$

$$= \begin{bmatrix} -2 & 5 & -3 \\ -8 & 17 & -10 \\ 5 & -10 & 6 \end{bmatrix}$$

Ans:-

$$(d) \quad A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{bmatrix}$$

(i)

We know:-

Ch equation:-

$$|A - \lambda I| = 0$$

$$\therefore A - \lambda I = \begin{bmatrix} 1-\lambda & 2 & 3 \\ 2 & 5-\lambda & 3 \\ 1 & 0 & 8-\lambda \end{bmatrix}$$

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Now:- $|A - \lambda I| = \begin{vmatrix} 1-\lambda & 2 & 3 \\ 2 & 5-\lambda & 3 \\ 1 & 0 & 8-\lambda \end{vmatrix} = 0$

$$= \lambda^3 - 14\lambda^2 + 46\lambda + 4 = 0$$

According to C-H theorem:-

$$A^3 - 14A^2 + 46A + I = 0 \quad \text{--- (1)}$$

$$A \cdot A = A^2 = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{bmatrix}$$

$$= \begin{bmatrix} 8 & 12 & 33 \\ 15 & 29 & 45 \\ 7 & 2 & 67 \end{bmatrix}$$

$$A^2 \cdot A = A^3 = \begin{bmatrix} 8 & 12 & 33 \\ 15 & 29 & 45 \\ 7 & 2 & 67 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{bmatrix}$$

$$= \begin{bmatrix} 65 & 76 & 324 \\ 118 & 175 & 402 \\ 80 & 28 & 567 \end{bmatrix}$$

Put All the value in eq (1)

L.H.S:-

$$= \begin{bmatrix} 65 & 76 & 324 \\ 118 & 175 & 402 \\ 80 & 28 & 567 \end{bmatrix} - \begin{bmatrix} 112 & 168 & 462 \\ 210 & 406 & 630 \\ 146 & 28 & 998 \end{bmatrix} + \begin{bmatrix} 46 & 0 & 198 \\ 0 & 46 & 0 \\ 0 & 0 & 46 \end{bmatrix}$$

$$+ \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= 0$$

$$= \text{R.H.S}$$

(Proved)

ii

Multiply A^{-1} in the C-H theorem

$$A^{-1}(A^3 - 14A^2 + 46A + I) = 0$$

$$A^2 - 14A + 46AA^{-1} + IA^{-1} = 0$$

$$A^{-1} = 14A - A^2 - 46I$$

$$= \begin{bmatrix} 14 & 28 & 42 \\ 28 & 70 & 42 \\ 14 & 0 & 112 \end{bmatrix} - \begin{bmatrix} 8 & 12 & 33 \\ 15 & 29 & 45 \\ 7 & 2 & 67 \end{bmatrix} - \begin{bmatrix} 46 & 0 & 0 \\ 0 & 46 & 0 \\ 0 & 0 & 46 \end{bmatrix}$$

$$= \begin{bmatrix} -40 & 16 & 9 \\ 13 & -5 & -3 \\ 5 & -2 & -1 \end{bmatrix} \quad \text{Ans:-}$$

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Vector Spaces:-

2-Dimensional Vector:-

⇒ The set of all order pairs of real numbers is called 2-Dimensional Vector space

Ex:- $(1, 2); (-1, 3), (\sqrt{3}, 0)$

3-Dimensional Vector:-

⇒ The set of all order triplets of real numbers is called 3-Dimensional vector space.

Ex:- $(1, 2, 3), (-1, 1, 0)$

in Vector Space there are two operation:-

1. Addition

2. Scalar Multiplication

Linear Combination:-

$\Rightarrow$ Let V be a Vector space $u_1, u_2, \dots, u_n \in V$,
 $\alpha_1, \alpha_2, \dots, \alpha_n$ scalars.

$$\text{if } \boxed{\alpha_1 u_1 + \alpha_2 u_2 + \dots + \alpha_n u_n = u \in V}$$

then u is a Linear Combination of $u_1, u_2, \dots, u_n$

Example:-

$\Rightarrow$ Write the vector $u = (1, -2, 5)$ as a Linear Combination of the vector $u_1 = (1, 1, 1)$, $u_2 = (1, 2, 1)$, $u_3 = (2, -1, 1)$

$\Rightarrow$ Let

$$\alpha_1 u_1 + \alpha_2 u_2 + \alpha_3 u_3 = u$$

$$\therefore \alpha_1 (1, 1, 1) + \alpha_2 (1, 2, 1) + \alpha_3 (2, -1, 1) = (1, -2, 5)$$

Now,

Corresponding Components:-

$$\alpha_1 + \alpha_2 + 2\alpha_3 = 1$$

$$\alpha_1 + 2\alpha_2 - \alpha_3 = -2$$

$$\alpha_1 + 3\alpha_2 + \alpha_3 = 5$$

∴ Augmented Matrix:-

$$= \left[\begin{array}{ccc|c} 1 & 1 & 2 & 1 \\ 1 & 2 & -1 & -2 \\ 1 & 3 & 1 & 5 \end{array} \right] \begin{array}{l} \tilde{r}_2 = r_2 - r_1 \\ \tilde{r}_3 = r_3 - r_1 \end{array}$$

$$= \left[\begin{array}{ccc|c} 1 & 1 & 2 & 1 \\ 0 & 1 & -3 & -3 \\ 0 & 0 & 2 & 4 \end{array} \right] \tilde{r}_3 = r_3 - 2r_2$$

$$= \left[\begin{array}{ccc|c} 1 & 1 & 2 & 1 \\ 0 & 1 & -3 & -3 \\ 0 & 0 & 5 & 10 \end{array} \right] \tilde{r}_4 = r_1 - r_2$$

$$= \left[\begin{array}{ccc|c} 1 & 0 & 5 & 4 \\ 0 & 1 & -3 & -3 \\ 0 & 0 & 5 & 10 \end{array} \right] \begin{array}{l} \tilde{r}_4 = r_1 - r_3 \\ \tilde{r}_2 = r_2 + 3r_3 \end{array}$$

$$= \left[\begin{array}{ccc|c} 1 & 0 & 0 & -6 \\ 0 & 1 & 0 & 9 \\ 0 & 0 & 1 & 2 \end{array} \right]$$

∴ Linear Combination $u = -6u_1 + 9u_2 + 2u_3$

Ans:-

Linearly Dependent & Independent

Let,

V be a vector space. And,

$u_1, u_2, \dots, u_n \in V$

$\alpha_1, \alpha_2, \dots, \alpha_n$ Scalars.

if

$$\alpha_1 u_1 + \alpha_2 u_2 + \dots + \alpha_n u_n = 0$$

the solution are all zeros, the $u_1, u_2, \dots, u_n$ are called linearly independent. otherwise linearly dependent.

Examples 1:-

$\Rightarrow$ Test whether the vector $u_1 = (1, 0, 1)$, $u_2 = (-3, 2, 6)$, $u_3 = (4, 5, -2)$ are linearly dependent or independent.

$\Rightarrow$ Let,

$$u_1 \alpha_1 + u_2 \alpha_2 + u_3 \alpha_3 = 0$$

$$\therefore \alpha_1 (1, 0, 1) + \alpha_2 (-3, 2, 6) + \alpha_3 (4, 5, -2) = 0$$

Now,

we can write from above eqⁿ:-

$$\alpha_1 - 3\alpha_2 + 4\alpha_3 = 0$$

$$0\alpha_1 + 2\alpha_2 + 5\alpha_3 = 0$$

$$\alpha_1 + 6\alpha_2 - 2\alpha_3 = 0$$

∴ Augmented Matrix:-

$$\left[\begin{array}{ccc|c} 1 & -3 & 4 & 0 \\ 0 & 2 & 5 & 0 \\ 1 & 6 & -2 & 0 \end{array} \right]$$

$$R_3 = R_4 - R_1$$

$$= \left[\begin{array}{ccc|c} 1 & -3 & 4 & 0 \\ 0 & 2 & 5 & 0 \\ 0 & 0 & -6 & 0 \end{array} \right]$$

$$R_3 = 2R_3 - 2 \times R_2$$

$$= \left[\begin{array}{ccc|c} 1 & -3 & 4 & 0 \\ 0 & 2 & 5 & 0 \\ 0 & 0 & -52 & 0 \end{array} \right]$$

$$\alpha_1 = 0$$

$$\alpha_2 = 0$$

$$\alpha_3 = 0$$

∴ Consistent & determinant

∴ the Vector are Linearly independent

if:- $\left[\begin{array}{ccc|c} 1 & 2 & 3 & 0 \\ 0 & 1 & 4 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$

then

$$\alpha_3 = a$$

$$\alpha_2 = -4a$$

$$\alpha_1 = -2(-4a) - 3a = 5a$$

So, that, the Vector are Linearly Dependent.

#Sample-Exercise: 4.3

Q:- Write the vector u as a linear combination of the vectors u_1, u_2, u_3 where.

$$\text{I: } u = (5, 0, 5), u_1 = (1, -1, 3), u_2 = (2, 1, 4), u_3 = (3, 2, 5)$$

$\Rightarrow$ Let

$$\alpha_1 u_1 + \alpha_2 u_2 + \alpha_3 u_3 = u$$

$$\alpha_1 (1, -1, 3) + \alpha_2 (2, 1, 4) + \alpha_3 (3, 2, 5) = (5, 0, 5)$$

Now,

Corresponding Component:-

$$\alpha_1 + 2\alpha_2 + 3\alpha_3 = 5$$

$$-\alpha_1 + \alpha_2 + 2\alpha_3 = 0$$

$$3\alpha_1 + 4\alpha_2 + 5\alpha_3 = 5$$

$\therefore$ Augmented Matrix:-

$$= \left[\begin{array}{ccc|c} 1 & 2 & 3 & 5 \\ -1 & 1 & 2 & 0 \\ 3 & 4 & 5 & 5 \end{array} \right] \quad \begin{array}{l} \tilde{r}_2 = r_2 + r_1 \\ \tilde{r}_3 = r_3 - 3r_1 \end{array}$$

$$= \left[\begin{array}{ccc|c} 1 & 2 & 3 & 5 \\ 0 & 3 & 5 & 14 \\ 0 & -2 & -4 & -10 \end{array} \right] \quad \begin{array}{l} \tilde{r}_3 = r_3 + r_2 \\ \tilde{r}_1 = 3r_1 - 2r_2 \end{array}$$

$$= \left[\begin{array}{ccc|c} 1 & 2 & 3 & 5 \\ 0 & 3 & 5 & 14 \\ 0 & 0 & -2 & -2 \end{array} \right] \quad \tilde{r}_1 = 3r_1 - 2r_2$$

There are mistakes in the solutions to these math problems 1-8. But the process is okay. You can carefully do the multiplication and solve the problem.

$$= \left[\begin{array}{ccc|c} 1 & 0 & -1 & -13 \\ 0 & 3 & 5 & 14 \\ 0 & 0 & -2 & -2 \end{array} \right] \quad \tilde{r}_1 = 2r_4 - r_3$$

$$= \left[\begin{array}{ccc|c} 2 & 0 & 0 & -24 \\ 0 & 3 & 5 & 14 \\ 0 & 0 & -2 & -2 \end{array} \right] \quad \tilde{r}_2 = 2r_2 + 5r_3$$

$$= \left[\begin{array}{ccc|c} 2 & 0 & 0 & -24 \\ 0 & 3 & 0 & 18 \\ 0 & 0 & -2 & -2 \end{array} \right] \quad \begin{aligned} r_1 &= \frac{r_1}{2} \\ r_2 &= \frac{r_2}{3} \\ r_3 &= \frac{r_3}{-2} \end{aligned}$$

$$= \left[\begin{array}{ccc|c} 1 & 0 & 0 & -12 \\ 0 & 1 & 0 & 6 \\ 0 & 0 & 1 & 1 \end{array} \right]$$

$\therefore$ Linear Combination $u = -12u_1 + 6u_2 + u_3$

Ans:-

② $u = (6, 20, 2), u_1 = (1, 2, 3), u_2 = (1, 3, -2), u_3 = (1, 4, 1)$

$\Rightarrow$ Let

$$d_1 u_1 + d_2 u_2 + d_3 u_3 = u$$

$$\therefore d_1 (1, 2, 3) + d_2 (1, 3, -2) + d_3 (1, 4, 1) = (6, 20, 2)$$

Now,

Corresponding Components:-

$$d_1 + d_2 + d_3 = 6$$

$$2d_1 + 3d_2 + 4d_3 = 20$$

$$3d_1 - 2d_2 + d_3 = 2$$

$\therefore$ Augmented Matrix:-

$$= \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 2 & 3 & 4 & 20 \\ 3 & -2 & 1 & 2 \end{array} \right] \quad \begin{array}{l} \tilde{r}_2 = r_2 - 2r_1 \\ \tilde{r}_3 = r_3 - 3r_1 \end{array}$$

$$= \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 8 \\ 0 & -5 & -2 & -16 \end{array} \right] \quad \tilde{r}_3 = r_3 + 5r_2$$

$$= \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 8 \\ 0 & 0 & 8 & 24 \end{array} \right] \quad \tilde{r}_1 = r_1 - r_2$$

$$= \left[\begin{array}{ccc|c} 1 & 0 & -1 & -2 \\ 0 & 1 & 2 & 8 \\ 0 & 0 & 8 & 24 \end{array} \right] \quad \begin{array}{l} \tilde{r}_1 = 8r_1 + r_3 \\ \tilde{r}_2 = 4r_2 - r_3 \end{array}$$

$$= \left[\begin{array}{ccc|c} 1 & 0 & 0 & 8 \\ 0 & 1 & 0 & 8 \\ 0 & 0 & 8 & 24 \end{array} \right] \quad \tilde{r}_3 = \frac{r_3}{8}$$

$$= \left[\begin{array}{ccc|c} 1 & 0 & 0 & 8 \\ 0 & 1 & 0 & 8 \\ 0 & 0 & 1 & 3 \end{array} \right]$$

$\therefore$ Linear Combination $u = 8u_1 + 8u_2 + 3u_3$

Ans:-

Q:- Test whether the following Vectors are linearly Independent or dependent.

④ $u_1 = (2, -1, 4), u_2 = (3, 6, 2), u_3 = (2, 10, -4)$

⇒ We know:-

For linearly dependency or Independence:-

$$d_1 u_1 + d_2 u_2 + d_3 u_3 = 0$$

$$\therefore d_1(2, -1, 4) + d_2(3, 6, 2) + d_3(2, 10, -4) = 0$$

so, the corresponding equation:-

$$2d_1 + 3d_2 + 2d_3 = 0$$

$$-d_1 + 6d_2 + 10d_3 = 0$$

$$4d_1 + 2d_2 - 4d_3 = 0$$

Now, the Augmented matrix:-

$$= \left[\begin{array}{ccc|c} 2 & 3 & 2 & 0 \\ -1 & 6 & 10 & 0 \\ 4 & 2 & -4 & 0 \end{array} \right] \begin{array}{l} \tilde{r}_2 = 2r_2 + r_1 \\ \tilde{r}_3 = r_3 - 2r_1 \end{array}$$

$$= \left[\begin{array}{ccc|c} 2 & 3 & 2 & 0 \\ 0 & 15 & 22 & 0 \\ 0 & -4 & -8 & 0 \end{array} \right] \tilde{r}_3 = 15r_3 + 4r_2$$

$$= \left[\begin{array}{ccc|c} 2 & 3 & 2 & 0 \\ 0 & 15 & 22 & 0 \\ 0 & 0 & -32 & 0 \end{array} \right]$$

$\therefore$ Consistent & determinant.

$$\therefore d_3 = 0$$

$$d_2 = 0$$

$$d_1 = 0$$

So, the matrix is Linearly Independent.

Ans 1

(5) $u_1 = (3, 0, 1, -1), u_2 = (2, -1, 0, 1), u_3 = (1, 1, 1, -2)$

$\Rightarrow$ We know—

For Linearly Dependence or Independence:

$$a_1 u_1 + a_2 u_2 + a_3 u_3 = 0$$

$$\therefore d_1(3, 0, 1, -1) + d_2(2, -1, 0, 1) + d_3(1, 1, 1, -2) = 0$$

Now, the corresponding equation—

$$3d_1 + 2d_2 + d_3 = 0$$

$$0 - d_2 + d_3 = 0$$

$$d_1 + 0 + d_3 = 0$$

$$-d_1 + d_2 - 2d_3 = 0$$

So, the Augmented Matrix

$$= \left[\begin{array}{ccc|c} 3 & 2 & 1 & 0 \\ 0 & -1 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ -1 & 1 & -2 & 0 \end{array} \right] \quad \hat{r}_4 = r_4 + r_3$$

$$= \left[\begin{array}{ccc|c} 3 & 2 & 1 & 0 \\ 0 & -1 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & -1 & 0 \end{array} \right] \quad \begin{array}{l} \hat{r}_3 = 3r_3 - r_1 \\ \hat{r}_4 = r_4 + r_2 \end{array}$$

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$$= \left[\begin{array}{ccc|c} 3 & 2 & 1 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & -2 & 2 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right] \quad \tilde{r}_3 = r_3 - 2r_2$$

$$= \left[\begin{array}{ccc|c} 3 & 2 & 1 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

here, r_3 is free variable

Let, $r_3 = a$

$$r_2 = a$$

$$r_1 = \frac{-2a - a}{3}$$

$$= -a$$

$\therefore$ the matrix is linearly dependent!

Ans:-

⑥ $u_1 = (1, -1, 2), u_2 = (3, -5, 1), u_3 = (2, 7, 8), u_4 = (-1, 1, 1)$

We know:-

For linearly dependency or independency:-

$$a_1 u_1 + a_2 u_2 + a_3 u_3 + a_4 u_4 = 0$$

$$\therefore (1, -1, 2)a_1 + (3, -5, 1)a_2 + (2, 7, 8)a_3 + (-1, 1, 1)a_4 = 0$$

∴ the corresponding equation:-

$$1x_1 + 2x_2 + 3x_3 - x_4 = 0$$

$$-1x_1 - 5x_2 + 7x_3 + 1x_4 = 0$$

$$2x_1 + x_2 + 8x_3 + x_4 = 0$$

Now:- the Augmented Matrix

$$= \left[\begin{array}{cccc|c} 1 & 2 & 3 & -1 & 0 \\ -1 & -5 & 7 & 1 & 0 \\ 2 & 1 & 8 & 1 & 0 \end{array} \right] \begin{array}{l} \tilde{r}_2 = r_2 + r_1 \\ \tilde{r}_3 = r_3 - 2r_1 \end{array}$$

$$= \left[\begin{array}{cccc|c} 1 & 2 & 3 & -1 & 0 \\ 0 & -3 & 10 & 0 & 0 \\ 0 & -3 & 2 & 3 & 0 \end{array} \right] \tilde{r}_3 = r_3 - r_2$$

$$= \left[\begin{array}{cccc|c} 1 & 2 & 3 & -1 & 0 \\ 0 & -3 & 10 & 0 & 0 \\ 0 & 0 & -8 & 3 & 0 \end{array} \right]$$

hence,

x_4 is free Variable

Let $x_4 = a$

$$-8x_3 + 3x_4 = 0$$

$$x_3 = \frac{-3x_4}{-8} = \frac{3a}{8}$$

$$-3x_2 + 10x_3 = 0$$

$$x_2 = \frac{5a}{4}$$

$$\left. \begin{array}{l} x_1 + 2x_2 + 3x_3 - x_4 = 0 \\ -x_1 = \left(\frac{5a}{2} + \frac{9}{8} - a \right) \\ x_1 = \frac{-2a}{8} \end{array} \right\}$$

∴ the matrix is linearly Dependent.

⑦ $u_1 = (2, 1, 2)$, $u_2 = (0, 1, -1)$, $u_3 = (4, 3, 3)$

$\Rightarrow$ we know:-

$$\alpha_1 u_1 + \alpha_2 u_2 + \alpha_3 u_3 = 0$$

$$\therefore \alpha_1(2, 1, 2) + \alpha_2(0, 1, -1) + \alpha_3(4, 3, 3) = 0$$

$\therefore$ Corresponding Eqⁿ:-

$$2\alpha_1 + 0 + 4\alpha_3 = 0$$

$$\alpha_1 + \alpha_2 + 3\alpha_3 = 0$$

$$2\alpha_1 - \alpha_2 + 3\alpha_3 = 0$$

Now Augmented Matrix:-

$$= \left[\begin{array}{ccc|c} 2 & 0 & 4 & 0 \\ 1 & 1 & 3 & 0 \\ 2 & -1 & 3 & 0 \end{array} \right] \begin{array}{l} \tilde{r}_1 = 2r_2 - r_1 \\ \tilde{r}_3 = r_3 - r_2 \end{array}$$

$$= \left[\begin{array}{ccc|c} 2 & 0 & 4 & 0 \\ 0 & 2 & 2 & 0 \\ 0 & -1 & -1 & 0 \end{array} \right] \begin{array}{l} \tilde{r}_2 = r_2 + 2r_3 \end{array}$$

$$= \left[\begin{array}{ccc|c} 2 & 0 & 4 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & -1 & -1 & 0 \end{array} \right] r_2 \leftrightarrow r_3$$

$$2 \left[\begin{array}{ccc|c} 2 & 0 & 4 & 0 \\ 0 & -1 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

here, α_3 is free Variable.

Let, $\alpha_3 = a$

$$\alpha_2 = -a$$

$$\alpha_1 = -2a$$

$\therefore$ The matrix is Linearly Dependent.

⑧ $u_1 = (2, 0, -1), u_2 = (1, 1, 0), u_3 = (0, -1, 1)$

We know:-

$$d_1 u_1 + d_2 u_2 + d_3 u_3 = 0$$

$$d_1(2, 0, -1) + d_2(1, 1, 0) + d_3(0, -1, 1) = 0$$

∴ Corresponding eqn:-

$$2d_1 + d_2 + 0 = 0$$

$$0 + d_2 - d_3 = 0$$

$$-d_1 + 0 + d_3 = 0$$

Now, Augmented Matrix:-

$$= \left[\begin{array}{ccc|c} 2 & 1 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ -1 & 0 & 1 & 0 \end{array} \right] \quad \tilde{r}_3 = 2r_3 + r_1$$

$$= \left[\begin{array}{ccc|c} 2 & 1 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 1 & 1 & 0 \end{array} \right] \quad \tilde{r}_3 = r_3 - r_2$$

$$= \left[\begin{array}{ccc|c} 2 & 1 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & -2 & 0 \end{array} \right]$$

$$\therefore d_3 = 0$$

$$d_2 = 0$$

$$d_1 = 0$$

the matrix is linearly Independent.

Ans:-

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Coordinate Systems
Chaos

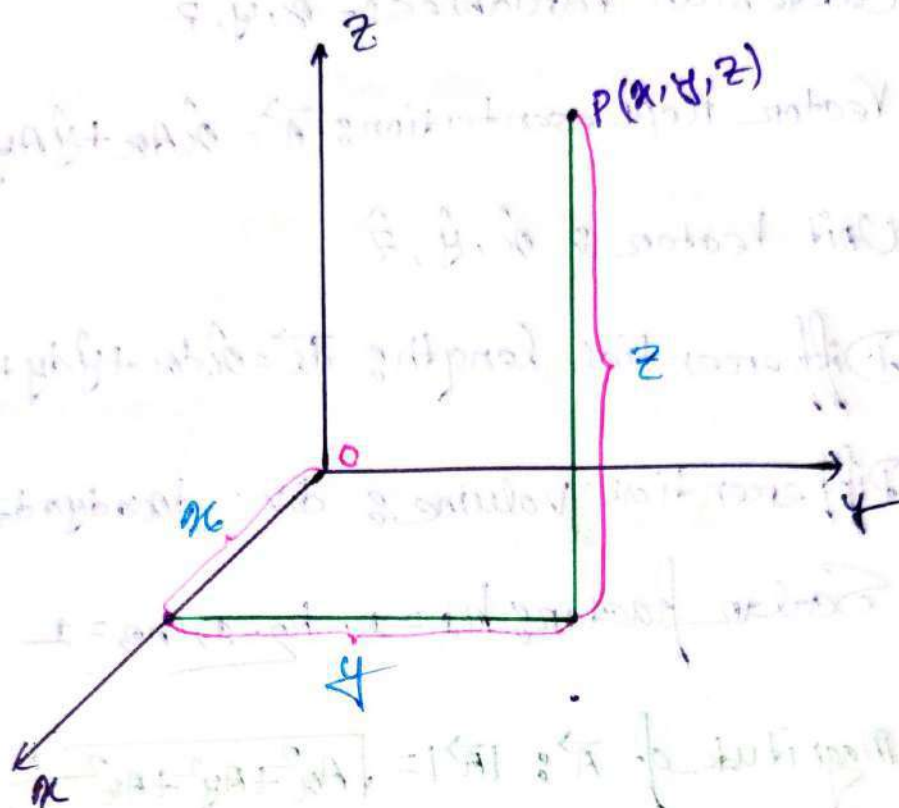
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Cartesian Coordinates:-



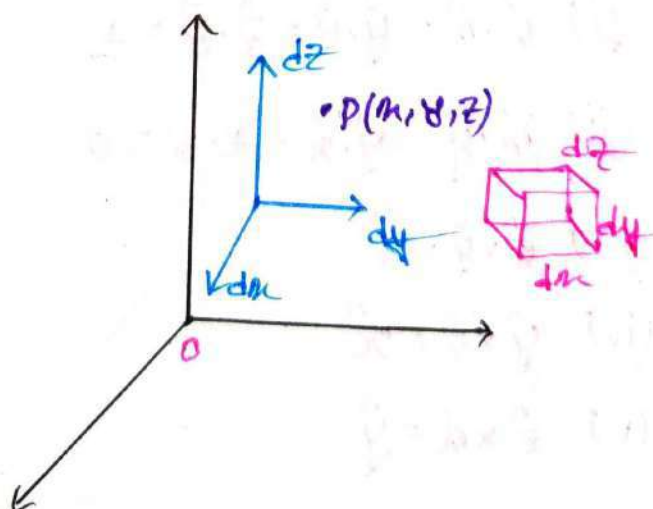
where;
now

$$-\infty < x < \infty$$

$$-\infty < y < \infty$$

$$-\infty < z < \infty$$

Now:-



So,

1. Cartesian Variables:- x, y, z
2. Vector representation: $\vec{A} = \hat{x}A_x + \hat{y}A_y + \hat{z}A_z$
3. Unit Vector: $\hat{x}, \hat{y}, \hat{z}$
4. Differential length: $d\vec{r} = \hat{x}dx + \hat{y}dy + \hat{z}dz$
5. Differential volume: $dV = dx \times dy \times dz$
6. Scalar factors: $h_1 = 1, h_2 = 1, h_3 = 1$
7. Magnitude of $\vec{A}$: $|\vec{A}| = \sqrt{A_x^2 + A_y^2 + A_z^2}$
8. position Vector $\vec{OP} = \hat{x}x + \hat{y}y + \hat{z}z$ for $P(x, y, z)$
9. Base Vector properties:
 - (i) $\hat{x} \cdot \hat{x} = \hat{y} \cdot \hat{y} = \hat{z} \cdot \hat{z} = 1$
 - (ii) $\hat{x} \cdot \hat{y} = \hat{y} \cdot \hat{x} = \hat{z} \cdot \hat{x} = 0$
 - (iii) $\hat{x} \times \hat{y} = \hat{z}$
 - (iv) $\hat{y} \times \hat{z} = \hat{x}$
 - (v) $\hat{z} \times \hat{x} = \hat{y}$
10. Dot product: $\vec{A} \cdot \vec{B} = A_x B_x + A_y B_y + A_z B_z$

11 Cross product: $\vec{A} \times \vec{B} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}$

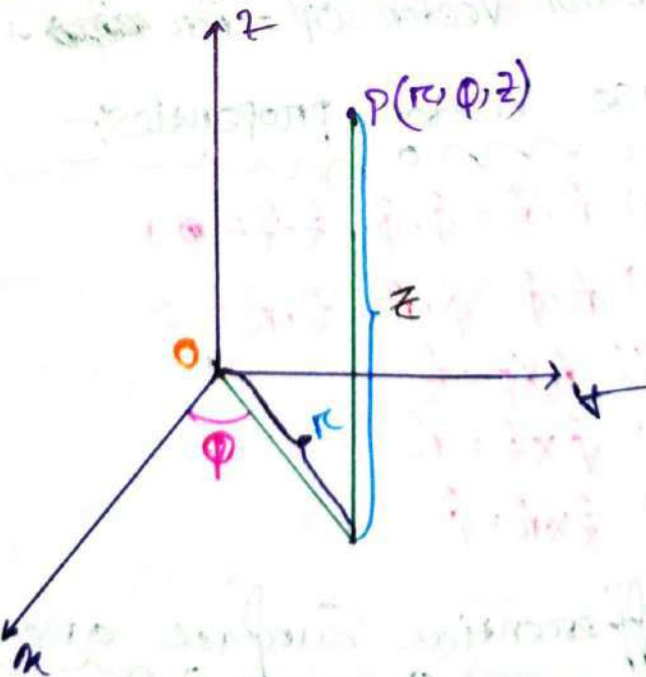
$$= \hat{x}(A_y B_z - A_z B_y) - \hat{y}(A_x B_z - A_z B_x) + \hat{z}(A_x B_y - A_y B_x)$$

12 Differential Surface area: $d\vec{S}_x = \hat{x} dy dz$

$$d\vec{S}_y = \hat{y} dz dx$$

$$d\vec{S}_z = \hat{z} dx dy$$

Cylindrical Coordinates:-



Where,

(i) $0 \leq r < \infty$

(ii) $0 \leq \phi < 2\pi$

(iii) $-\infty \leq z \leq \infty$

So,(i) Cylindrical Variable ρ, ϕ, z (2) Vector representation: $\vec{A} = \hat{r}A_r + \hat{\phi}A_\phi + \hat{z}A_z$ (3) Unit Vector $\hat{r}, \hat{\phi}, \hat{z}$ (4) Differential Length: $d\vec{r} = \hat{r}dr + \hat{\phi}r d\phi + \hat{z}dz$ (5) Differential Volume: $dV = dr \times r d\phi \times dz$ (6) Scalar factor: $h_1 = 1$
 $h_2 = r$
 $h_3 = 1$ (7) Magnitude $|\vec{A}| = \sqrt{A_r^2 + A_\phi^2 + A_z^2}$ (8) position Vector $\vec{OP} = r\hat{r} + z\hat{z}$ for $P(\rho, \phi, z)$

(9) Base Vector properties:-

(i) $\hat{r} \cdot \hat{r} = \hat{\phi} \cdot \hat{\phi} = \hat{z} \cdot \hat{z} = 1$

(ii) $\hat{r} \cdot \hat{\phi} = \hat{\phi} \cdot \hat{r} = \hat{z} \cdot \hat{r} = 0$

(iii) $\hat{r} \times \hat{\phi} = \hat{z}$

(iv) $\hat{\phi} \times \hat{z} = \hat{r}$

(v) $\hat{z} \times \hat{r} = \hat{\phi}$

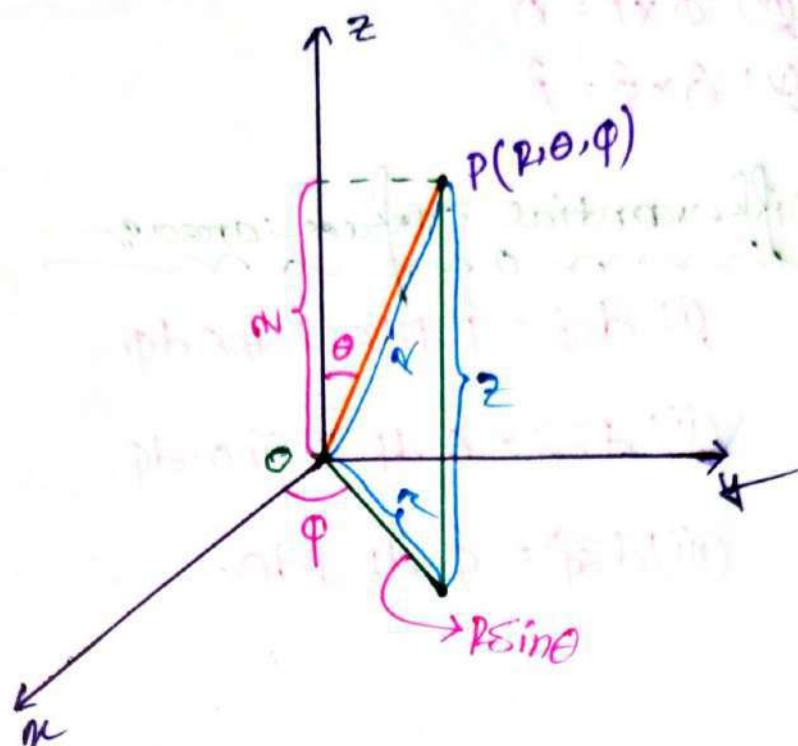
(10) Differential Surface area:-

(i) $d\vec{S}_r = \hat{r} \cdot r d\phi dz$

(ii) $d\vec{S}_\phi = \hat{\phi} \cdot dr dz$

(iii) $d\vec{S}_z = \hat{z} \cdot dr \cdot r d\phi$

Spherical Coordinates:-



Sci

1. Spherical Variables: R, θ, ϕ

2. Vector Representation: $\vec{A} = \hat{R}A_R + \hat{\theta}A_\theta + \hat{\phi}A_\phi$

3. Unit Vectors: $\hat{R}, \hat{\theta}, \hat{\phi}$

4. Differential Length: $d\vec{r} = \hat{R}dR + \hat{\theta}Rd\theta + \hat{\phi}R\sin\theta \cdot d\phi$

5. Differential Volume: $dv = dR \times R d\theta \times R \sin\theta d\phi$

6. Scalar factor: $h_1 = 1$
 $h_2 = R$
 $h_3 = R \sin\theta$

7. Magnitude: $\vec{A}$: $|\vec{A}| = \sqrt{A_R^2 + A_\theta^2 + A_\phi^2}$

8. Position Vector: $\vec{OP} = \hat{R}R$ for $P(R, \theta, \phi)$

9. Base Vector Properties:-

(i) $\hat{R} \cdot \hat{R} = \hat{\theta} \cdot \hat{\theta} = \hat{\phi} \cdot \hat{\phi} = 1$

(ii) $\hat{R} \cdot \hat{\theta} = \hat{\theta} \cdot \hat{R} = \hat{\phi} \cdot \hat{R} = 0$

$$(iii) \hat{r} \times \hat{\theta} = \hat{\phi}$$

$$(iv) \hat{\theta} \times \hat{r} = -\hat{\theta}$$

$$(v) \hat{\theta} \times \hat{\phi} = \hat{r}$$

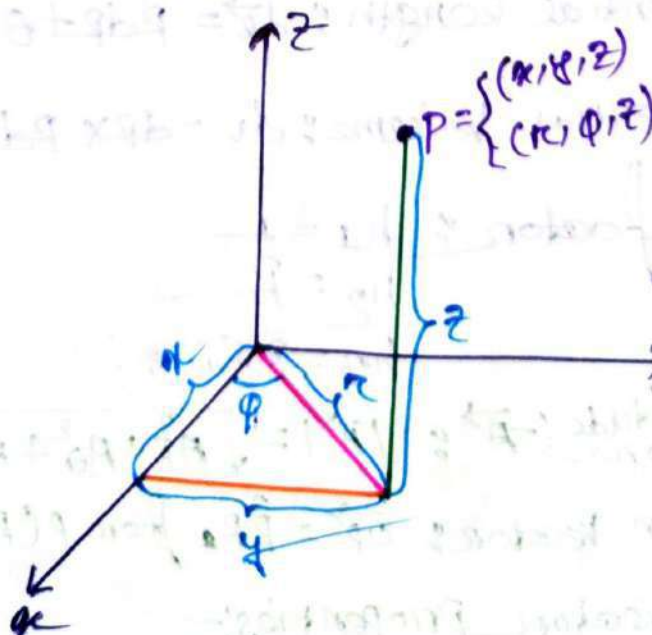
10. Differential Surface area:-

$$(i) d\vec{S}_r = \hat{r} \, r \, d\theta \, r \sin\theta \, d\phi$$

$$(ii) d\vec{S}_\theta = \hat{\theta} \, dr \, r \sin\theta \, d\phi$$

$$(iii) d\vec{S}_\phi = \hat{\phi} \, dr \, r \, d\theta$$

Relation between Cartesian
And Cylindrical Coordinates:-



We know:-

$$x = r \cos \phi$$

$$y = r \sin \phi$$

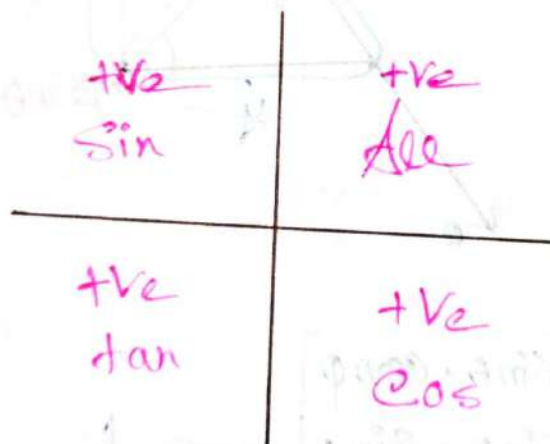
$$z = z$$

$$\therefore r = \sqrt{x^2 + y^2}$$

$$\tan \phi = \frac{y}{x}$$

$$\therefore \phi = \tan^{-1} \left(\frac{y}{x} \right)$$

Now:-



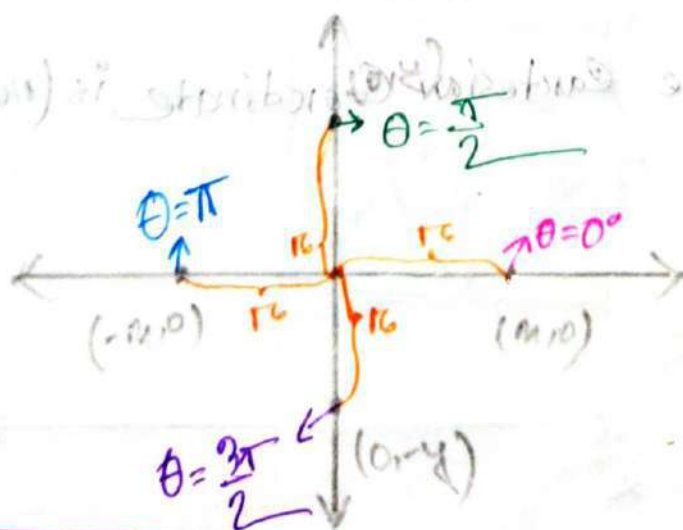
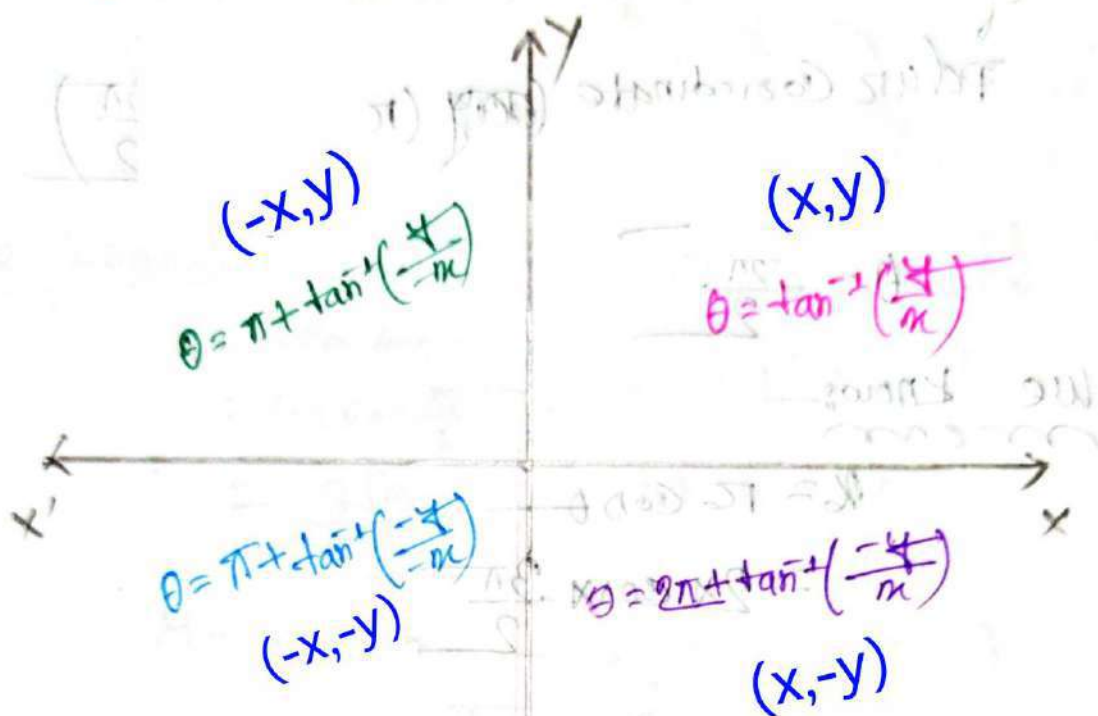
(i) $\tan \phi = \frac{y}{x}$ 1st coordinate

(ii) $\tan \phi = \tan (\pi - \phi)$ 2nd coordinate

(iii) $\tan \phi = \tan (\pi + \phi)$ 3rd coordinate

(iv) $\tan \phi = \tan (2\pi - \phi)$ 4th coordinate.

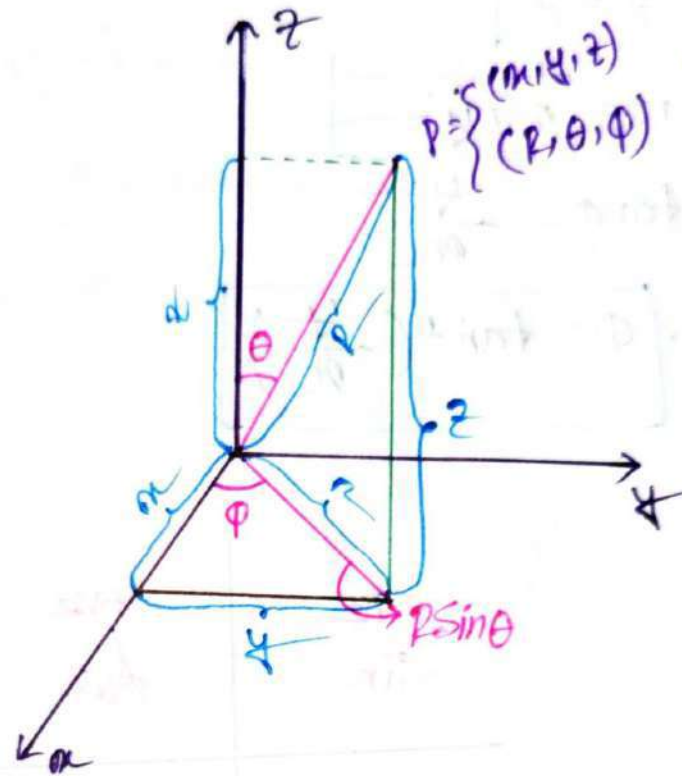
Standard Position of Angles in the Coordinate plane:-



* Polar Coordinate $\equiv (r, \theta)$

* Cartesian Coordinates $\equiv (x, y)$

Relation between Cartesian And Spherical Coordinates



Now:-

$$x = r \sin \theta \cdot \cos \phi$$

$$y = r \sin \theta \cdot \sin \phi$$

$$z = r \cos \theta$$

$$r = \sqrt{x^2 + y^2 + z^2}$$

$$\theta = \tan^{-1} \left(\frac{\sqrt{x^2 + y^2}}{z} \right)$$

$$\phi = \tan^{-1} \left(\frac{y}{x} \right)$$

$$x^2 + y^2 = r^2 \sin^2 \theta$$

$$z^2 = r^2 \cos^2 \theta$$

$$x^2 + y^2 + z^2 = r^2$$

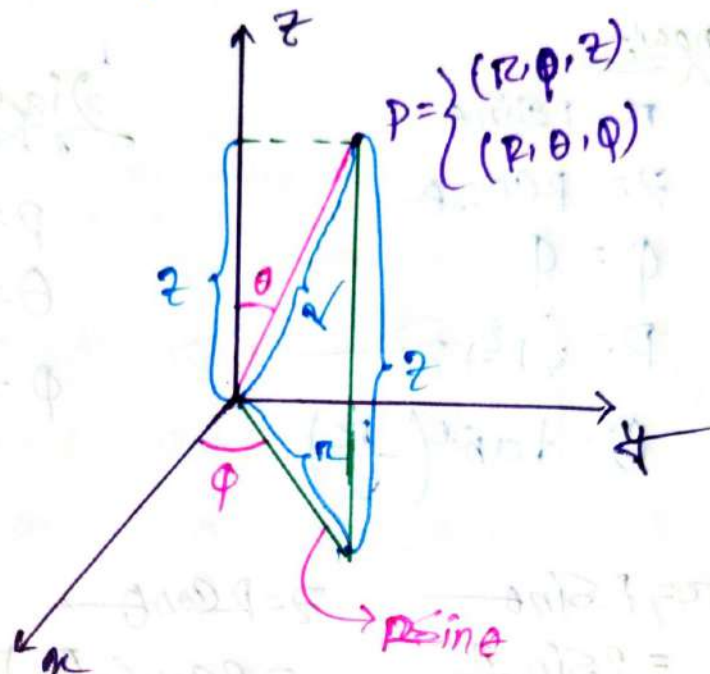
$$r \sin \theta = \sqrt{x^2 + y^2}$$

$$z = r \cos \theta$$

$$\therefore \tan \theta = \frac{\sqrt{x^2 + y^2}}{z}$$

$$\theta = \pi - \theta$$

Relation between Spherical And cylindrical coordinate:-



Notes:-

$$r = R \sin \theta$$

$$z = R \cos \theta$$

$$\phi = \phi$$

$$R = \sqrt{r^2 + z^2}$$

$$\theta = \tan^{-1} \left(\frac{r}{z} \right)$$

Example 1:-
mon

⇒ Transform $(\sqrt{2}, \frac{3\pi}{4}, 3)$ from cylindrical coordinate to Cartesian coordinates.

⇒ we know:-
mon

$$x = r \cos \phi$$

$$y = r \sin \phi$$

$$z = z$$

is given:-
mon

$$r = \sqrt{2}$$

$$\phi = \frac{3\pi}{4}$$

$$z = 3$$

Sol

$$x = r \cos \phi$$

$$= \sqrt{2} \cos\left(\frac{3\pi}{4}\right)$$

$$= -1$$

$$y = r \sin \phi$$

$$= \sqrt{2} \sin\left(\frac{3\pi}{4}\right)$$

$$= 1$$

$$z = z$$

$$= 3$$

So, the Cartesian coordinates:-
mon

$$(x, y, z) = (-1, 1, 3)$$

Ans:-

Example 2:-
mon

⇒ Transform $(1, 0, \sqrt{3})$ from Cartesian to spherical

⇒ we know:-
mon

$$x = \rho \sin \theta \cdot \cos \phi$$

$$y = \rho \sin \theta \cdot \sin \phi$$

$$z = \rho \cos \theta$$

$$\left\{ \begin{array}{l} \rho = \sqrt{x^2 + y^2 + z^2} \\ \theta = \tan^{-1} \left(\frac{\sqrt{x^2 + y^2}}{z} \right) \\ \phi = \tan^{-1} \left(\frac{y}{x} \right) \end{array} \right.$$

$\Rightarrow$ is given:—

$$x = 1$$

$$y = 0$$

$$z = \sqrt{3}$$

So,

$$\begin{aligned} R &= \sqrt{x^2 + y^2 + z^2}, \quad \theta = \tan^{-1} \left(\frac{\sqrt{x^2 + y^2}}{z} \right) \\ &= \sqrt{1^2 + 0^2 + (\sqrt{3})^2} \\ &= 2 \end{aligned}$$

$$\theta = \tan^{-1} \left(\frac{\sqrt{1^2 + 0^2}}{\sqrt{3}} \right)$$

$$= \frac{\pi}{6}$$

$$\begin{aligned} \phi &= \tan^{-1} \left(\frac{y}{x} \right) \\ &= \tan^{-1} \left(\frac{0}{1} \right) \\ &= 0 \end{aligned}$$

So,

The spherical points

$$(R, \theta, \phi) = \left(2, \frac{\pi}{6}, 0 \right)$$

Example 8:- Transform $(2, \frac{\pi}{6}, 0)$ from Spherical to cylindrical.

$\Rightarrow$ We know:-

$$r = \rho \sin \theta$$

$$z = \rho \cos \theta$$

$$\phi = \phi$$

$$\rho = \sqrt{r^2 + z^2}$$

$$\theta = \tan^{-1}\left(\frac{r}{z}\right)$$

2 is given:-

$$\rho = 2$$

$$\theta = \frac{\pi}{6}$$

$$\phi = 0$$

So

$$r = \rho \sin \theta$$

$$= 2 \sin \frac{\pi}{6}$$

$$= 1$$

$$z = \rho \cos \theta$$

$$= 2 \cos\left(-\frac{\pi}{6}\right)$$

$$= \sqrt{3}$$

$$\phi = \phi$$

$$= 0$$

So, the cylindrical point $(r, \phi, z) = (1, 0, \sqrt{3})$

Ans:-

Transformation of a Vector from Cartesian to cylindrical Coordinate

⇒ We know

$$\vec{A} = A_x \hat{a}_x + A_y \hat{a}_y + A_z \hat{a}_z$$

Now

$$\begin{bmatrix} A_r \\ A_\phi \\ A_z \end{bmatrix} = \begin{bmatrix} \cos\phi & \sin\phi & 0 \\ -\sin\phi & \cos\phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix}$$

$$\therefore A_r = A_x \cos\phi + A_y \sin\phi \quad \text{--- (i)}$$

$$A_\phi = -A_x \sin\phi + A_y \cos\phi \quad \text{--- (ii)}$$

$$A_z = A_z \quad \text{--- (iii)}$$

Transformation of a Vector from cylindrical to Cartesian Coordinate

⇒ We know

$$\vec{A} = A_r \hat{a}_r + A_\phi \hat{a}_\phi + A_z \hat{a}_z$$

Now

$$\begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix} = \begin{bmatrix} \cos\phi & -\sin\phi & 0 \\ \sin\phi & \cos\phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} A_r \\ A_\phi \\ A_z \end{bmatrix}$$

$$\therefore A_x = A_r \cos\phi - A_\phi \sin\phi$$

$$A_y = A_r \sin\phi + A_\phi \cos\phi$$

$$A_z = A_z$$

Transform of a Vector from Spherical to Cartesian Coordinate

we know:-

$$\vec{A} = A_R \hat{a}_R + A_\theta \hat{a}_\theta + A_\phi \hat{a}_\phi$$

Now:-

$$\begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix} = \begin{bmatrix} \sin\theta \cos\phi & \cos\theta \cos\phi & -\sin\phi \\ \sin\theta \sin\phi & \cos\theta \sin\phi & \cos\phi \\ \cos\theta & -\sin\theta & 0 \end{bmatrix} \begin{bmatrix} A_R \\ A_\theta \\ A_\phi \end{bmatrix}$$

$$\therefore A_x = A_R \sin\theta \cos\phi + A_\theta \cos\theta \cos\phi - A_\phi \sin\phi$$

$$A_y = A_R \sin\theta \sin\phi + A_\theta \cos\theta \sin\phi + A_\phi \cos\phi$$

$$A_z = A_R \cos\theta - A_\theta \sin\theta$$

Transform of a Vector from Cartesian Coordinate to Spherical Coordinate

we know:-

$$\vec{A} = A_x \hat{a}_x + A_y \hat{a}_y + A_z \hat{a}_z$$

Now:-

$$\begin{bmatrix} A_R \\ A_\theta \\ A_\phi \end{bmatrix} = \begin{bmatrix} \sin\theta \cos\phi & \sin\theta \sin\phi & \cos\theta \\ \cos\theta \cos\phi & \cos\theta \sin\phi & -\sin\theta \\ -\sin\phi & \cos\phi & 0 \end{bmatrix} \begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix}$$

$$A_R = A_x \sin\theta \cos\phi + A_y \sin\theta \sin\phi + A_z \cos\theta$$

$$A_\theta = A_x \cos\theta \cos\phi + A_y \cos\theta \sin\phi - A_z \sin\theta$$

$$A_\phi = -A_x \sin\phi + A_y \cos\phi$$

Ex Example 1: — Transform a vector $\vec{A} = y\hat{a}_x - x\hat{a}_y + z\hat{a}_z$ into cylindrical coordinates.

⇒ we know:-

$$\vec{A} = A_x \hat{a}_x + A_y \hat{a}_y + A_z \hat{a}_z$$

So,

$$A_x = y$$

$$A_y = -x$$

$$A_z = z$$

Now,

$$A_r = A_x \cos \phi + A_y \sin \phi = y \cos \phi + (-x) \sin \phi \quad \dots (1)$$

$$A_\phi = -A_x \sin \phi + A_y \cos \phi = -y \sin \phi - x \cos \phi \quad \dots (1')$$

$$A_z = A_z = z \quad \dots (1'')$$

Again we know:-

$$x = r \cos \phi$$

$$y = r \sin \phi$$

$$z = z$$

put all the value of x, y, z in eq (1):

$$\therefore A_r = r \sin \phi \times \cos \phi - r \cos \phi \times \sin \phi = 0$$

$$A_\phi = -r \sin \phi \times \sin \phi - r \cos \phi \times \cos \phi = -r(\sin^2 \phi + \cos^2 \phi) = -r$$

$$A_z = z$$

So,

$$\vec{A} = A_r \hat{a}_r + A_\phi \hat{a}_\phi + A_z \hat{a}_z$$

$$= -r \hat{a}_\phi + z \hat{a}_z$$

Ans:-

Example 2: Transform the vector $\vec{A} = \hat{r} \cos \phi + \hat{\theta} \sin \phi + \hat{\phi} \sin^2 \theta$ at the point $(3, \frac{\pi}{2}, \pi)$ to cylindrical coordinates.

$\Rightarrow$ is given:-

$$R = 3$$

$$\theta = \frac{\pi}{2}$$

$$\phi = \pi$$

$\Rightarrow$ we know:-

$$r = R \sin \theta$$

$$= 3 \sin\left(\frac{\pi}{2}\right)$$

$$= 3$$

$$z = R \cos \theta$$

$$= 3 \cos\left(\frac{\pi}{2}\right)$$

$$= 0$$

$$\phi = \phi$$

$$= \pi$$

So, we get:-

$$\left(3, \frac{\pi}{2}, \pi\right) = (3, \pi, 0)$$

Now:-

$$A_r = \cos \phi$$

$$= \cos(\pi)$$

$$= -1$$

$$A_\theta = \sin \phi$$

$$= \sin(\pi)$$

$$= 0$$

$$A_\phi = \sin^2 \theta$$

$$= \sin^2\left(\frac{\pi}{2}\right)$$

$$= 1$$

And we know:-

$$A_r = A_r \sin \theta + A_\theta \cos \theta = -1$$

$$A_\phi = A_\phi = 1$$

$$A_z = A_r \cos \theta - A_\theta \sin \theta = 0$$

$$\therefore \vec{A} = -\hat{r} + \hat{\phi}$$

#Sample Exercise: 5.1

①

Q:- Convert the coordinates of the following points from Cartesian to cylindrical and spherical coordinates.

① $P_1 = (-1, \sqrt{3}, -2\sqrt{3})$

⇒ For cylindrical—

we know—

$$x = r \cos \phi$$

$$y = r \sin \phi$$

$$z = z$$

$$r = \sqrt{x^2 + y^2}$$

$$\phi = \tan^{-1}\left(\frac{y}{x}\right)$$

$$= \tan^{-1}\left(\pi - \frac{y}{x}\right) \text{ For 2nd Coordinate.}$$

$$= \pi + \tan^{-1}\left(\frac{y}{-x}\right)$$

$$r = \sqrt{x^2 + y^2}$$

$$= \sqrt{(-1)^2 + (\sqrt{3})^2}$$

$$= 2$$

$$\phi = \pi + \tan^{-1}\left(\frac{y}{-x}\right)$$

$$= \pi - \frac{\pi}{3}$$

$$= \frac{2\pi}{3}$$

$$z = z$$

$$= -2\sqrt{3}$$

in given—

$$x = -1$$

$$y = \sqrt{3}$$

$$z = -2\sqrt{3}$$

So,

the cylindrical point $(r, \phi, z) = (2, \frac{2\pi}{3}, -2\sqrt{3})$

Ans:-

For spherical:-

We know:-

$$x = \rho \sin \theta \cdot \cos \phi$$

$$y = \rho \sin \theta \cdot \sin \phi$$

$$z = \rho \cos \theta$$

$$\rho = \sqrt{x^2 + y^2 + z^2}$$

$$\theta = \tan^{-1} \left(\frac{\sqrt{x^2 + y^2}}{z} \right) + \pi$$

$$\phi = \pi + \tan^{-1} \left(\frac{y}{x} \right)$$

Is given:-

$$x = -1$$

$$y = \sqrt{3}$$

$$z = -2\sqrt{3}$$

Now:-

$$\rho = \sqrt{x^2 + y^2 + z^2}$$

$$= \sqrt{(-1)^2 + (\sqrt{3})^2 + (-2\sqrt{3})^2}$$

$$= 4$$

$$\theta = \tan^{-1} \left(\frac{\sqrt{x^2 + y^2}}{z} \right) + \pi$$

$$= \tan^{-1} \left(\frac{\sqrt{(-1)^2 + (\sqrt{3})^2}}{-2\sqrt{3}} \right) + \pi$$

$$= -\frac{\pi}{6} + \pi$$

$$= \frac{5\pi}{6}$$

$$\therefore \phi = \pi + \tan^{-1} \left(\frac{y}{x} \right)$$

$$= \pi + \tan^{-1} \left(\frac{\sqrt{3}}{-1} \right)$$

$$= \frac{2\pi}{3}$$

$$\therefore (\rho, \theta, \phi) = \left(4, \frac{5\pi}{6}, \frac{2\pi}{3} \right)$$

Ans:-

⑥ $P_2 = (4, 0, -4)$

⇒ Is given:-

$x = 4$

$y = 0$

$z = -4$

For Cylindrical:-

we know:-

$x = r \cos \phi$

$y = r \sin \phi$

$z = z$

$r = \sqrt{x^2 + y^2}$

$\phi = \tan^{-1}\left(\frac{y}{x}\right)$

Now:-

$r = \sqrt{x^2 + y^2} = \sqrt{4^2 + 0^2} = 4$

$\phi = \tan^{-1}\left(\frac{y}{x}\right) = \tan^{-1}\left(\frac{0}{4}\right) = 0$

$z = z = -4$

$(r, \phi, z) = (4, 0, -4)$

Ans:-

For Spherical:-

we know:-

$x = \rho \sin \theta \cos \phi$

$y = \rho \sin \theta \sin \phi$

$z = \rho \cos \theta$

$\rho = \sqrt{x^2 + y^2 + z^2}$

$\theta = \tan^{-1}\left(\frac{\sqrt{x^2 + y^2}}{z}\right) + \pi$ for $z < 0$

$\phi = \tan^{-1}\left(\frac{y}{x}\right)$

Now,

$\rho = \sqrt{x^2 + y^2 + z^2} = 4\sqrt{2}$

$\theta = \pi + \tan^{-1}\left(\frac{\sqrt{x^2 + y^2}}{z}\right) = \frac{3\pi}{4}$

$\phi = \tan^{-1}\left(\frac{y}{x}\right) = 0$

$\therefore (\rho, \theta, \phi) = (4\sqrt{2}, \frac{3\pi}{4}, 0)$

Ans:-

① $P_0 = (\sqrt{8}, -\sqrt{8}, 4)$

is given:-

$$x = \sqrt{8}$$

$$y = -\sqrt{8}$$

$$z = 4$$

For Cylindrical:-

we know:-

$$x = r \cos \phi$$

$$y = r \sin \phi$$

$$z = z$$

$$r = \sqrt{x^2 + y^2}$$

$$\phi = 2\pi + \tan^{-1}\left(\frac{y}{x}\right) \rightarrow \text{with}$$

Now:-

$$r = \sqrt{x^2 + y^2} = \sqrt{(\sqrt{8})^2 + (-\sqrt{8})^2} = 4$$

$$\phi = 2\pi + \tan^{-1}\left(\frac{-\sqrt{8}}{\sqrt{8}}\right) = \frac{7\pi}{4}$$

$$z = z = 4$$

$$\therefore (r, \phi, z) = (4, \frac{7\pi}{4}, 4)$$

For Spherical:-

we know:-

$$x = r \sin \theta \cos \phi$$

$$y = r \sin \theta \sin \phi$$

$$z = r \cos \theta$$

$$r = \sqrt{x^2 + y^2 + z^2}$$

$$\theta = \tan^{-1}\left(\frac{\sqrt{x^2 + y^2}}{z}\right)$$

$$\phi = \tan^{-1}\left(\frac{y}{x}\right) + 2\pi$$

Now:-

$$r = \sqrt{x^2 + y^2 + z^2} = 4\sqrt{2}$$

$$\theta = \tan^{-1}\left(\frac{\sqrt{x^2 + y^2}}{z}\right) = \frac{\pi}{4}$$

$$\phi = \tan^{-1}\left(\frac{-\sqrt{8}}{\sqrt{8}}\right) + 2\pi$$

$$= \frac{-\pi}{4} + 2\pi$$

$$= \frac{7\pi}{4}$$

$$\therefore (r, \theta, \phi) = (4\sqrt{2}, \frac{\pi}{4}, \frac{7\pi}{4})$$

Ans:-

Q. Convert the coordinates of the following point from cylindrical to cartesian and spherical coordinates.

(a) $P_1 = (2, \frac{2\pi}{3}, 2\sqrt{3})$

$\Rightarrow$ Is given:-

$$r = 2$$

$$\theta = \frac{2\pi}{3}$$

$$z = 2\sqrt{3}$$

For Cartesian:-

we know:-

$$x = r \cos \theta$$

$$y = r \sin \theta$$

$$z = z$$

$$r = \sqrt{x^2 + y^2}$$

$$\theta = \tan^{-1}\left(\frac{y}{x}\right)$$

Now:-

$$x = r \cos \theta = 2 \cos \frac{2\pi}{3} = -1$$

$$y = r \sin \theta = 2 \sin \frac{2\pi}{3} = \sqrt{3}$$

$$z = z = 2\sqrt{3}$$

$$\therefore (x, y, z) = (-1, \sqrt{3}, 2\sqrt{3})$$

Ans:-

For Spherical:-

we know:-

$$r = \rho \sin \theta$$

$$z = \rho \cos \theta$$

$$\phi = \phi$$

$$\rho = \sqrt{r^2 + z^2}$$

$$\theta = \tan^{-1}\left(\frac{r}{z}\right)$$

Now:-

$$\rho = \sqrt{r^2 + z^2} = 4$$

$$\theta = \tan^{-1}\left(\frac{2}{2\sqrt{3}}\right) = \frac{\pi}{6}$$

$$\phi = \phi = \frac{2\pi}{3}$$

$$\therefore (\rho, \theta, \phi) = (4, \frac{\pi}{6}, \frac{2\pi}{3})$$

Ans:-

⑥ $P_2 = (\sqrt{3}, 0, -1)$

It is given:-

$$r = \sqrt{3}$$

$$\phi = 0^\circ$$

$$z = -1$$

For Cartesian:-

we know:-

$$x = r \cos \phi$$

$$y = r \sin \phi$$

$$z = z$$

$$r = \sqrt{x^2 + y^2}$$

$$\phi = \tan^{-1}\left(\frac{y}{x}\right)$$

sol

$$x = \sqrt{3} \cos 0^\circ = \sqrt{3}$$

$$y = \sqrt{3} \sin 0^\circ = 0$$

$$z = z = -1$$

$$\therefore (x, y, z) = (\sqrt{3}, 0, -1)$$

Ans:-

For Spherical:-

we know:-

$$r = r \cos \theta$$

$$r = r \sin \theta$$

$$\phi = \phi$$

$$r = \sqrt{x^2 + y^2}$$

$$\theta = \tan^{-1}\left(\frac{r}{z}\right) + \pi$$

sol

$$r = \sqrt{(-1)^2 + (\sqrt{3})^2} = 2$$

$$\theta = \tan^{-1}\left(\frac{\sqrt{3}}{-1}\right) + \pi = \frac{2\pi}{3}$$

$$\phi = \phi = 0$$

$$\therefore (r, \theta, \phi) = \left(2, \frac{2\pi}{3}, 0\right)$$

Ans:-

③. $A_3 = (4\sqrt{3}, \pi, -4)$

is given:-

$$r = 4\sqrt{3}$$

$$\phi = \pi$$

$$z = -4$$

For Cartesian:-

we know:-

$$y = r \sin \phi$$

$$x = r \cos \phi$$

$$z = z$$

$$r = \sqrt{x^2 + y^2}$$

$$\phi = \tan^{-1}\left(\frac{y}{x}\right)$$

sol

$$x = 4\sqrt{3} \times \cos \pi = -4\sqrt{3}$$

$$y = 4\sqrt{3} \sin \pi = 0$$

$$z = z = -4$$

$$\therefore (x, y, z) = (-4\sqrt{3}, 0, -4)$$

Ans:-

For Spherical:-

we know:-

$$r = r \sin \theta$$

$$z = r \cos \theta$$

$$\phi = \phi$$

$$r = \sqrt{r^2 + z^2}$$

$$\theta = \pi + \tan^{-1}\left(\frac{r}{z}\right)$$

sol

$$r = \sqrt{(4\sqrt{3})^2 + (-4)^2} = 8$$

$$\theta = \pi + \tan^{-1}\left(\frac{4\sqrt{3}}{-4}\right) = \frac{2\pi}{3}$$

$$\phi = \phi = \pi$$

$$\therefore (r, \theta, \phi) = (8, \frac{2\pi}{3}, \pi)$$

Ans:-

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Q3. Convert the coordinates of the following points from spherical to Cartesian and cylindrical coordinate:

Q. $P_1 = (3, \frac{3\pi}{4}, \frac{5\pi}{3})$

is given:-

$$\rho = 3$$

$$\theta = \frac{3\pi}{4}$$

$$\phi = \frac{5\pi}{3}$$

For Cartesian:-

we know:-

$$x = \rho \cos \theta \cos \phi$$

$$y = \rho \sin \theta \cos \phi$$

$$z = \rho \sin \theta \sin \phi$$

$$\rho = \sqrt{x^2 + y^2 + z^2}$$

$$\phi = \tan^{-1} \left(\frac{y}{x} \right)$$

sol $x = \frac{3}{\sqrt{2}} \cos \frac{5\pi}{3} = \frac{3\sqrt{2}}{4}$

$$y = \frac{3}{\sqrt{2}} \sin \left(\frac{5\pi}{3} \right) = -\frac{3\sqrt{6}}{4}$$

$$z = z = -\frac{3}{\sqrt{2}}$$

$$\therefore (x, y, z) = \left(\frac{3\sqrt{2}}{4}, -\frac{3\sqrt{6}}{4}, -\frac{3}{\sqrt{2}} \right)$$

Ans:-

For Cylindrical:-

we know:-

$$\rho = \rho \cos \theta$$

$$r = \rho \sin \theta$$

$$\phi = \phi$$

$$\rho = \sqrt{r^2 + z^2}$$

$$\theta = \tan^{-1} \left(\frac{r}{z} \right)$$

sol

$$r = 3 \sin \left(\frac{3\pi}{4} \right) = \frac{3}{\sqrt{2}}$$

$$z = 3 \cos \left(\frac{3\pi}{4} \right) = -\frac{3}{\sqrt{2}}$$

$$\phi = \phi = \frac{5\pi}{3}$$

$$\therefore (r, \phi, z) = \left(\frac{3}{\sqrt{2}}, \frac{5\pi}{3}, -\frac{3}{\sqrt{2}} \right)$$

Ans:-

$$(b) P_2 = (4, \frac{\pi}{2}, \frac{\pi}{4})$$

is given:-

$$R = 4$$

$$\theta = \frac{\pi}{2}$$

$$\phi = \frac{\pi}{4}$$

For Cylindrical:-

we know:-

$$r = R \sin \theta$$

$$z = R \cos \theta$$

$$\phi = \phi$$

$$R = \sqrt{r^2 + z^2}$$

$$\theta = \tan^{-1}\left(\frac{r}{z}\right)$$

Sol $r = 4 \sin \frac{\pi}{2} = 4$

$$z = 4 \cos \frac{\pi}{2} = 0$$

$$\phi = \phi = \frac{\pi}{4}$$

$$\therefore (r, \phi, z) = (4, \frac{\pi}{4}, 0)$$

Ans:-

For Cartesian:-

we know:-

$$x = R \cos \theta \cos \phi$$

$$y = R \sin \theta \cos \phi$$

$$z = z$$

$$R = \sqrt{x^2 + y^2 + z^2}$$

$$\phi = \tan^{-1}\left(\frac{y}{x}\right)$$

Sol $x = 4 \cos(\frac{\pi}{2}) \cos(\frac{\pi}{4}) = 0$

$$y = 4 \sin(\frac{\pi}{2}) \cos(\frac{\pi}{4}) = 2\sqrt{2}$$

$$z = z = 0$$

$$\therefore (x, y, z) = (0, 2\sqrt{2}, 0)$$

Ans:-

Q) $P_3 = (1, \frac{\pi}{4}, \frac{\pi}{2})$

is given:-

$$R = 1$$

$$\theta = \frac{\pi}{4}$$

$$\phi = \frac{\pi}{2}$$

For cylindrical:-

we know:-

$$r = R \sin \theta$$

$$z = R \cos \theta$$

$$\phi = \phi$$

$$R = \sqrt{r^2 + z^2}$$

$$\theta = \tan^{-1} \left(\frac{r}{z} \right)$$

sol
 $r = 1 \sin \left(\frac{\pi}{4} \right) = \frac{1}{\sqrt{2}}$

$$z = 1 \cos \left(\frac{\pi}{4} \right) = \frac{1}{\sqrt{2}}$$

$$\phi = \phi = \frac{\pi}{2}$$

$$\therefore (r, \phi, z) = \left(\frac{1}{\sqrt{2}}, \frac{\pi}{2}, \frac{1}{\sqrt{2}} \right)$$

Ans:-

For Cartesian:-

we know:-

$$x = R \sin \theta \cdot \cos \phi$$

$$y = R \sin \theta \cdot \sin \phi$$

$$z = R \cos \theta$$

$$R = \sqrt{x^2 + y^2 + z^2}$$

$$\theta = \tan^{-1} \left(\frac{\sqrt{x^2 + y^2}}{z} \right)$$

$$\phi = \tan^{-1} \left(\frac{y}{x} \right)$$

sol

$$x = 1 \sin \left(\frac{\pi}{4} \right) \cdot \cos \left(\frac{\pi}{2} \right) = 0$$

$$y = 1 \sin \left(\frac{\pi}{4} \right) \cdot \sin \left(\frac{\pi}{2} \right) = \frac{1}{\sqrt{2}}$$

$$z = 1 \cos \left(\frac{\pi}{4} \right) = \frac{1}{\sqrt{2}}$$

$$\therefore (x, y, z) = \left(0, \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right)$$

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The Gradient

Section

Ch 06

Table of Differentiations

$$\underline{1.} \quad \frac{d}{dx}(x^n) = nx^{n-1}$$

$$\underline{2.} \quad \frac{d}{dx}(e^x) = e^x$$

$$\underline{3.} \quad \frac{d}{dx}(e^{mx}) = me^{mx}$$

$$\underline{4.} \quad \frac{d}{dx}(e^{-x}) = -e^{-x}$$

$$\underline{5.} \quad \frac{d}{dx}(e^{-mx}) = -me^{-mx}$$

$$\underline{6.} \quad \frac{d}{dx}(\sin x) = \cos x$$

$$\underline{7.} \quad \frac{d}{dx}(\cos x) = -\sin x$$

$$\underline{8.} \quad \frac{d}{dx}(uv) = u \frac{d}{dx}(v) + v \frac{d}{dx}(u)$$

$$\underline{9.} \quad \frac{d}{dx}\left(\frac{u}{v}\right) = \frac{v \frac{d}{dx}(u) - u \frac{d}{dx}(v)}{v^2}$$

$$\underline{10.} \quad \frac{d}{dx} \sqrt{x} = \frac{1}{2\sqrt{x}}$$

$$\underline{11.} \quad \frac{d}{dx} \ln(x) = \frac{1}{x}$$

$$\underline{12.} \quad \frac{d}{dx} \tan x = \sec^2 x$$

$$\underline{12.} \quad \frac{d}{dx} \tan x = \sec^2 x$$

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The gradient Vector: (grad)

we know:-

$$\text{del operator; } \vec{\nabla} = \frac{\hat{u}_1}{h_1} \times \frac{\partial}{\partial u_1} + \frac{\hat{u}_2}{h_2} \times \frac{\partial}{\partial u_2} + \frac{\hat{u}_3}{h_3} \times \frac{\partial}{\partial u_3}$$

is given:-

The gradient of a scalar function: $T(u_1, u_2, u_3)$ by

$$\text{grad } T = \vec{\nabla} \cdot T = \frac{\hat{u}_1}{h_1} \times \frac{\partial T}{\partial u_1} + \frac{\hat{u}_2}{h_2} \times \frac{\partial T}{\partial u_2} + \frac{\hat{u}_3}{h_3} \times \frac{\partial T}{\partial u_3}$$

Example 1:- Find the gradient of $T(R, \theta, \phi) = R \cos \theta \cdot \sin \phi$ at the point $(2, \frac{\pi}{2}, \frac{\pi}{4})$

⇒ we know:-

The spherical c.s:-

$$h_1 = 1$$

$$h_2 = R$$

$$h_3 = R \sin \theta$$

Now:-

$$\text{gradient } T = \vec{\nabla} \cdot T = \frac{\hat{R}}{1} \times \frac{\partial}{\partial R} (R \cos \theta \cdot \sin \phi) + \frac{\hat{\theta}}{R} \times \frac{\partial}{\partial \theta} (R \cos \theta \cdot \sin \phi) + \frac{\hat{\phi}}{R \sin \theta} \times \frac{\partial}{\partial \phi} (R \cos \theta \cdot \sin \phi)$$

$$= \hat{R} \cos \theta \cdot \sin \phi + \hat{\theta} (-\sin \theta \cdot \sin \phi) + \hat{\phi} \cos \theta \cdot \cos \phi$$

So, At the point $(2, \frac{\pi}{2}, \frac{\pi}{4})$ $\text{grad } T = -\hat{\theta} \times \frac{1}{\sqrt{2}}$

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Example 2:- Find gradient of

$$T(r, \phi, z) = \frac{z \cos \phi}{(1+r^2)}$$

at point $(1, \pi, 2)$.

$\Rightarrow$ we know:-

The cylindrical c.s:-

$$h_1 = 1$$

$$h_2 = r\phi$$

$$h_3 = 1$$

Now:-

$$\text{The gradient of } T = \vec{\nabla} \cdot T = \frac{\hat{r}}{1} \times \frac{\partial}{\partial r} \left(\frac{z \cos \phi}{(1+r^2)} \right) + \frac{\hat{\phi}}{r} \times \frac{\partial}{\partial \phi} \left(\frac{z \cos \phi}{(1+r^2)} \right) + \frac{\hat{z}}{1} \times \frac{\partial}{\partial z} \left(\frac{z \cos \phi}{(1+r^2)} \right)$$

$$= \hat{r} \times \frac{2rz \cos \phi}{(1+r^2)^2} + \hat{\phi} \times \frac{1}{r} \times \frac{-z \sin \phi}{1+r^2} + \hat{z} \times \frac{\cos \phi}{1+r^2}$$

$$= -\hat{r} \times \frac{2rz \cos \phi}{(1+r^2)^2} - \hat{\phi} \times \frac{1}{r} \times \frac{z \sin \phi}{1+r^2} + \hat{z} \times \frac{\cos \phi}{1+r^2}$$

So, At the point $(1, \pi, 2)$ grad $T = \hat{r} - \hat{z} \times \frac{1}{2}$

Ans:-

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Example 3:- Find the gradient of $T(x, y, z) = \frac{xyz}{x^2+y^2+z^2}$ at the Point $(1, 1, 1)$.

$\Rightarrow$ we know:-

The Cartesian e.s:-

$$h_1 = 1$$

$$h_2 = 1$$

$$h_3 = 1$$

Now:-

$$\begin{aligned} \text{The gradient of } T &= \vec{\nabla} \cdot T = \hat{i} \times \frac{\partial}{\partial x} \left(\frac{xyz}{x^2+y^2+z^2} \right) + \hat{j} \times \frac{\partial}{\partial y} \left(\frac{xyz}{x^2+y^2+z^2} \right) + \hat{k} \times \frac{\partial}{\partial z} \left(\frac{xyz}{x^2+y^2+z^2} \right) \\ &= \hat{i} \frac{(x^2+y^2+z^2)yz - xyz \times 2x}{(x^2+y^2+z^2)^2} + \hat{j} \frac{(x^2+y^2+z^2)xz - xyz \times 2y}{(x^2+y^2+z^2)^2} \\ &\quad + \hat{k} \frac{(x^2+y^2+z^2)xy - xyz \times 2z}{(x^2+y^2+z^2)^2} \\ &= \hat{i} \frac{(x^2+y^2+z^2)yz - 2x^2yz}{(x^2+y^2+z^2)^2} + \hat{j} \frac{(x^2+y^2+z^2)xz - 2xy^2z}{(x^2+y^2+z^2)^2} + \hat{k} \frac{(x^2+y^2+z^2)xy - 2xyz^2}{(x^2+y^2+z^2)^2} \end{aligned}$$

At the point $(1, 1, 1)$ grad $T = \frac{1}{8}(\hat{i} + \hat{j} + \hat{k})$

Ans:-

Sample exercise 6.1:-

①

Q:- Find the gradient of the following scalar functions at the indicated points.

(a) $T(x, y, z) = 2xyz^2 + yz^2 - 5\frac{y}{z}$ at the point $(0, 2, -1)$

⇒ we know:-
now

for Cartesian c.s:-

$$h_1 = 1$$

$$h_2 = 1$$

$$h_3 = 1$$

So, the gradient of $T = \nabla T$

$$= \hat{x} \times \frac{\partial}{\partial x} (2xyz^2 + yz^2 - 5\frac{y}{z}) + \hat{y} \times \frac{\partial}{\partial y} (2xyz^2 + yz^2 - 5\frac{y}{z}) + \hat{z} \times \frac{\partial}{\partial z} (2xyz^2 + yz^2 - 5\frac{y}{z})$$

$$\Rightarrow \hat{x} (6xyz^2 + 2xy^2 - 0) + \hat{y} (2xz^2 + 2yz - 5\frac{1}{z}) + \hat{z} (2xz + 0 - 5 \times \frac{-y}{z^2})$$

$$= \hat{x} (6xyz^2 + 2xy^2) + \hat{y} (2xz^2 + 2yz - 5 \times \frac{1}{z}) + \hat{z} (2xz + \frac{5y}{z^2})$$

Now

At the point $(0, 2, -1)$ grad $T = 5\hat{y} + 10\hat{z}$

Ans

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⑥ $T(r, \phi, z) = \frac{z + \sin \phi}{r}$ at the point $(2, \frac{\pi}{4}, \frac{2\pi}{3})$

⇒ We know:-

The cylindrical c.s:-

$$h_1 = 1$$

$$h_2 = r$$

$$h_3 = 1$$

Now:-

The gradient of $T = \vec{\nabla} \cdot T$

$$\begin{aligned} &= \hat{r} \times \frac{\partial}{\partial r} \left(\frac{z + \sin \phi}{r} \right) + \frac{\hat{\phi}}{r} \times \frac{\partial}{\partial \phi} \left(\frac{z + \sin \phi}{r} \right) + \hat{z} \times \frac{\partial}{\partial z} \left(\frac{z + \sin \phi}{r} \right) \\ &= -\hat{r} \times \frac{z + \sin \phi}{r^2} + \frac{\hat{\phi}}{r} \times \frac{\cos \phi}{r} + \hat{z} \times \frac{1}{r} \end{aligned}$$

At the point $(2, \frac{\pi}{4}, 1)$ Grad $T = \frac{1}{2} \hat{z}$

Ans:-

⑦ $T(r, \theta, \phi) = r^2 \cos \phi \sin \theta$ at the point $(2, \frac{\pi}{4}, \frac{2\pi}{3})$

⇒ We know:-

The spherical c.s:-

$$h_1 = 1, h_2 = r, h_3 = r \cdot \sin \theta$$

Now:-

The gradient of $T = \vec{\nabla} \cdot T$

$$\begin{aligned} &= \hat{r} \times \frac{\partial}{\partial r} (r^2 \cos \phi \sin \theta) + \frac{\hat{\theta}}{r} \times \frac{\partial}{\partial \theta} (r^2 \cos \phi \sin \theta) + \frac{\hat{\phi}}{r \sin \theta} \times \frac{\partial}{\partial \phi} (r^2 \cos \phi \sin \theta) \\ &= \hat{r} \times 2r \cos \phi \sin \theta + \hat{\theta} \times \frac{1}{r} \times r^2 \cos \phi \cos \theta - \hat{\phi} \times \frac{1}{r \sin \theta} \times r^2 \sin \phi \sin \theta \\ &= \hat{r} \times 2r \sin \theta \cos \phi + \hat{\theta} \times r \cos \theta \cos \phi - \hat{\phi} r \sin \phi \end{aligned}$$

At the point $(2, \frac{\pi}{4}, \frac{2\pi}{3})$ Grad $\nabla T = -\sqrt{2} \hat{r} - \frac{1}{\sqrt{2}} \hat{\theta} - \sqrt{3} \hat{\phi}$

Ans:-

$$\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0 \right) \text{ unit vector at } (0,0,0) \text{ is } \frac{1}{\sqrt{2}} \hat{i} + \frac{1}{\sqrt{2}} \hat{j} \quad (1)$$

Directional Derivatives!

$$\left(\frac{x^2}{2}, \frac{y^2}{2}, z \right) \text{ unit vector at } (1,1,1) \text{ is } \frac{1}{\sqrt{3}} \hat{i} + \frac{1}{\sqrt{3}} \hat{j} + \frac{1}{\sqrt{3}} \hat{k} \quad (2)$$

$$\text{unit vector at } (1,1,1) \text{ is } \frac{1}{\sqrt{3}} \hat{i} + \frac{1}{\sqrt{3}} \hat{j} + \frac{1}{\sqrt{3}} \hat{k}$$

$$\text{unit vector at } (1,1,1) \text{ is } \frac{1}{\sqrt{3}} \hat{i} + \frac{1}{\sqrt{3}} \hat{j} + \frac{1}{\sqrt{3}} \hat{k}$$

$$\left(\frac{x^2}{2}, \frac{y^2}{2}, z \right) \text{ unit vector at } (1,1,1) \text{ is } \frac{1}{\sqrt{3}} \hat{i} + \frac{1}{\sqrt{3}} \hat{j} + \frac{1}{\sqrt{3}} \hat{k} \quad (3)$$

$$\text{unit vector at } (1,1,1) \text{ is } \frac{1}{\sqrt{3}} \hat{i} + \frac{1}{\sqrt{3}} \hat{j} + \frac{1}{\sqrt{3}} \hat{k}$$

$$\text{unit vector at } (1,1,1) \text{ is } \frac{1}{\sqrt{3}} \hat{i} + \frac{1}{\sqrt{3}} \hat{j} + \frac{1}{\sqrt{3}} \hat{k}$$

$$\text{unit vector at } (1,1,1) \text{ is } \frac{1}{\sqrt{3}} \hat{i} + \frac{1}{\sqrt{3}} \hat{j} + \frac{1}{\sqrt{3}} \hat{k}$$

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Directional Derivatives (D.D)

⇒ D.D is the rate of change of a scalar function $T(u_1, u_2, u_3)$ at a point P along a particular direction $\vec{a}$ is determined

by $D.D = (\vec{\nabla} \cdot T)_P \cdot \hat{a}$

where:-
 $\hat{a} = \frac{\vec{a}}{|\vec{a}|}$

Example 1:- Find the D.D of $T(r, \theta, \phi) = \frac{1}{r} \sin^2 \theta$ at the point $(5, \frac{\pi}{4}, \frac{\pi}{2})$ in the direction $\hat{r}$.

⇒ We know:-

The spherical C.S:-

$$h_1 = 1$$

$$h_2 = r$$

$$h_3 = r \sin \theta$$

Now:-

$$\begin{aligned} \vec{\nabla} \cdot T &= \hat{r} \times \frac{\partial}{\partial r} \left(\frac{1}{r} \sin^2 \theta \right) + \frac{\hat{\theta}}{r} \times \frac{\partial}{\partial \theta} \left(\frac{1}{r} \sin^2 \theta \right) + \frac{\hat{\phi}}{r \sin \theta} \times \frac{\partial}{\partial \phi} \left(\frac{1}{r} \sin^2 \theta \right) \\ &= -\frac{\hat{r}}{r^2} \sin^2 \theta + \frac{\hat{\theta}}{r} \times \frac{1}{r} \times 2 \sin \theta \cos \theta + 0 \end{aligned}$$

∴ At the point $(5, \frac{\pi}{4}, \frac{\pi}{2})$, $\vec{\nabla} T = -\frac{1}{50} \hat{r} + \frac{1}{25} \hat{\theta}$

$$\begin{aligned} \therefore D.D &= (\vec{\nabla} T)_P \cdot \hat{r} \\ &= \left(-\frac{1}{50} \hat{r} + \frac{1}{25} \hat{\theta} \right) \cdot \hat{r} \\ &= -\frac{1}{50} \end{aligned}$$

Since:-
 $\hat{a} \cdot \hat{a} = 1$
 $\hat{a} \cdot \hat{b} = 0$

Sample Exercise 6.1

Q. Find the D.D of $T(x, y, z) = x^2y - xz$ at the Point $(1, 0, 2)$ in the direction $\vec{d} = \hat{x} - 2\hat{y} - 6\hat{z}$

⇒ we know

The Cartesian C.S.:-

$$h_1 = 1$$

$$h_2 = 1$$

$$h_3 = 1$$

Now:-

$$\begin{aligned}\vec{\nabla} T &= \hat{x} \times \frac{\partial}{\partial x}(x^2y - xz) + \hat{y} \times \frac{\partial}{\partial y}(x^2y - xz) + \hat{z} \times \frac{\partial}{\partial z}(x^2y - xz) \\ &= \hat{x} \times (2xy - z) + \hat{y} \times (x^2) + \hat{z} \times (-x) \\ &= \hat{x} \times (2xy - z) + \hat{y}x^2 - \hat{z}x\end{aligned}$$

So, At the point $(1, 0, 2)$ $\nabla T = -2\hat{x} + \hat{y} - \hat{z}$

Now,

The unit vector of $\vec{d} = \frac{\vec{r}}{|\vec{r}|} = \frac{\hat{x} - 2\hat{y} - 6\hat{z}}{\sqrt{1^2 + 2^2 + 6^2}} = \frac{1}{\sqrt{41}}(\hat{x} - 2\hat{y} - 6\hat{z})$

∴ D.D = $\nabla T \cdot \vec{d}$

$$\begin{aligned}&= (-2\hat{x} + \hat{y} - \hat{z}) \cdot \left(\frac{\hat{x}}{\sqrt{41}} - \frac{2\hat{y}}{\sqrt{41}} - \frac{6\hat{z}}{\sqrt{41}} \right) \\ &= -\frac{2}{\sqrt{41}} - \frac{2}{\sqrt{41}} + \frac{6}{\sqrt{41}} \\ &= \frac{2}{\sqrt{41}}\end{aligned}$$

Ans:-

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⑥ Find the D.D of $T(r, \phi, z) = r^3 \cos \phi$ at the point $(2, \frac{\pi}{4}, 1)$ in the direction $\hat{r}$.

⇒ We know:-

The cylindrical co-ords:-

$$h_1 = 1$$

$$h_2 = r$$

$$h_3 = 1$$

Now:-

$$\begin{aligned} \nabla \cdot T &= \hat{r} \times \frac{\partial}{\partial r} (r^3 \cos \phi) + \frac{\hat{\phi}}{r} \times \frac{\partial}{\partial \phi} (r^3 \cos \phi) + \hat{z} \times \frac{\partial}{\partial z} (r^3 \cos \phi) \\ &= \hat{r} \times 3r^2 \cos \phi + \hat{\phi} \times \frac{1}{r} \times (-r^3 \sin \phi) + \hat{z} \times 0 \\ &= \hat{r} \times 3r^2 \cos \phi - \hat{\phi} r^2 \sin \phi \end{aligned}$$

∴ At the point $(2, \frac{\pi}{4}, 1)$ $\nabla T = 6\sqrt{2} \hat{r} - 2\sqrt{2} \hat{\phi}$

∴ D.D = $\nabla T \cdot \hat{r}$

$$= (6\sqrt{2} \hat{r} - 2\sqrt{2} \hat{\phi}) \cdot \hat{r}$$

$$= 6\sqrt{2}$$

Ans:-

Since
 $\hat{r} \cdot \hat{r} = 1$
 $\hat{r} \cdot \hat{\phi} = 0$

(c) Find the D.D of $T(R, \theta, \phi) = \frac{1}{R} \cos^2 \theta$ at the point $(1, \frac{\pi}{4}, \frac{\pi}{2})$ in the direction $\hat{R} - \hat{\theta}$

$\Rightarrow$ we know:-

The Spherical co-ords:-

$$h_1 = 1$$

$$h_2 = R$$

$$h_3 = R \sin \theta$$

Now:-

$$\vec{\nabla} \cdot T = \hat{R} \times \frac{\partial}{\partial R} \left(\frac{1}{R} \cos^2 \theta \right) + \frac{\hat{\theta}}{R} \times \frac{\partial}{\partial \theta} \left(\frac{1}{R} \cos^2 \theta \right) + \frac{\hat{\phi}}{R \sin \theta} \times \frac{\partial}{\partial \phi} \left(\frac{1}{R} \cos^2 \theta \right)$$

$$= -\hat{R} \times \frac{\cos^2 \theta}{R^2} + \hat{\theta} \times \frac{1}{R^2} \times (-2 \cos \theta \cdot \sin \theta) + 0$$

$$= -\hat{R} \times \frac{\cos^2 \theta}{R^2} - \hat{\theta} \times \frac{1}{R^2} \times 2 \cos \theta \cdot \sin \theta$$

So At the point $(1, \frac{\pi}{4}, \frac{\pi}{2})$, $\nabla T = -\frac{1}{2} \hat{R} - \hat{\theta}$

Now:-

The Unit vector of $\hat{R} - \hat{\theta} = \frac{\hat{R} - \hat{\theta}}{\sqrt{1^2 + 1^2}} = \frac{\hat{R}}{\sqrt{2}} - \frac{\hat{\theta}}{\sqrt{2}}$

$$\therefore D.D = \nabla T \cdot (\hat{R} - \hat{\theta})$$

$$= \left(-\frac{1}{2} \hat{R} - \hat{\theta} \right) \cdot \left(\frac{\hat{R}}{\sqrt{2}} - \frac{\hat{\theta}}{\sqrt{2}} \right)$$

$$= -\frac{1}{2\sqrt{2}} + \frac{1}{\sqrt{2}}$$

Ans:-

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#Divergence

Cardi

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Divergence:-

Let,

$$A = \hat{u}_1 A_1(u_1, u_2, u_3) + \hat{u}_2 A_2(u_1, u_2, u_3) + \hat{u}_3 A_3(u_1, u_2, u_3)$$

So,

$$\text{div } A = \nabla \cdot A = \frac{1}{h_1 h_2 h_3} \left[\frac{\partial}{\partial u_1} (A_1 h_2 h_3) + \frac{\partial}{\partial u_2} (h_1 A_2 h_3) + \frac{\partial}{\partial u_3} (h_1 h_2 A_3) \right]$$

if $\nabla \cdot A = 0$ then A is called solenoidal.

Curl:-

Let,

$$A = \hat{u}_1 A_1(u_1, u_2, u_3) + \hat{u}_2 A_2(u_1, u_2, u_3) + \hat{u}_3 A_3(u_1, u_2, u_3)$$

So,

$$\text{Curl } A = \nabla \times A = \frac{1}{h_1 h_2 h_3} \begin{vmatrix} \hat{u}_1 h_2 h_3 & \hat{u}_2 h_2 h_3 & \hat{u}_3 h_2 h_3 \\ \frac{\partial}{\partial u_1} & \frac{\partial}{\partial u_2} & \frac{\partial}{\partial u_3} \\ h_1 A_1 & h_2 A_2 & h_3 A_3 \end{vmatrix}$$

if $\text{curl } A = \nabla \times A = 0$ then A is called Conservative or irrotational.

Example: 1:- Determine the divergence and curl. Also check each of the following vector fields solenoidal, Conservative or both.

① $A = x^2\hat{i} + y^2\hat{j}$

⇒ we know:-

The Cartesian C.S:-

$$h_1 = 1$$

$$h_2 = 1$$

$$h_3 = 1$$

Now:-

$$\begin{aligned} \text{div } A &= \frac{1}{1 \times 1 \times 1} \left[\frac{\partial}{\partial x}(x^2) + \frac{\partial}{\partial y}(y^2) + 0 \right] \\ &= 2x + 2y \\ &= 4x \end{aligned}$$

And:-

$$\text{Curl } A = \frac{1}{1 \times 1 \times 1} \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2 & y^2 & 0 \end{vmatrix}$$

$$\begin{aligned} &= \hat{i} \left(0 - \frac{\partial}{\partial z}(y^2) \right) - \hat{j} \left(0 - \frac{\partial}{\partial z}(x^2) \right) + \hat{k} \left(\frac{\partial}{\partial x}(y^2) - \frac{\partial}{\partial y}(x^2) \right) \\ &= 0 - 0 + \hat{k} \times 0 \\ &= \hat{k} \times 0 \end{aligned}$$

Since $\text{div } A \neq 0$, $\text{curl } A \neq 0$

So A is not solenoidal or not Conservative

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$$(6) A = \hat{r} \frac{\sin \phi}{r^2} + \hat{\phi} \frac{\cos \phi}{r^2}$$

We know:-

The cylindrical C.S:-

$$h_1 = 1$$

$$h_2 = r$$

$$h_3 = 1$$

Now:-

$$\text{div } A = \nabla \cdot A = \frac{1}{1 \cdot r \cdot 1} \left[\frac{\partial}{\partial r} \times \frac{\sin \phi}{r^2} \times r \times 1 + \frac{\partial}{\partial \phi} \times \frac{\cos \phi}{r^2} \times 1 \times 1 \right]$$

$$= \frac{1}{r} \left[-\frac{\sin \phi}{r^2} - \frac{\sin \phi}{r^2} \right]$$

$$= -\frac{\sin \phi}{r^3} - \frac{\sin \phi}{r^3}$$

$$= -\frac{2 \sin \phi}{r^3}$$

$$\text{Curl } A = \nabla \times A = \frac{1}{1 \cdot r \cdot 1} \begin{vmatrix} \hat{r} & \hat{\phi} & \hat{z} \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial \phi} & \frac{\partial}{\partial z} \\ \frac{\sin \phi}{r^2} & \frac{\cos \phi}{r^2} & 0 \end{vmatrix}$$

$$= \left[\hat{z} \left(\frac{\partial \cos \phi / r^2}{\partial r} - \frac{\partial \sin \phi / r^2}{\partial \phi} \right) \right] \times \frac{1}{r}$$

$$= \left(-\frac{\cos \phi}{r^2} - \frac{\cos \phi}{r^2} \right) \times \hat{z} \times \frac{1}{r}$$

$$= -\hat{z} \frac{2 \cos \phi}{r^3}$$

Since $\text{div} \neq 0$, $\text{curl} \neq 0$

So A is not Solenoidal or not Conservative.

$$\textcircled{c} A = \hat{r}(R e^{-R})$$

⇒ we know:-

The Spherical c.s:-

$$h_1 = 1$$

$$h_2 = R$$

$$h_3 = R \sin \theta$$

Now:-

$$\text{div } A = \nabla \cdot A = \frac{1}{h_1 h_2 h_3} \left[\frac{\partial}{\partial u_1} A_1 h_2 h_3 + \frac{\partial}{\partial u_2} h_1 A_2 h_3 + \frac{\partial}{\partial u_3} h_1 h_2 A_3 \right]$$

$$= \frac{1}{R^2 \sin \theta} \left[\frac{\partial}{\partial R} (R e^{-R} \cdot R^2 \sin \theta) \right]$$

$$= \frac{1}{R^2 \sin \theta} \left[-R^3 \sin \theta \times e^{-R} + e^{-R} 2R^2 \sin \theta \right]$$

$$= \frac{1}{R^2 \sin \theta} \times e^{-R} \times R^2 \sin \theta [(2 - R)]$$

$$= e^{-R} (2 - R)$$

$$\text{Curl } A = \nabla \times A = \frac{1}{h_1 h_2 h_3} \begin{vmatrix} \hat{u}_1 & \hat{u}_2 & \hat{u}_3 \\ \frac{\partial}{\partial u_1} & \frac{\partial}{\partial u_2} & \frac{\partial}{\partial u_3} \\ h_1 A_1 & h_2 A_2 & h_3 A_3 \end{vmatrix}$$

$$= \frac{1}{R^2 \sin \theta} \begin{vmatrix} \hat{r} & \hat{\theta} & \hat{\phi} \\ \frac{\partial}{\partial R} & \frac{\partial}{\partial \theta} & \frac{\partial}{\partial \phi} \\ R e^{-R} & 0 & 0 \end{vmatrix}$$

$$= 0$$

Since $\text{div } A \neq 0$ But $\text{Curl } A = 0$

So, A is not Solenoidal but Conservative.

#Laplacian
Operator

#Formula:—

$$\underline{\underline{1.}} \int dx = x + C$$

$$\underline{\underline{2.}} \int 1 dx = x + C$$

$$\underline{\underline{3.}} \int a dx = ax + C$$

$$\underline{\underline{4.}} \int x^n \cdot dx = \frac{x^{n+1}}{n+1} + C$$

$$\underline{\underline{5.}} \int e^x \cdot dx = e^x + C$$

$$\underline{\underline{6.}} \int \frac{1}{x} \cdot dx = \ln|x| + C$$

$$\underline{\underline{7.}} \int \frac{1}{\sqrt{x}} \cdot dx = 2\sqrt{x} + C$$

$$\underline{\underline{8.}} \int a^x \cdot dx = \frac{a^x}{\ln a} + C$$

$$\underline{\underline{9.}} \int \sin x \cdot dx = -\cos x + C$$

$$\underline{\underline{10.}} \int \cos x \cdot dx = \sin x + C$$

$$\underline{\underline{11.}} \int \sec^2 x \cdot dx = \tan x + C$$

$$\underline{\underline{12.}} \int \csc^2 x \cdot dx = -\cot x + C$$

$$\underline{\underline{13.}} \int \sec x \cdot (\tan x) \cdot dx = \sec x + C$$

$$\underline{\underline{14.}} \int \csc x \cdot (\cot x) \cdot dx = -\csc x + C$$

$$\underline{\underline{15.}} \int \frac{1}{1+x^2} \cdot dx = \tan^{-1} x + C$$

$$\underline{\underline{16.}} \int \frac{1}{\sqrt{1-x^2}} \cdot dx = \sin^{-1} x + C$$

$$\underline{\underline{17.}} \int \frac{1}{|x| \cdot \sqrt{x^2-1}} \cdot dx = \sec^{-1} x + C$$

$$\underline{\underline{18.}} \int \frac{1}{a^2+x^2} dx = \frac{1}{a} + \tan^{-1} \frac{x}{a} + C$$

$$\underline{\underline{19.}} \int \frac{1}{\sqrt{a^2-x^2}} \cdot dx = \sin^{-1} \frac{x}{a} + C$$

$$\underline{\underline{20.}} \int \cosh x \cdot dx = \sinh x + C$$

$$\underline{\underline{21.}} \int \sinh x \cdot dx = +\cosh x + C$$

$$\underline{\underline{22.}} \int e^{mx} \cdot dx = \frac{1}{m} e^{mx} + C$$

$$\underline{\underline{23.}} \int \sin mx \cdot dx = -\frac{1}{m} \cos mx + C$$

$$\underline{\underline{24.}} \int e^{\frac{mx}{n}} \cdot dx = \frac{n}{m} e^{\frac{mx}{n}} + C$$

$$\underline{\underline{25.}} \int \operatorname{cosec} u \cdot du = -\ln |\operatorname{cosec} u + \cot u| + C$$

$$\textcircled{26.} \int \sec u \cdot du = \ln |\sec u + \tan u| + C$$

$$\textcircled{27.} \int \sec 2u \cdot du = \frac{1}{2} \ln |\sec 2u + \tan 2u| + C$$

#Laplacian Operator:-

$\Rightarrow$ The differential operator $\nabla^2 = \nabla \cdot \nabla$ is called Laplacian operator.

$\Rightarrow \nabla \cdot (\nabla T) = \nabla^2 T(u_1, u_2, u_3) = 0$ is called Laplace Equation.

$\Rightarrow$ The Laplace of scalar function T in different coordinate system are define as follows.

① In Cartesian Coordinates:-

$$\nabla^2 T = \frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2}$$

② In cylindrical coordinates:-

$$\nabla^2 T = \frac{1}{r} \times \frac{\partial}{\partial r} \left(r \times \frac{\partial T}{\partial r} \right) + \frac{1}{r^2} \times \frac{\partial^2 T}{\partial \phi^2} + \frac{\partial^2 T}{\partial z^2}$$

③ In spherical coordinates:-

$$\nabla^2 T = \frac{1}{r^2} \times \frac{\partial}{\partial r} \left(r^2 \frac{\partial T}{\partial r} \right) + \frac{1}{r \sin \theta} \times \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial T}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \left(\frac{\partial^2 T}{\partial \phi^2} \right)$$

Ex Example :- Find the Laplacian of the scalar function $T = \frac{3}{x^2+y^2}$

⇒ We know:-

$$\nabla^2 T = \frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2}$$

Now, $\frac{\partial T}{\partial x} = \frac{\partial}{\partial x} \times \frac{3}{x^2+y^2} = \frac{-6x}{(x^2+y^2)^2}$

$$\therefore \frac{\partial^2 T}{\partial x^2} = \frac{-6(x^2+y^2)^2 + 12x(x^2+y^2) \times 2x}{(x^2+y^2)^4}$$

$$= \frac{24x^2(x^2+y^2) - 6(x^2+y^2)^2}{(x^2+y^2)^4}$$

$$= \frac{(x^2+y^2)^2 \{24x^2 - 6(x^2+y^2)\}}{(x^2+y^2)^4}$$

$$= \frac{18x^2 - 6y^2}{(x^2+y^2)^3}$$

And:-

$$\frac{\partial T}{\partial y} = \frac{\partial}{\partial y} \times \frac{3}{x^2+y^2} = \frac{-6y}{(x^2+y^2)^2}$$

$$\therefore \frac{\partial^2 T}{\partial y^2} = \frac{-6(x^2+y^2)^2 + 6 \times 2 \times y \times 2y(x^2+y^2)}{(x^2+y^2)^4}$$

$$= \frac{(x^2+y^2)^2 \{24y^2 - 6x^2 - 6y^2\}}{(x^2+y^2)^4}$$

$$= \frac{-6x^2 + 18y^2}{(x^2+y^2)^3}$$

$$\therefore \nabla^2 T = \frac{18x^2 - 6y^2 - 6x^2 + 18y^2}{(x^2+y^2)^3}$$

$$= \frac{12}{(x^2+y^2)^2}$$

Example: 2:- Find the Laplacian for $T = 4yz^2$

we know:-

$$\nabla^2 T = \frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2}$$

So, $\frac{\partial T}{\partial x} = \frac{\partial}{\partial x} (4yz^2) = 0$

$\therefore \frac{\partial^2 T}{\partial x^2} = 0$

And:- $\frac{\partial T}{\partial y} = 8yz^2$

$\therefore \frac{\partial^2 T}{\partial y^2} = 8z^2$

Also:- $\frac{\partial T}{\partial z} = 8yz$

$\therefore \frac{\partial^2 T}{\partial z^2} = 8y$

$\therefore \nabla^2 T = 8z^2 + 8y$

Example: 3:- Test whether: $A = \hat{x}(y^2 \cos x + z^3) + \hat{y}(2y \sin x - y) + \hat{z}(3xz^2 + 2)$ is a conservative force field. If conservative find the scalar potential T such that $A = \nabla T$. Hence find the work done in moving an object from $(0, 1, -1)$ to $(\frac{\pi}{2}, -1, 2)$

$\Rightarrow$ we know:-
anon

For Conservative $\text{Curl } \vec{A} = \nabla \times \vec{A} = 0$

$$\underline{\text{So,}} \quad \text{Curl } \vec{A} = \vec{\nabla} \times \vec{A} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y^2 \cos x + z^3 & 2y \sin x + y & 3xz^2 + 2 \end{vmatrix}$$

$$= \hat{x}(0-0) - \hat{y}(3z^2-3z^2) + \hat{z}(2y \cos x - 2y \cos x) \\ = 0$$

So, $\vec{A}$ is Conservative.

$\therefore$ Let,

$T(x, y, z)$ be a scalar potential of $\vec{A}$

$$\therefore \vec{A} = \vec{\nabla} \cdot T$$

$$\begin{aligned} T &= \int \vec{A} \cdot d\vec{r} \\ &= \int (y^2 \cos x + z^3) dx + (2y \sin x + y) dy \\ &\quad + (3xz^2 + 2) dz \end{aligned} \quad \left. \begin{array}{l} \text{where,} \\ d\vec{r} = \hat{x} dx + \hat{y} dy + \hat{z} dz \end{array} \right\}$$

$$= \int [y^2 \sin x + z^3 x + \frac{2}{2} y^2 \sin x - \frac{1}{2} y^2 + \frac{3}{3} xz^3 + 2z]$$

$$= \int d[y^2 \sin x + z^3 x + 2z - \frac{y^2}{2} + c]$$

$$= y^2 \sin x + z^3 x - \frac{y^2}{2} + 2z + c$$

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So,

$$\text{work done} = \int_{P_1}^{P_2} \vec{A} \cdot d\vec{r} = T(P_2) - T(P_1)$$

Now,

$$T\left(\frac{\pi}{2}, -1, 2\right) = 1 + 8 \times \frac{\pi}{2} - \frac{1}{2} + 4 + c$$

$$T(0, 1, -1) = -\frac{1}{2} - 2 + c$$

So,

$$\begin{aligned} \text{work done w.d} &= T\left(\frac{\pi}{2}, -1, 2\right) - T(0, 1, -1) \\ &= 1 + 4\pi - \frac{1}{2} + 4 + c + \frac{1}{2} + 2 - c \\ &= 7 + 4\pi \text{ unit} \end{aligned}$$

Ans:-

Example: 4:- Test whether $A = \hat{r}(\sin\theta \cdot \cos\phi) + \hat{\theta}(\cos\theta \cdot \cos\phi) + \hat{\phi}(-\sin\phi)$ is a conservative force field. If conservative, find the scalar potential T such that $A = \nabla T$. Hence find the work done in moving an object in this field from $(0, \frac{\pi}{4}, \frac{\pi}{6})$ to $(1, \frac{\pi}{2}, \pi)$

⇒ We know —
mon

For Conservative curve $A = 0$

$$\begin{aligned}\therefore \nabla \times A &= \frac{1}{r^2 \sin \theta} \begin{vmatrix} \hat{r} & \hat{\theta} & \hat{\phi} r \sin \theta \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial \theta} & \frac{\partial}{\partial \phi} \\ \sin \theta \cdot \cos \phi & \cos \theta \cdot \cos \phi & r \sin \theta \cdot (-\sin \phi) \end{vmatrix} \\ &= \frac{1}{r^2 \sin \theta} \left[\hat{r}(-\cos \theta \cdot \sin \phi + \cos \theta \cdot \sin \phi) - \hat{\theta}(-\sin \theta \cdot \sin \phi + \sin \theta \cdot \sin \phi) \right. \\ &\quad \left. + \hat{\phi} r \sin \theta (\cos \theta \cdot \cos \phi - \cos \theta \cdot \cos \phi) \right] \\ &= \frac{1}{r^2 \sin \theta} \left[\hat{r}(0) - \hat{\theta}(0) + \hat{\phi} r \sin \theta (0) \right] \\ &= 0\end{aligned}$$

so, $\vec{A}$ is conservative.

Let

$T(r, \theta, \phi)$ be a scalar potential of A , $\vec{A} = \nabla T$

$$\therefore T = \int \vec{A} \cdot d\vec{r}$$

$$\begin{aligned}&= \int [\hat{r}(\sin \theta \cdot \cos \phi) + \hat{\theta}(-\sin \phi) + \hat{\phi}(\cos \theta \cdot \cos \phi)] \cdot [\hat{r}dr + \hat{\theta} \cdot r d\theta + \hat{\phi} r \sin \theta d\phi] \\ &= \int [(\sin \theta \cdot \cos \phi)dr + (\cos \theta \cdot \cos \phi) \cdot r d\theta + (-\sin \phi) r \sin \theta \cdot d\phi] \\ &= \int [r \sin \theta \cdot \cos \phi + \sin \theta \cdot \cos \phi \cdot r + r \cos \phi \cdot \sin \theta] \\ &= \int d[r \sin \theta \cdot \cos \phi + c] \\ &= r \sin \theta \cdot \cos \phi + c\end{aligned}$$

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Now - Work done = $\int_{P_1}^{P_2} \vec{A} \cdot d\vec{r} = T(P_2) - T(P_1) \quad \text{--- (1)}$

So, $T_1(0, \frac{\pi}{4}, \frac{\pi}{3}) = 0$

$T_2(1, \frac{\pi}{2}, \pi) = -1$

$$\left. \begin{aligned} T_1 &= r \sin \theta \cdot \cos \phi \\ &= 0 \times \sin\left(\frac{\pi}{4}\right) \cos\left(\frac{\pi}{3}\right) = 0 \\ T_2 &= r \sin \theta \cdot \cos \phi \\ &= 1 \times \sin\left(\frac{\pi}{2}\right) \cdot \cos(\pi) \\ &= -1 \end{aligned} \right\}$$

Put the values in eq (1) $\Rightarrow$

$\therefore W \cdot d = -1 - 0$
 $= -1 \text{ Unit}$

Ans:-

Example 5:- Test whether $\vec{A} = \hat{r}(\cos \phi) + \hat{\phi}(-\sin \phi)$ is a conservative force field, if conservative, find the scalar potential T such that $\vec{A} = \nabla T$. Hence find the work done in moving an object in this field from $(0, \pi, -1)$ to $(1, \pi, 2)$

$\Rightarrow$ we know:-

For Conservative, $\text{Curl } \vec{A} = \nabla \times \vec{A} = 0$

$$\begin{aligned} \therefore \nabla \times \vec{A} &= \begin{vmatrix} \hat{r} & \hat{\phi} & \hat{z} \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial \phi} & \frac{\partial}{\partial z} \\ \cos \phi & -\sin \phi & 0 \end{vmatrix} \times \frac{1}{r} \\ &= \frac{1}{r} [\hat{r}(0-0) - \hat{\phi}(0-0) + \hat{z}(-\sin \phi + \sin \phi)] \\ &= 0 \end{aligned}$$

So, The vector $\vec{A}$ is conservative.

Let $T(r, \phi, z)$ is a scalar potential of A , $\vec{A} = \vec{\nabla} \cdot T$

$$\therefore T = \int \vec{A} \cdot d\vec{r}$$

$$= \int [\hat{r}(\cos\phi) + \hat{\phi}(-\sin\phi)] \cdot [\hat{r} dr + \hat{\phi} r d\phi + \hat{z} dz]$$

$$= \int [r \cos\phi \cdot dr - r \sin\phi \cdot d\phi]$$

$$= \int [r \cos\phi + r \cos\phi]$$

$$= \int d(r \cos\phi + c)$$

$$= r \cos\phi + c$$

Now

$$\text{work done, } W_d = \int_{P_1}^{P_2} \vec{A} \cdot d\vec{r} = TP_2 - TP_1$$

$$\therefore TP_2 = r \cos\phi = 1 \times \cos(2 \times 180) = -1$$

$$TP_1 = r \cos\phi = 0 \times \cos(2 \times 180) = 0$$

$$\therefore W_d = T(P_2) - T(P_1)$$

$$= -1 - 0$$

$$= -1 \text{ Unit}$$

Ans:-

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Gauss's Divergence
Theorem

Gauss's - Divergence Theorem :-

Surface integral $\int \text{div } \vec{A} \cdot d\vec{v} = \oint_S \vec{A} \cdot d\vec{s}$

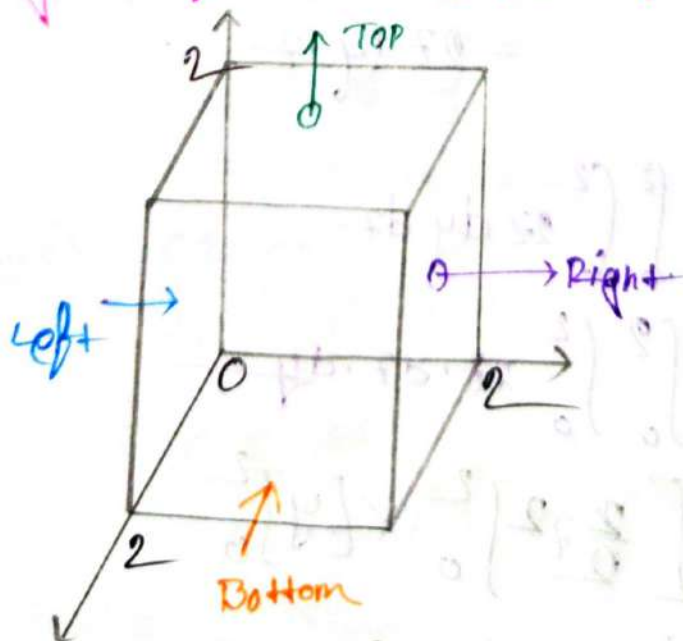
Example 1:-

3) For the vector field $A = x^2\hat{x} - y^2\hat{y} - z^2\hat{z}$ verify the divergence theorem by computing

(a) The total outward flux flowing through the surface of a cube center at the origin and with sides equal to 2 units each and parallel to the Cartesian axes.

(b) The integral of $\vec{v} \cdot \vec{A}$ over the cube's volume.

⇒



$\Rightarrow$ we know:-

In Cartesian C.S:-

the differential length

$$d\vec{r} = \hat{x} \cdot dx + \hat{y} \cdot dy + \hat{z} \cdot dz$$

For Surface:-

$$x = 2$$

The differential surface area: $(d\vec{s})$

$$d\vec{s}_x = \hat{x} \cdot dy \cdot dz$$

It is given:-

$$\vec{A} = \hat{x} \cdot xz - \hat{y} \cdot yz - \hat{z} \cdot xy$$

$$\begin{aligned} \therefore \vec{A} \cdot d\vec{s} &= xz \cdot dy \cdot dz \\ &= 2z \cdot dy \cdot dz \end{aligned}$$

[$\because x=2$]

$$\begin{aligned} \therefore S_{\text{font}} &= \int_0^2 \int_0^2 2z \cdot dy \cdot dz \\ &= \int_0^2 \int_0^2 2z \cdot dz \cdot dy \\ &= \left[\frac{2z^2}{2} \right]_0^2 \times [y]_0^2 \\ &= [4 \times 2] = 8 \end{aligned}$$

Back Surface:-

$$x=0$$

The differential surface area:-

$$d\vec{s} = -\hat{x} \cdot dy dz \quad [\because x=0]$$

$$\therefore \vec{A} \cdot d\vec{s} = -xz \cdot dy dz$$
$$= 0$$

$$\therefore S_{\text{back}} = 0$$

Right - Surface:-

$$y=2$$

The D.S.A: (ds)

$$d\vec{s}_y = \hat{y} \cdot dx dz$$

$$\therefore \vec{A} \cdot d\vec{s} = -y z^2 dx dz$$

$$[\because y=2]$$

$$= -2z^2 dx dz$$

$$\therefore S_{\text{right}} = \int_0^2 \int_0^2 -2z^2 \cdot dz \cdot dx$$

$$= -2 \times \left[\frac{1}{3} z^3 \right]_0^2 \left[x \right]_0^2$$

$$= -\frac{32}{3}$$

Left-Surface:-

$$y=0$$

∴ The D.S.A (dS):-

$$d\vec{S} = -\hat{y} \cdot dx dz$$

$$\therefore \vec{A} \cdot d\vec{S} = +yz^2 \cdot dx dz \quad [\because y=0]$$

$$= 0$$

$$\therefore S_{\text{left}} = 0$$

Top-Surface:-

$$z=2$$

∴ The D.S.A:-

$$d\vec{S} = \hat{z} \cdot dx dy$$

$$\therefore \vec{A} \cdot d\vec{S} = -xy \, dx dy$$

So

$$S_{\text{Top}} = \int_0^2 \int_0^2 -xy \, dx \, dy$$

$$= -\left[\frac{1}{2}x^2\right]_0^2 \times \left[\frac{1}{2}y^2\right]_0^2$$

$$= -4$$

Bottom surface:-
 $z=0$

$\therefore$ The D.S.A:-

$$d\vec{s} = -\hat{z} \cdot dxdy$$

$$\therefore \vec{A} \cdot d\vec{s} = +xy dxdy \quad [\because z=0]$$

$$\begin{aligned} \text{So, } S_{\text{bottom}} &= \int_0^2 \int_0^2 xy \, dxdy \\ &= \left[\frac{1}{2} x^2 \right]_0^2 \times \left[\frac{1}{2} y^2 \right]_0^2 \\ &= 4 \end{aligned}$$

$$\begin{aligned} \therefore \int \vec{A} \cdot d\vec{s} &= S_{\text{front}} + S_{\text{back}} + S_{\text{top}} + S_{\text{bottom}} + S_{\text{left}} + S_{\text{right}} \\ &= 8 - \frac{8}{3} + 4 - 4 + 0 + 0 \\ &= -\frac{8}{3} \end{aligned}$$

Now:-

Volume Integral:-

We know:-

For Cartesian co-ords:-

$$dv = dxdydz$$

$$\begin{aligned} \therefore \text{div } \vec{A} = \vec{\nabla} \cdot \vec{A} &= \frac{\partial}{\partial x}(Ax) + \frac{\partial}{\partial y}(Ay) + \frac{\partial}{\partial z}(Az) \\ &= \frac{\partial}{\partial x}(xz) + \frac{\partial}{\partial y}(-yz^2) + \frac{\partial}{\partial z}(-xy) \\ &= z - z^2 \end{aligned}$$

$$\begin{aligned}
 \therefore \int \operatorname{div} \vec{A} \cdot dV &= \int_0^2 \int_0^2 \int_0^2 (z - z^2) dx dy dz \\
 &= [x]_0^2 \times [y]_0^2 \times \left[\frac{1}{2} z^2 - \frac{1}{3} z^3 \right]_0^2 \\
 &= 2 \times 2 \times \left(2 - \frac{8}{3} \right) \\
 &= -\frac{8}{3}
 \end{aligned}$$

$$\therefore \int \operatorname{div} \vec{A} \cdot dV = \int \vec{A} \cdot d\vec{S} = -\frac{8}{3}$$

So, Divergence - Theorem Verified :-

Ans:-

Example 2:-

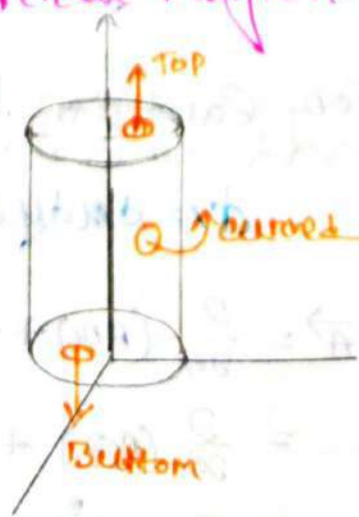
⇒ For a vector function $A = r^2 \hat{r} + z \hat{z}$, verify for the circular cylindrical region enclosed by $r = 5$, $z = 0$, $z = 4$.

⇒ Given:-

$$r = 5$$

$$z = 0$$

$$z = 4$$



We know:-

For Cylindrical co-ord.

$$d\vec{r} = \hat{r} \cdot dr + \hat{\phi} \cdot r d\phi + \hat{z} \cdot dz$$

Top-Surface:-

$$d\vec{s} = \hat{z} \cdot dr \cdot r \cdot d\phi$$

$$\therefore \vec{A} \cdot d\vec{s} = 2z \cdot r \cdot dr \cdot d\phi$$

$$= 2 \times 4 \cdot r \cdot dr \cdot d\phi$$

$$[\because z=4]$$

$$= 8r dr \cdot d\phi$$

$$\therefore S_{\text{Top}} = \int_0^{2\pi} \int_0^5 8r dr d\phi$$

$$= 8 \left[\frac{1}{2} r^2 \right]_0^5 \left[\phi \right]_0^{2\pi}$$

$$= 100 \times 2\pi$$

$$= 200\pi$$

Bottom Surface:-

$$z=0$$

$$d\vec{s} = -\hat{z} \cdot dr \cdot r \cdot d\phi$$

$$\therefore \vec{A} \cdot d\vec{s} = -2zr \cdot dr \cdot d\phi$$

$$= 0$$

$$[\because z=0]$$

$$\therefore S_{\text{Bottom}} = 0$$

For Curved Surface:-

$$R = 5$$

$$d\vec{s} = \hat{r} \cdot r d\phi \cdot dz$$

$$\vec{A} \cdot d\vec{s} = r^2 \cdot d\phi \cdot dz$$

$$= 5^2 \cdot d\phi \cdot dz$$

$$= 125 d\phi \cdot dz$$

$$\therefore \text{Surface} = \int_0^{2\pi} \int_0^4 125 d\phi \cdot dz$$

$$= 125 [\phi]_0^{2\pi} \times [z]_0^4$$

$$= 1000\pi$$

$$\therefore \int \vec{A} \cdot d\vec{s} = 1000\pi + 200\pi$$

$$= 1200\pi$$

Now Volume Integral:-

$$dV = dr \cdot r d\phi \cdot dz$$

$$\text{div } \vec{A} = \vec{\nabla} \cdot \vec{A} = \frac{1}{r} \left(\frac{\partial}{\partial r} (r^2 \cdot r') + \frac{\partial}{\partial \phi} (0 \times 1 \times 1) + \frac{\partial}{\partial z} (1 \cdot r \cdot 2z) \right)$$

$$= \frac{1}{r} (3r^2 + 2r)$$

$$= 3r + 2$$

$$\begin{aligned}
 \int \operatorname{div} \vec{A} dV &= \int_{z=0}^4 \int_{\phi=0}^{2\pi} \int_{r=0}^5 (3r+2) r \cdot dr d\phi dz \\
 &= \int_0^4 \int_0^{2\pi} \int_0^5 (3r^2 + 2r) dr d\phi dz \\
 &= \left[3 \times \frac{1}{3} r^3 + 2 \times \frac{1}{2} r^2 \right]_0^5 \times \left[\phi \right]_0^{2\pi} \times \left[z \right]_0^4 \\
 &= 150 \times 2\pi \times 4 \\
 &= 1200\pi
 \end{aligned}$$

$$\therefore \oint_S \vec{A} \cdot d\vec{s} = \int \operatorname{div} \vec{A} \cdot dV = 1200\pi$$

(Verified)

Example 3:-

$\Rightarrow \vec{A} = r\vec{r}$ in the region between two concentric cylindrical surface defined by $r=1$ and $r=2$, with both the cylinders extending between $z=0$ and $z=5$ verify divergence theorem

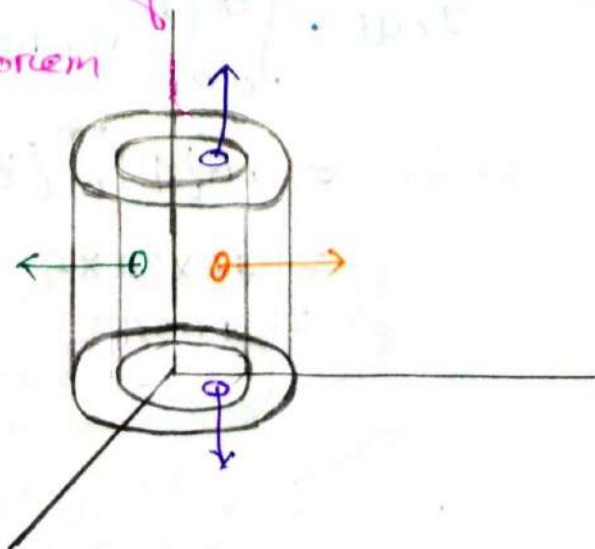
$\Rightarrow$ Is given:-

$$r=1$$

$$r=2$$

$$z=0$$

$$z=5$$



We know:-

For Cylindrical c.s:-

$$d\vec{r} = \hat{r} \cdot dr + \hat{\phi} \cdot r \cdot d\phi + \hat{z} \cdot dz$$

Top Surface:-

$$\therefore d\vec{s} = \hat{z} \cdot dr \cdot r \cdot d\phi$$

$$\vec{A} \cdot d\vec{s} = 0 \quad [\because z=5]$$

$$\therefore I_{\text{top}} = 0$$

Bottom Surface:-

$$z=0$$

$$\therefore d\vec{s} = -\hat{z} \cdot dr \cdot r \cdot d\phi$$

$$\vec{A} \cdot d\vec{s} = 0$$

$$I_{\text{bot}} = 0$$

Outer-Surface:-

$$r=2$$

$$d\vec{s} = \hat{r} \cdot r \cdot d\phi \cdot dz$$

$$\therefore \vec{A} \cdot d\vec{s} = r^4 \cdot d\phi \cdot dz$$

$$= 16 d\phi \cdot dz \quad [\because r=2]$$

$$I_{\text{out}} = \int_0^{2\pi} \int_0^5 16 \cdot d\phi \cdot dz$$

$$= 16 [\phi]_0^{2\pi} [z]_0^5$$

$$= 16 \times 2\pi \times 5$$

$$= 160\pi$$

Inner-Surface

$$d\vec{s} = -\hat{r} \cdot d\tau \cdot r d\phi$$

$$\vec{A} \cdot d\vec{s} = -r^4 \cdot d\phi \cdot d\tau$$

$$= -d\phi \cdot d\tau \quad [\because r=1]$$

$$I_{\text{inn}} = - \int_0^{2\pi} \int_0^5 d\tau \cdot d\phi$$

$$= -[\phi]_0^{2\pi} \times [\tau]_0^5$$

$$= -5 \times 2\pi$$

$$= -10\pi$$

$$\therefore \oint d\vec{s} \cdot \vec{A} = 160\pi - 10\pi$$

$$= 150\pi$$

Now:-

Surface Integral:-

$$dv = dr \cdot r \cdot d\phi \cdot d\tau$$

$$\oint \text{div} \vec{A} \cdot dv = \int \frac{1}{r} \left\{ \frac{\partial}{\partial r} (r^2 \times r \times 1) \right\} \cdot dv$$

$$= \frac{1}{r} (4r^3)$$

$$= 4r^2$$

$$\therefore \int \text{div} \vec{A} \cdot dv = \int_{r=1}^{r=2} \int_{\phi=0}^{2\pi} \int_{\tau=0}^5 4r^2 \cdot r \cdot dr \cdot d\phi \cdot d\tau$$

$$= \left[4 \times \frac{1}{4} r^4 \right]_1^2 [\phi]_0^{2\pi} [\tau]_0^5$$

$$= 15 \times 2\pi \times 5$$

$$= 150\pi$$

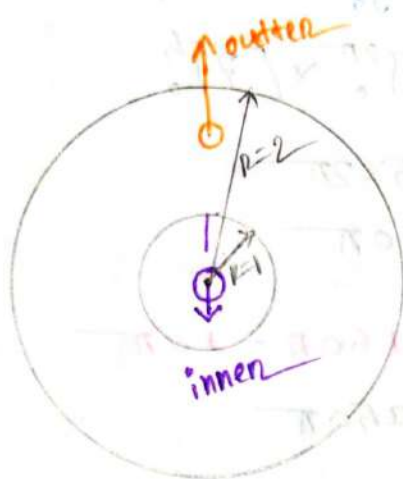
$$\therefore \oint d\vec{s} \cdot \vec{A} = \int \text{div} \vec{A} \cdot dv = 150\pi$$

(verified)

Example 4:-

$\Rightarrow \vec{A} = \hat{r} \cdot 3r^2$, Evaluate both sides of the divergence for the region enclosed between spherical shells defined by $r=1$ and $r=2$

$\Rightarrow$ In given:-



We know:-

Spherical co-ords:-

$$d\vec{r} = \hat{r} \cdot dr + \hat{\theta} \cdot r d\theta + \hat{\phi} \cdot r \sin\theta \cdot d\phi$$

$$h_1 = 1$$

$$h_2 = r$$

$$h_3 = r \sin\theta$$

outer-Surface:-

$$r=2$$

$$d\vec{s} = \hat{r} \times r d\theta \times r \sin\theta \cdot d\phi$$

$$\therefore \vec{A} \cdot d\vec{s} = 3r^3 d\theta \cdot r \cdot \sin\theta \cdot d\phi$$

$$= 3r^4 d\theta \cdot \sin\theta \cdot d\phi$$

$$= 3 \times 2^4 \sin\theta \cdot d\theta \cdot d\phi$$

$$= 48 \sin\theta \cdot d\theta \cdot d\phi$$

$$[\because r=2]$$

$$I_{\text{outer}} = \int_{\theta=0}^{\pi} \int_{\phi=0}^{2\pi} 48 \cdot \sin \theta \cdot d\theta \cdot d\phi$$

$$= -48 [\cos \theta]_0^{\pi} \times [\phi]_0^{2\pi}$$

$$= -48(-1-1) \times 2\pi$$

$$= 192\pi$$

Inner-Surface:-

$$R=1$$

$$d\vec{s} = -\hat{r} \cdot R^2 \sin \theta \cdot d\theta \cdot d\phi$$

$$\vec{A} \cdot d\vec{s} = -3R^4 \sin \theta \cdot d\theta \cdot d\phi$$

$$= -3 \sin \theta \cdot d\theta \cdot d\phi \quad [\because R=1]$$

$$I_{\text{inner}} = \int_{\phi=0}^{2\pi} \int_{\theta=0}^{\pi} -3 \sin \theta \cdot d\theta \cdot d\phi$$

$$= -3 [-\cos \theta]_0^{\pi} \times [\phi]_0^{2\pi}$$

$$= -3(1+1) \times 2\pi$$

$$= -12\pi$$

$$\therefore \oint \vec{A} \cdot d\vec{s} = 192\pi - 12\pi$$

$$= 180\pi$$

now Surface Integral:-
for spherical c.s:-

$$dv = dr \cdot r d\theta \cdot r \sin\theta \cdot d\phi$$

$$\therefore \text{div } \vec{A} = \vec{\nabla} \cdot \vec{A} = \left\{ \frac{1}{r^2 \sin\theta} \right\} \frac{\partial}{\partial r} (3r^2 \cdot r \cdot r \cdot \sin\theta) \Bigg\}$$

$$= \frac{1}{r^2 \sin\theta} \times 12r^3 \cdot \sin\theta$$

$$= 12r$$

$$\therefore \int \text{div } \vec{A} \cdot dv = \int_{\phi=0}^{2\pi} \int_{\theta=0}^{\pi} \int_{r=1}^2 12r^3 \cdot \sin\theta \cdot d\theta \cdot d\phi$$

$$= 12 \left[\frac{1}{4} r^4 \right]_1^2 \times [-\cos\theta]_0^{\pi} \times [\phi]_0^{2\pi}$$

$$= 12 [16 - 1] \times [1 + 1] \times 2\pi$$

$$= 180\pi$$

$$\therefore \int \text{div } \vec{A} \cdot dv = \oint \vec{A} \cdot d\vec{s} = 180\pi$$

(verified)

#Stoke's Theorem!

Stokes's Theorem:-

$$\int_S (\nabla \times \vec{A}) \cdot d\vec{s} = \oint_C \vec{A} \cdot d\vec{r}$$

Example :-

$$\Rightarrow \vec{A} = \hat{x}(2x^2 + y^2) + \hat{y}(xy - y^2)$$

(a) find $\oint_C \vec{A} \cdot d\vec{r}$ around the triangular contour.

(b) find $\int_S (\nabla \times \vec{A}) \cdot d\vec{s}$ over triangular area.

(c) Verify Stoke's theorem

(d) Can $\vec{A}$ be expressed as gradient of a scalar? Explain.

$\Rightarrow$ we know:-

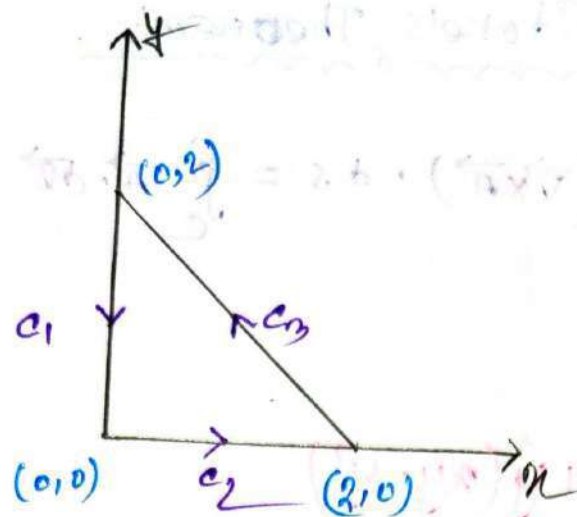
Differential length $d\vec{r} = \hat{x} \cdot dx + \hat{y} \cdot dy + \hat{z} \cdot dz$

Is given:-

$$\vec{A} = \hat{x}(2x^2 + y^2) + \hat{y}(xy - y^2)$$

$$\therefore \vec{A} \cdot d\vec{r} = (2x^2 + y^2)dx + (xy - y^2)dy$$

$$\frac{\partial f}{\partial x} = \frac{\partial}{\partial x}(2x^2 + y^2) = 4x$$



Along c_1 :-

$$x = 0$$

$$\therefore dx = 0$$

So, $\vec{A} \cdot d\vec{r} = -y^2 \cdot dy$ where $y \rightarrow 2 \text{ to } 0$

$$\therefore I_{c1} = - \int_2^0 y^2 \cdot dy$$

$$= - \left[\frac{1}{3} y^3 \right]_2^0$$

$$= \frac{8}{3}$$

Along c_2 :-

$$y = 0$$

$$\therefore dy = 0$$

So, $\vec{A} \cdot d\vec{r} = 2x^2 \cdot dx$ where $x \rightarrow 0 \text{ to } 2$

$$\therefore I_{c2} = \int_0^2 2x^2 \cdot dx$$

$$= \frac{2}{3} [x^3]_0^2 = \frac{16}{3}$$

Along CB:-

$$\therefore y = 2 - x$$

$$dy = -dx$$

$$\frac{x}{2} + \frac{y}{2} = 1$$

$$x + y = 2$$

$$y = 2 - x$$

$$\therefore \vec{A} \cdot d\vec{r} = (2x^2 + y^4) dx + (xy - y^2) dy$$

$$= (2x^2 + (2-x)^2) dx + (x(2-x) - (2-x)^2) (-dx)$$

$$= [2x^2 + (2-x)^2 - 2x + x^2 + (2-x)^2] dx$$

$$= [2x^2 - 2x + x^2 + 2(2-x)^2] dx$$

$$= [3x^2 - 2x + 2(2-x)^2] dx$$

$$\therefore I_{C3} = \int_2^0 [3x^2 - 2x + 2(2-x)^2] dx$$

$$= \left[\frac{3}{3} x^3 - \frac{1}{2} \times 2 x^2 + \frac{2(2-x)^3}{-3} \right]_2^0$$

$$= -8 + 4 - \frac{1}{3}(8 - 0)$$

$$= -\frac{28}{3}$$

$$= \frac{d}{dx} (a+bx)^n$$

$$= \frac{(a+bx)^{n+1}}{(n+1) \times b}$$

$$\therefore \oint \vec{A} \cdot d\vec{r} = I_{C1} + I_{C2} + I_{C3}$$

$$= \frac{8}{3} + \frac{16}{3} - \frac{28}{3}$$

$$= -\frac{4}{3}$$

Now $\vec{r} \times \vec{A} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2+y^2 & xy-y^2 & 0 \end{vmatrix}$

$$= \hat{k}(y-2xy)$$

$$= -\hat{k}y$$

$$\therefore d\vec{S} = \hat{k} \cdot dx dy$$

So $(\vec{r} \times \vec{A}) \cdot d\vec{S} = -y dx dy$

$$\therefore \int (\vec{r} \times \vec{A}) \cdot d\vec{S} = - \int_0^2 \int_0^{2-x} y dy dx$$

$$= - \int_0^2 \left[\frac{1}{2} y^2 \right]_0^{2-x} dx$$

$$= - \int_0^2 \frac{1}{2} (2-x)^2 dx$$

$$= - \frac{1}{2} \left[\frac{(2-x)^3}{-3} \right]_0^2$$

$$= -\frac{4}{3}$$

$$\therefore \int (\vec{r} \times \vec{A}) \cdot d\vec{S} = \oint d\vec{r} \cdot \vec{A} = -\frac{4}{3}$$

proven

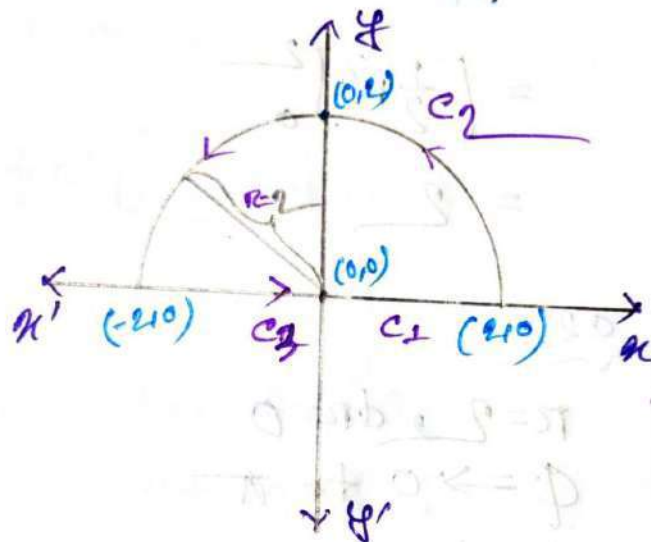
problem 2:-

⇒ Assume $\vec{A} = \hat{r} \cdot r \cos \phi + \hat{\phi} \sin \phi$

(a) find $\oint_C \vec{A} \cdot d\vec{r}$ over the semicircular

(b) find $\int_S (\nabla \times \vec{A}) \cdot d\vec{s}$ over the surface of the semicircle

(c) verify Stoke's theorem.



⇒ we know

for cylindrical coordinate system:-

$$d\vec{r} = \hat{r} dr + \hat{\phi} \cdot r \cdot d\phi + \hat{z} \cdot dz$$

$$h_1 = 1$$

$$h_2 = r$$

$$h_3 = 1$$

$$\vec{A} = \hat{r} \cdot r \cdot \cos \phi + \hat{\phi} \cdot \sin \phi$$

$$\therefore \vec{A} \cdot d\vec{r} = r \cdot \cos \phi dr + \sin \phi \cdot r \cdot d\phi$$

Is given:-

Along C_1 :-
note

$$\phi = 0$$

$$\therefore d\phi = 0$$

$$\therefore \vec{A} \cdot d\vec{r} = r \cdot dr \quad \text{where } r \Rightarrow 0 \text{ to } 2$$

$$I_{C_1} = \int_0^2 r \cdot dr$$

$$= \left[\frac{1}{2} r^2 \right]_0^2$$

$$= 2$$

Along C_2 :-
note

$$r = 2, dr = 0$$

$$\phi \Rightarrow 0 \text{ to } \pi$$

$$\therefore \vec{A} \cdot d\vec{r} = 2 \sin \phi \cdot d\phi$$

$$I_{C_2} = \int_0^\pi 2 \sin \phi \cdot d\phi$$

$$= -2 [\cos \phi]_0^\pi$$

$$= -2(-1 - 1)$$

$$= 4$$

Along ϕ axis

$$\phi = \pi$$

$$\therefore d\phi = 0$$

$$\therefore \vec{A} \cdot d\vec{r} = r \cos\phi \, dr \quad \text{where } r \rightarrow 2 \text{ to } 0$$

$$= -r \, dr \quad [\text{since } \phi = \pi]$$

$$\therefore I_{\phi} = \int_2^0 -r \, dr$$

$$= -\left[\frac{1}{2}r^2\right]_2^0$$

$$= 2$$

$$\therefore \oint \vec{A} \cdot d\vec{r} = I_{\phi_1} + I_{\phi_2} + I_{\phi_3}$$

$$= 2 + 4 + 2$$

$$= 8$$

Now

$$\vec{\nabla} \times \vec{A} = \begin{vmatrix} \hat{r} & r\hat{\phi} & \hat{z} \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial \phi} & \frac{\partial}{\partial z} \\ r\cos\phi & r\sin\phi & 0 \end{vmatrix} \times \frac{1}{r}$$

$$= \hat{z} (\sin\phi + r\sin\phi) \times \frac{1}{r}$$

$$= \hat{z} (1+r) \times \frac{1}{r} \sin\phi$$

$$\therefore d\vec{S} = \frac{1}{r} \cdot dr \cdot r \, d\phi$$

$$\underline{\text{Sol}} \quad (\vec{\nabla} \times \vec{A}) \cdot d\vec{z} = \frac{1}{2} r^2 (1+r) \sin \phi \cdot dr d\phi$$

$$= (1+r) \sin \phi \cdot dr d\phi$$

$$\int_S (\vec{\nabla} \times \vec{A}) \cdot d\vec{z} = \int_0^\pi \int_0^2 (1+r) \sin \phi \cdot dr \cdot d\phi$$

$$= \left[r + \frac{r^2}{2} \right]_0^2 \times -[\cos \phi]_0^\pi$$

$$= 4 \times (-1 - 1)$$

$$= 2 \times 4$$

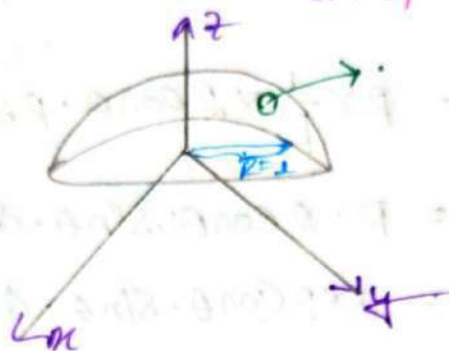
$$= 8$$

$$\therefore \int_S (\vec{\nabla} \times \vec{A}) \cdot d\vec{z} = \oint \vec{A} \cdot d\vec{r} = 8$$

(verified)

Example:-

$\Rightarrow \vec{A} = \hat{r} \cos \theta + \hat{\phi} \sin \theta$. Verify Stokes's theorem by evaluating it on the hemisphere of unit radius.



We know:-

for spherical co-ords:-

$$d\vec{r} = \hat{r} \cdot dr + \hat{\theta} \cdot r \cdot d\theta + \hat{\phi} \cdot r \sin \theta \cdot d\phi$$

$$h_1 = 1$$

$$h_2 = r$$

$$h_3 = r \sin \theta$$

$$\therefore \vec{\nabla} \times \vec{A} = \frac{1}{r^2 \sin \theta} \begin{vmatrix} \hat{r} & \hat{\theta} & \hat{\phi} \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial \theta} & \frac{\partial}{\partial \phi} \\ \cos \theta & r \cdot 0 & r \cdot \sin \theta \cdot \sin \theta \end{vmatrix}$$

$$= \frac{1}{r^2 \sin \theta} \left\{ \hat{r} (r \times 2 \sin \theta \cdot \cos \theta) - r \hat{\theta} (\sin^2 \theta - 0) + \hat{\phi} r (+\sin^2 \theta) \right\}$$

$$= \hat{r} \times \frac{1}{r} 2 \cos \theta - \hat{\theta} \times \frac{1}{r} \sin \theta + \hat{\phi} \frac{1}{r} \sin \theta$$

$$= \frac{1}{r} (\hat{r} 2 \cos \theta - \hat{\theta} \sin \theta + \hat{\phi} \sin \theta)$$

$$\therefore d\vec{s} = \hat{r} \cdot R d\theta \cdot R \sin\theta \cdot d\phi$$

$$\begin{aligned} \therefore (\vec{\nabla} \times \vec{A}) \cdot d\vec{s} &= R \times \frac{1}{R} \times 2 \cos\theta \cdot R \sin\theta \cdot d\theta \cdot d\phi \\ &= R \cdot 2 \cos\theta \cdot \sin\theta \cdot d\theta \cdot d\phi \\ &= 2 \cos\theta \cdot \sin\theta \cdot d\theta \cdot d\phi \\ &= \sin 2\theta \cdot d\theta \cdot d\phi \end{aligned}$$

$$\begin{aligned} \therefore \int_S (\vec{\nabla} \times \vec{A}) \cdot d\vec{s} &= \int_{\phi=0}^{2\pi} \int_{\theta=0}^{\frac{\pi}{2}} \sin 2\theta \cdot d\theta \cdot d\phi \\ &= \left[\phi \right]_0^{2\pi} \times \left[-\frac{\cos 2\theta}{2} \right]_0^{\frac{\pi}{2}} \\ &= 2\pi \times -\frac{1}{2} (\cos \pi - 1) \\ &= 2\pi \end{aligned}$$

Now for line integrals

We know:-

Spherical c.s:-

$$d\vec{r} = \hat{r} \cdot dr + \hat{\theta} \cdot r \cdot d\theta + \hat{\phi} \cdot r \cdot \sin\theta \cdot d\phi$$

$$\therefore \vec{A} \cdot d\vec{r} = \cos\theta \cdot dr + r \sin^2\theta \cdot d\phi$$

here,

$$r = 1$$

$$\therefore dr = 0$$

$$\theta = \frac{\pi}{2}$$

$$\therefore \vec{A} \cdot d\vec{x} = d\phi$$

$$\oint_C \vec{A} \cdot d\vec{x} = \int_0^{2\pi} d\phi$$

$$= [\phi]_0^{2\pi}$$

$$= 2\pi$$

$$\therefore \int_S (\nabla \times \vec{A}) \cdot d\vec{S} = \oint_C \vec{A} \cdot d\vec{x} = 2\pi$$

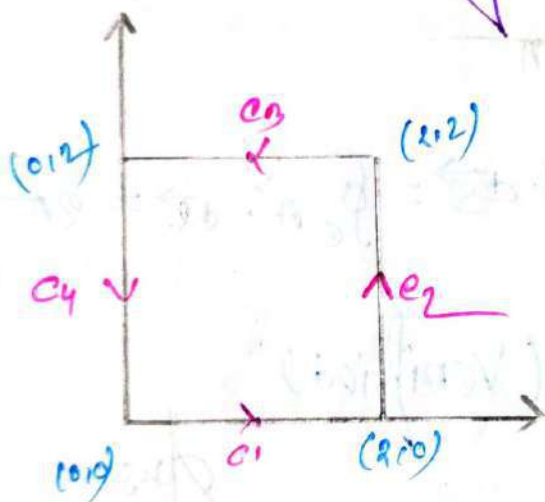
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Ans:-

Sample exercise

3. Assume that a vector field $A = \hat{x}(x^2 - y^2) + \hat{y}(x^2 - xy)$.

- Find $\oint_C \vec{A} \cdot d\vec{r}$ around the rectangular contour.
- Find $\int_S (\nabla \times A) \cdot d\vec{s}$ over the rectangular area.
- Verify Stokes's theorem.
- Can A be expressed as gradient of a scalar?



$\Rightarrow$ We know -

Diff. length $d\vec{r} = \hat{x} \cdot dx + \hat{y} \cdot dy + \hat{z} \cdot dz$

In given -

$$\vec{A} = \hat{x}(x^2 - y^2) + \hat{y}(x^2 - xy)$$

$$\therefore \vec{A} \cdot d\vec{r} = (x^2 - y^2) dx + (x^2 - xy) dy$$

TOPIC NAME : _____

DAY : _____

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Along C_1 :-

$$y=0$$

$$\therefore dy=0 \text{ where } x \rightarrow 0 \text{ to } 2$$

$$\therefore \vec{A} \cdot d\vec{r} = x^2 \cdot dx$$

$$\therefore I_{C_1} = \int_0^2 x^2 \cdot dx$$

$$= \left[\frac{1}{3} x^3 \right]_0^2$$

$$= \frac{8}{3}$$

Along C_2 :-

$$x=2$$

$$dx=0 \text{ where } y \rightarrow 0 \text{ to } 2$$

$$\therefore \vec{A} \cdot d\vec{r} = (4-2y) dy$$

$$\therefore I_{C_2} = \int_0^2 (4-2y) dy$$

$$= [4y]_0^2 - 2 \left[\frac{1}{2} y^2 \right]_0^2$$

$$= 8 - 4$$

$$= 4$$

Along C_3 :-

$$y=2$$

$$\therefore dy=0 \text{ where } x \rightarrow 2 \text{ to } 0$$

$$\therefore \vec{A} \cdot d\vec{r} = (x^2-4) dx$$

$$\therefore I_{C_3} = \int_2^0 (x^2-4) dx$$

$$= \left[\frac{1}{3} x^3 \right]_2^0 - [4x]_2^0$$

$$= -\frac{8}{3} + 8$$

$$= \frac{16}{3}$$

Along C_4 :-

$$x=0$$

$$\therefore dx=0 \text{ where } y \rightarrow 2 \text{ to } 0$$

$$\therefore \vec{A} \cdot d\vec{r} = 0$$

$$\therefore I_{C_4} = 0$$

$$\therefore \oint \vec{A} \cdot d\vec{r} = I_{C_1} + I_{C_2} + I_{C_3} + I_{C_4}$$

$$= \frac{8}{3} + 4 + \frac{16}{3}$$

$$= 12$$

Now

$$\vec{r} \times \vec{A} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ (x^2+y) & (x^2-xy) & 0 \end{vmatrix}$$

$$= \hat{z} \{ (2x-y) - (-2y) \}$$

$$= \hat{z} (2x+y)$$

$$\therefore d\vec{r} = \hat{z} \cdot dx dy$$

$$\text{Sol} \int (\vec{r} \times \vec{A}) \cdot d\vec{r} = (2x+y) \cdot dx dy$$

$$\therefore \int (\vec{r} \times \vec{A}) \cdot d\vec{r} = \int_0^2 \int_0^2 (2x+y) dx dy$$

$$= \int_0^2 \left[\frac{1}{2} \times 2x^2 + xy \right]_0^2 dy$$

$$= \int_0^2 [4 + 2y] dy$$

$$= [4y + y^2]_0^2$$

$$= 8 + 4$$

$$= 12$$

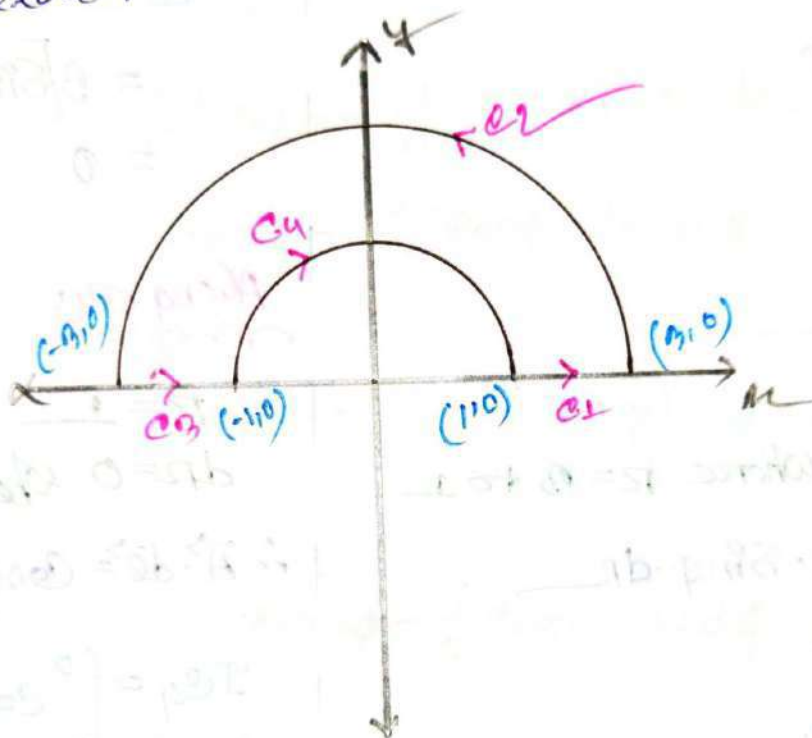
$$\therefore \oint \vec{A} \cdot d\vec{r} = \int (\vec{r} \times \vec{A}) \cdot d\vec{r} = 12$$

(Verified)

4. Assume that a vector $\vec{A} = \hat{r} r \sin \phi + \hat{\phi} r \cos \phi$.

(a) Find $\int_S (\vec{\nabla} \times \vec{A}) \cdot d\vec{S}$ over the surface of the semicircles.

(b) Find $\oint_C \vec{A} \cdot d\vec{r}$ over the semicircular contour shown below.



$\Rightarrow$ Is given —
mon

$$\vec{A} = \hat{r} r \sin \phi + \hat{\phi} r \cos \phi$$

We know —
mon

for cylindrical coordinate system:

$$d\vec{r} = \hat{r} dr + \hat{\phi} r d\phi + \hat{z} dz$$

$$h_r = 1$$

$$h_\phi = r$$

$$h_z = 1$$

$$\therefore \vec{A} \cdot d\vec{r} = r \sin \phi dr + r \cos \phi \cdot d\phi$$

Along e_1 :-

$$\phi = 0$$

$$\therefore d\phi = 0 \text{ where } r = 0 \text{ to } 3$$

$$\begin{aligned} \therefore \vec{A} \cdot d\vec{r} &= r \sin \phi \cdot dr \\ &= r \sin 0 \cdot dr \\ &= 0 \end{aligned}$$

$$\therefore I_{e_1} = 0$$

Along e_3 :-

$$\phi = \pi$$

$$d\phi = 0 \text{ where } r = 3 \text{ to } 1$$

$$\begin{aligned} \therefore \vec{A} \cdot d\vec{r} &= r \sin \phi \cdot dr \\ &= 0 \end{aligned}$$

$$\therefore I_{e_3} = 0$$

Along e_2 :-

$$r = 3$$

$$\therefore dr = 0 \text{ where } \phi \rightarrow 0 \text{ to } \pi$$

$$\therefore \vec{A} \cdot d\vec{r} = 3 \cos \phi \cdot d\phi$$

$$I_{e_2} = \int_0^\pi 3 \cos \phi \cdot d\phi$$

$$= 3 [\sin \phi]_0^\pi$$

$$= 0$$

Along e_4 :-

$$r = 1$$

$$dr = 0 \text{ where } \phi \rightarrow \pi \text{ to } 0$$

$$\therefore \vec{A} \cdot d\vec{r} = \cos \phi \cdot d\phi$$

$$I_{e_4} = \int_\pi^0 \cos \phi \cdot d\phi$$

$$= [\sin \phi]_\pi^0$$

$$= 0 - \sin \pi$$

$$= 0$$

$$\therefore \oint \vec{A} \cdot d\vec{r} = I_{e_1} + I_{e_2} + I_{e_3} + I_{e_4}$$

$$= 0 + 0 + 0 + 0$$

$$= 0$$

Now! $\vec{\nabla} \times \vec{A} = \begin{vmatrix} \hat{r} & r\hat{\phi} & \hat{z} \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial \phi} & \frac{\partial}{\partial z} \\ r\sin\phi & r\cos\phi & 0 \end{vmatrix} \times \frac{1}{r}$

$$= \hat{z} (\cos\phi - r\cos\phi) \times \frac{1}{r}$$

$$\therefore d\vec{S} = \hat{z} \cdot r dr d\phi$$

$$\begin{aligned} \text{So } (\vec{\nabla} \times \vec{A}) \cdot d\vec{S} &= \left(\frac{1}{r} \cos\phi - \frac{1}{r^2} \cos\phi \right) \cdot r dr d\phi \\ &= (\cos\phi - \frac{1}{r} \cos\phi) \cdot dr \cdot d\phi \end{aligned}$$

$$\begin{aligned} \therefore \int (\vec{\nabla} \times \vec{A}) \cdot d\vec{S} &= \int_0^{\pi} \int_1^3 (\cos\phi - \frac{1}{r} \cos\phi) dr d\phi \\ &= \int_0^{\pi} \int_1^3 [\cos\phi - \frac{1}{r} \cos\phi] dr d\phi \\ &= \int_1^3 \int_0^{\pi} (\cos\phi - \frac{1}{r} \cos\phi) d\phi dr \\ &= \int_1^3 [\sin\phi - \frac{1}{r} \sin\phi]_0^{\pi} dr \\ &= \int_1^3 (0 - 0) dr \\ &= [0 \cdot r]_1^3 \\ &= 0 \end{aligned}$$

$$\therefore \oint \vec{A} \cdot d\vec{r} = \int (\vec{\nabla} \times \vec{A}) \cdot d\vec{S} = 0$$

(verified)

Q. Using
System of Differential
Eigenvalue &
Eigen Vector:

Example:-

⇒ Solve the following system of Differential equation using eigenvalues and eigenvectors.

$$\begin{cases} \dot{x}_1(t) = -1.5x_1(t) + 0.5x_2(t) \\ \dot{x}_2(t) = x_1(t) - x_2(t) \end{cases}$$

with $x_1(0) = 5$

$x_2(0) = 4$

$$\Rightarrow \underline{\text{Let}}, \begin{pmatrix} \dot{x}_1(t) \\ \dot{x}_2(t) \end{pmatrix} = \begin{pmatrix} -1.5 & 0.5 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} x_1(t) \\ x_2(t) \end{pmatrix} \quad \begin{pmatrix} x_1(0) \\ x_2(0) \end{pmatrix} = \begin{pmatrix} 5 \\ 4 \end{pmatrix}$$

In Matrix form:-

$$\dot{x}(t) = AX(t) \Rightarrow x(t) = Cve^{\lambda t}$$

where:-

λ = eigenvalue of A

v = eigenvector of A

We know:-

C.H. Matrix is $= A - \lambda I$

$$\begin{aligned} &= \begin{bmatrix} -1.5 & 0.5 \\ 1 & -1 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} \\ &= \begin{bmatrix} -1.5 - \lambda & 0.5 \\ 1 & -1 - \lambda \end{bmatrix} \end{aligned}$$

Now:- Q.H polynomial $= |A - \lambda I|$

$$= \begin{vmatrix} -1.5 - \lambda & 0.5 \\ 1 & -1 - \lambda \end{vmatrix}$$

$$= 1.5 + 1.5\lambda + \lambda + \lambda^2 = 0.5$$

$$= \lambda^2 + 2.5\lambda + 1$$

Now:-

Q.H Equation $\Rightarrow |A - \lambda I| = 0$

$$\Rightarrow \lambda^2 + 2.5\lambda + 1 = 0$$

$$\therefore \lambda = -\frac{1}{2}, -2$$

Now:-

The eigenvectors:-

for $\lambda = -\frac{1}{2}$,

we know:-

$$\Rightarrow (A - \lambda I)V = 0$$

$$= (A - \lambda I)$$

$$= \begin{bmatrix} -1.5 + \frac{1}{2} & 0.5 \\ 1 & -1 + \frac{1}{2} \end{bmatrix}$$

$$= \begin{bmatrix} -1 & 0.5 \\ 1 & -0.5 \end{bmatrix}$$

Now:-

$$[A|B] = \left[\begin{array}{cc|c} -1 & 0.5 & 0 \\ 1 & -0.5 & 0 \end{array} \right] \quad \tilde{r}_2 = r_2 + r_1$$

$$= \left[\begin{array}{cc|c} -1 & 0.5 & 0 \\ 0 & 0 & 0 \end{array} \right]$$

here, v_2 is free variable

So,

let, $v_2 = a$

$$\therefore v_1 = 0.5a$$

$$\therefore V_1 = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} 0.5a \\ a \end{pmatrix}$$

Now For $\lambda = -2$:-

$$= (A - \lambda I)$$

$$= \begin{bmatrix} -1.5+2 & 0.5 \\ 1 & -1+2 \end{bmatrix}$$

$$= \begin{bmatrix} 0.5 & 0.5 \\ 1 & 1 \end{bmatrix}$$

So $[A|B] = \left[\begin{array}{cc|c} 0.5 & 0.5 & 0 \\ 1 & 1 & 0 \end{array} \right] \quad \tilde{r}_2 = 2r_1 - r_2$

$$= \left[\begin{array}{cc|c} 0.5 & 0.5 & 0 \\ 0 & 0 & 0 \end{array} \right]$$

here, v_2 is free variable

So

let, $v_2 = b$
 $v_1 = -b \quad \therefore V_2 = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} -b \\ b \end{pmatrix}$

So,
$$\begin{pmatrix} x_1(t) \\ x_2(t) \end{pmatrix} = c_1 \begin{pmatrix} 0.5a \\ a \end{pmatrix} e^{-0.5t} + c_2 \begin{pmatrix} -b \\ b \end{pmatrix} e^{-2t} \quad \text{--- (i)}$$

Now putting the initial condition:—

$$c_1 \times 0.5a - c_2 b = 5 \quad \text{--- (ii)}$$

$$c_1 a + c_2 b = 4 \quad \text{--- (iii)}$$

$$1.5c_1 = 0$$

$$\therefore c_1 = -\frac{6}{a}$$

put the value of c_1 in eq (iii) =

$$\therefore c_2 = \frac{-2}{b}$$

Now:- from equation (i) $\Rightarrow$

$$\begin{pmatrix} x_1(t) \\ x_2(t) \end{pmatrix} = \frac{6}{a} \begin{pmatrix} 0.5a \\ a \end{pmatrix} e^{-0.5t} + \left(\frac{-2}{b}\right) \begin{pmatrix} -b \\ b \end{pmatrix} e^{-2t}$$

$$\therefore x_1(t) = 3e^{-0.5t} + 2e^{-2t}$$

$$x_2(t) = 6e^{-0.5t} - 2e^{-2t}$$

Ans:

Sample Exercise 4.2

Q. Solve the following system of differential equations using eigenvalue and eigenvector where, $\dot{x}_1(t) = \frac{dx_1}{dt}$
 $\dot{x}_2(t) = \frac{dx_2}{dt}$

$$(a) \begin{cases} \dot{x}_1(t) = x_1(t) + 2x_2(t) \\ \dot{x}_2(t) = 3x_1(t) + 2x_2(t) \end{cases} \quad \text{with } \begin{cases} x_1(0) = 0 \\ x_2(0) = -4 \end{cases}$$

$$\Rightarrow \begin{pmatrix} \dot{x}_1(t) \\ \dot{x}_2(t) \end{pmatrix} = \begin{pmatrix} 1 & 2 \\ 3 & 2 \end{pmatrix} \begin{pmatrix} x_1(t) \\ x_2(t) \end{pmatrix} \quad \begin{pmatrix} x_1(0) \\ x_2(0) \end{pmatrix} = \begin{pmatrix} 0 \\ -4 \end{pmatrix}$$

In matrix form:-

$$\dot{x}(t) = Ax(t) \Rightarrow x(t) = Cve^{At}$$

where,

λ = eigenvalue of A

v = eigenvector of A

we know-

$$\text{C.H. polynomial} = |A - \lambda I|$$

$$= \begin{vmatrix} 1-\lambda & 2 \\ 3 & 2-\lambda \end{vmatrix}$$

$$= 2 - \lambda - 2\lambda + \lambda^2 - 6$$

$$= \lambda^2 - 3\lambda - 4$$

$$\text{C.H. equation} \Rightarrow |A - \lambda I| = 0$$

$$\lambda^2 - 3\lambda - 4 = 0$$

$$\therefore \lambda = 4, -1$$

Now, we know:-

The eigenvector $V = (A - \lambda I)V = 0$

for $\lambda = -1$:-

$$= (A - \lambda I)$$

$$= \begin{bmatrix} 1+1 & 2 \\ 3 & 2+1 \end{bmatrix}$$

$$= \begin{bmatrix} 2 & 2 \\ 3 & 3 \end{bmatrix}$$

Now,

$$[A|B] = \left[\begin{array}{cc|c} 2 & 2 & 0 \\ 3 & 3 & 0 \end{array} \right] \quad R_2 = 3R_1 - 2R_2$$

$$= \left[\begin{array}{cc|c} 2 & 2 & 0 \\ 0 & 0 & 0 \end{array} \right]$$

So, v_2 is free variable

Let $v_2 = a$

$$\therefore v_1 = -a$$

$$\therefore V_1 = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} -a \\ a \end{pmatrix}$$

for $\lambda = 4$:-

$$= (A - \lambda I)$$

$$= \begin{bmatrix} 1-4 & 2 \\ 3 & 2-4 \end{bmatrix}$$

$$= \begin{bmatrix} -3 & 2 \\ 3 & -2 \end{bmatrix}$$

Now $[A|b] = \left[\begin{array}{cc|c} -3 & 2 & 0 \\ 3 & -2 & 0 \end{array} \right] \quad r_2 = r_2 + r_1$

$$= \left[\begin{array}{cc|c} -3 & 2 & 0 \\ 0 & 0 & 0 \end{array} \right]$$

So, v_2 is free variable.

Let, $v_2 = b$

$$\therefore v_1 = \frac{2b}{3}$$

$$\therefore V_2 = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} \frac{2b}{3} \\ b \end{pmatrix}$$

So, $\begin{pmatrix} x_1(t) \\ x_2(t) \end{pmatrix} = c_1 \begin{pmatrix} -a \\ a \end{pmatrix} e^{-4t} + c_2 \begin{pmatrix} \frac{2b}{3} \\ b \end{pmatrix} e^{4t} \quad \text{--- (I)}$

Now

putting the initial condition:-

$$-ac_1 + \frac{2b}{3}c_2 = 0$$

$$\Rightarrow -3ac_1 + 2bc_2 = 0 \quad \text{--- (I)}$$

$$\Rightarrow ac_1 + bc_2 = -4 \quad \text{--- (II)}$$

(I) - (II) \times 2 =

$$\begin{array}{r} -3ac_1 + 2bc_2 = 0 \\ 2ac_1 + 2bc_2 = -4 \times 2 \\ \hline -5ac_1 = 4 \times 2 \\ \therefore c_1 = \frac{-4 \times 2}{5a} \\ \therefore c_1 = \frac{-8}{5a} \end{array}$$

Put the value of c_1 in eq (II) =

$$\begin{aligned} -\frac{8}{5} + bc_2 &= -4 \\ bc_2 &= -4 + \frac{4 \times 2}{5} \\ c_2 &= \frac{-10}{5b} \end{aligned}$$

Now Form equation (1) $\Rightarrow$

$$\begin{pmatrix} \dot{x}_1(t) \\ \dot{x}_2(t) \end{pmatrix} = -\frac{8}{5} e^{-t} \begin{pmatrix} -a \\ a \end{pmatrix} - \frac{12}{5b} e^{4t} \begin{pmatrix} \frac{2b}{5} \\ b \end{pmatrix}$$

$$\therefore x_1(t) = +\frac{8}{5} e^{-t} - \frac{8}{5} e^{4t}$$

$$x_2(t) = -\frac{8}{5} e^{-t} - \frac{12}{5} e^{4t}$$

Ans.

$$\textcircled{b} \begin{cases} \dot{x}_1(t) = -5x_1(t) + x_2(t) \\ \dot{x}_2(t) = 4x_1(t) - 2x_2(t) \end{cases} \text{ with } \begin{cases} x_1(0) = 1 \\ x_2(0) = 2 \end{cases}$$

$$\Rightarrow \begin{pmatrix} \dot{x}_1(t) \\ \dot{x}_2(t) \end{pmatrix} = \begin{pmatrix} -5 & 1 \\ 4 & -2 \end{pmatrix} \begin{pmatrix} x_1(t) \\ x_2(t) \end{pmatrix} \quad \begin{pmatrix} x_1(0) \\ x_2(0) \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

In Matrix form:-

$$\dot{x}(t) = AX(t) \Rightarrow x(t) = Cve^{\lambda t}$$

we know:-

$$\text{C.H polynomial} = |A - \lambda I|$$

$$= \begin{vmatrix} -5-\lambda & 1 \\ 4 & -2-\lambda \end{vmatrix}$$

$$= 10 + 5\lambda + 2\lambda + \lambda^2 - 4$$

$$= \lambda^2 + 7\lambda + 6$$

$$\text{C.H Equation} \Rightarrow |A - \lambda I| = 0$$

$$\therefore \lambda^2 + 7\lambda + 6 = 0$$

$$\therefore \lambda = -1, -6$$

Now:- we know:-
non

The Eigenvector $\Rightarrow (A - \lambda I)V = 0$

for $\lambda = -1$:-
non

$$= A - \lambda I$$

$$= \begin{bmatrix} -5+1 & 1 \\ 4 & -2+1 \end{bmatrix}$$

$$= \begin{bmatrix} -4 & 1 \\ 4 & -1 \end{bmatrix}$$

so $[A|0] = \left[\begin{array}{cc|c} -4 & 1 & 0 \\ 4 & -1 & 0 \end{array} \right] \quad \tilde{r}_2 = r_2 + r_1$

$$= \left[\begin{array}{cc|c} -4 & 1 & 0 \\ 0 & 0 & 0 \end{array} \right]$$

so, v_2 is free variable.

let, $v_2 = a$

$$\therefore v_1 = \frac{a}{4} \quad \left(v_1 = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} \frac{a}{4} \\ a \end{pmatrix} \right)$$

For $\lambda = -6$:-
non

$$= A - \lambda I$$

$$= \begin{bmatrix} -5+6 & 1 \\ 4 & -2+6 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 1 \\ 4 & 4 \end{bmatrix}$$

so $[A|0] = \left[\begin{array}{cc|c} 1 & 1 & 0 \\ 4 & 4 & 0 \end{array} \right] \quad \tilde{r}_2 = 4r_1 - r_2$

$$= \left[\begin{array}{cc|c} 1 & 1 & 0 \\ 0 & 0 & 0 \end{array} \right]$$

so, v_2 is free variable.

Let $v_2 = b$
 $\therefore v_1 = -b$

$$\therefore V_2 = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} -b \\ b \end{pmatrix}$$

Now, $\begin{pmatrix} \dot{x}_1(t) \\ \dot{x}_2(t) \end{pmatrix} = c_1 \begin{pmatrix} \frac{a}{4} \\ a \end{pmatrix} e^{-t} + c_2 \begin{pmatrix} -b \\ b \end{pmatrix} e^{-6t}$ ——— (1)

Putting the initial condition:-

$$\frac{a}{4} c_1 - b c_2 = 1$$

$$\Rightarrow a c_1 - 4 b c_2 = 4 \quad \text{--- (II)}$$

$$\Rightarrow a c_1 + b c_2 = 2 \quad \text{--- (III)}$$

(II) - (III) =

$$\begin{array}{r} a c_1 - 4 b c_2 = 4 \\ (-) \quad a c_1 + b c_2 = 2 \\ \hline -5 b c_2 = 2 \\ \therefore c_2 = \frac{-2}{5b} \end{array}$$

Put the value of c_2 in eq (III)

$$\begin{array}{l} a c_1 + \left(\frac{-2}{5} \right) = 2 \\ a c_1 = 2 + \frac{2}{5} \\ \therefore c_1 = \frac{12}{5a} \end{array}$$

From equation (1) =

$$\begin{pmatrix} \dot{x}_1(t) \\ \dot{x}_2(t) \end{pmatrix} = \frac{12}{5a} \begin{pmatrix} \frac{a}{4} \\ a \end{pmatrix} e^{-t} + \frac{-2}{5b} \begin{pmatrix} -b \\ b \end{pmatrix} e^{-6t}$$

$$\therefore x_1(t) = \frac{3}{5} e^{-t} + \frac{2}{5} e^{-6t}$$

$$x_2(t) = \frac{12}{5} e^{-t} - \frac{2}{5} e^{-6t}$$

Ans:-

$$\textcircled{a} \begin{cases} \dot{x}_1(t) = x_2(t) \\ \dot{x}_2(t) = 1.5x_1(t) - 2.5x_2(t) \end{cases} \quad \text{with } x_1(0) = -4 \\ x_2(0) = 0$$

$$\Rightarrow \begin{pmatrix} \dot{x}_1(t) \\ \dot{x}_2(t) \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 1.5 & -2.5 \end{pmatrix} \begin{pmatrix} x_1(t) \\ x_2(t) \end{pmatrix} \quad \begin{pmatrix} x_1(0) \\ x_2(0) \end{pmatrix} = \begin{pmatrix} -4 \\ 0 \end{pmatrix}$$

In Matrix Form:-

$$(\dot{x}(t)) = AX(t) \Rightarrow x(t) = Ce^{\lambda t}$$

We know:-

$$\text{C.H polynomial} = |A - \lambda I|$$

$$= \begin{vmatrix} -\lambda & 1 \\ 1.5 & -2.5 - \lambda \end{vmatrix} = 2.5\lambda + \lambda^2 - 1.5$$

$$\therefore \text{C.H equation} \Rightarrow |A - \lambda I| = 0$$

$$\Rightarrow \lambda^2 + 2.5\lambda - 1.5 = 0$$

$$\therefore \lambda = -3, 0.5$$

for

Now:-

We know:-

$$\text{The eigenvector} \Rightarrow (A - \lambda I)V = 0$$

for

for

$$= (A - \lambda I)$$

$$= \begin{bmatrix} 3 & 1 \\ 1.5 & 0.5 \end{bmatrix}$$

$$\text{So, } [A|B] = \left[\begin{array}{cc|c} 3 & 1 & 6 \\ 1.5 & 0.5 & 0 \end{array} \right] \quad \tilde{r}_2 = 2r_2 - r_1$$

$$= \left[\begin{array}{cc|c} 0 & 0 & 6 \\ 1.5 & 0.5 & 0 \end{array} \right] \quad r_1 \leftrightarrow r_2$$

$$= \left[\begin{array}{cc|c} 1.5 & 0.5 & 6 \\ 0 & 0 & 0 \end{array} \right]$$

here v_2 is free variable.

let $v_2 = a$

$$v_1 = -\frac{a}{3}$$

$$\therefore V_1 = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} -\frac{a}{3} \\ a \end{pmatrix}$$

for $\lambda = 0.5$:-

$$= (A - \lambda I)$$

$$= \begin{bmatrix} -0.5 & 1 \\ 1.5 & -3 \end{bmatrix}$$

so $[A|B] = \left[\begin{array}{cc|c} -0.5 & 1 & 0 \\ 1.5 & -3 & 0 \end{array} \right] \quad R_2 = R_2 + 3R_1$

$$= \left[\begin{array}{cc|c} -0.5 & 1 & 0 \\ 0 & 0 & 0 \end{array} \right]$$

here v_2 is free variable

let $\therefore v_2 = b$

$$v_1 = 2b$$

$$\therefore V_2 = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} 2b \\ b \end{pmatrix}$$

$$\therefore \begin{pmatrix} x_1(t) \\ x_2(t) \end{pmatrix} = c_1 \begin{pmatrix} -\frac{a}{3} \\ a \end{pmatrix} e^{-0.5t} + c_2 \begin{pmatrix} 2b \\ b \end{pmatrix} e^{0.5t} \quad \text{--- (1)}$$

putting the initial conditions:-

$$-\frac{a}{3}c_1 + 2bc_2 = -4$$

$$\Rightarrow -ac_1 + 6bc_2 = -12 \quad \text{--- (II)}$$

$$\Rightarrow ac_1 + bc_2 = 0 \quad \text{--- (III)}$$

$$\textcircled{II} + \textcircled{III} =)$$

$$-ae_1 + 6be_2 = -12$$

$$ae_1 + be_2 = 0$$

$$7be_2 = -12$$

$$\therefore e_2 = -\frac{12}{7b}$$

put the value of e_2 in eq $\textcircled{II} =)$

$$ae_1 - \frac{12}{7} = 0$$

$$\therefore e_1 = \frac{12}{7a}$$

From equation $\textcircled{I} =)$

$$\begin{pmatrix} x_1(t) \\ x_2(t) \end{pmatrix} = \frac{12}{7a} \begin{pmatrix} -\frac{a}{3} \\ a \end{pmatrix} e^{-3t} + \left(\frac{-12}{7b} \right) \begin{pmatrix} 2b \\ b \end{pmatrix} e^{0.5t}$$

$$\therefore x_1(t) = -\frac{12}{7} e^{-3t} - \frac{6}{7} e^{0.5t}$$

$$x_2(t) = \frac{12}{7} e^{-3t} - \frac{12}{7} e^{0.5t}$$

Ans:-